LET US BREW AN AI MODEL

What Tiny Language Models Can Actually Learn When Given One Job



ZHIQI TAO

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Version 20261001

Zhiqi Tao

Building a small language model is relatively easy. Determining what it actually learned is the hard part.

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About the author. Zhiqi Tao works on AI systems, with a focus on building and evaluating small models in controlled settings. This book grew out of a running laboratory notebook: build a tiny model, give it one job, and find out precisely what it can and cannot learn — with the failure ledger as much a part of the result as the successes.

Companion repository and errata. The code, scripts, configurations, and checkpoints that reproduce every experiment in this book live in the companion repository (see Appendix C for the quick start and the release tag). An online errata page will be maintained for any corrections discovered after publication; report issues to zhiqi.tao@gmail.com.

Preface

This book is a controlled empirical investigation into what a small language model can actually learn when it is given exactly one narrowly defined, objectively testable job.

The task: read a natural-language programming request and produce a Python program, which is then executed against hidden tests.

Along the way we build a laboratory: a controlled world (a sandbox), a teacher (a semantic data generator), a judge (an execution-based evaluator), and rivals (simple baselines). The model is the instrument. The experiment is the subject. And the honest territory includes the moments the model falls flat — and the moment we discovered that our own measuring stick was broken, not the model.

Who this book is for

This is an accessible experimental case study, not a build-from-scratch transformer textbook. It is written for readers who are comfortable reading (or willing to skim) Python and who want to see how to set up a small, honest evaluation of a tiny model — and, more than that, how easily a number from such an experiment can be misread. No GPU or deep-learning background is required to follow the conceptual argument; a little Python helps with the code listings. Nothing here needs a large model or a large

machine — the whole investigation runs on a single-laptop CPU.

What the book claims (and what it does not)

We set the scope deliberately narrow, and we repeat it throughout. All the results are measurements on **one synthetic task** (natural-language request → single Python function), **one architecture family** (a small decoder-only transformer), **one training regime** (CPU, fixed budgets), and parameter counts up to roughly 30M. We do not claim a universal law about tiny models, and we do not claim to have identified the internal cause of the limits we observe. We claim only: *under this setup, here is what was measured, here is what did not move, and here is what remains open.* If a reader takes nothing else from the book, we hope it is the discipline of saying exactly what a measurement establishes and leaving the mechanism open when the controls have not identified it.

How to use the book

- **Read the body for the argument**. Each chapter opens with a concrete image before the mechanics.
- **Use the appendices for reproducibility**. Appendix A shows the generator; Appendix B is the failure ledger; Appendix C is the experiment registry that ties every number to a run ID, config, and artifact.
- **Run it yourself if you like**. The scripts, configs, and a release tag are in the companion repository (see Appendix C). On this CPU-only setup, the pilot trains in minutes; the larger models take longer.

The honest payoff is not "small models are inadequate. It is a precise map of where the boundary sits under our tests — and the many ways a promising-looking number can be an artifact of how we measured it.

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PART I.

The Question

CHAPTER 1.

Let Us Brew an AI Model

A tiny model is given one job. It reads a sentence — *"write a function that takes two numbers, a and b, and returns their sum"* — and it writes back the right Python:

```
[] def solve(a, b): return a + b
```

We run the hidden tests. It passes. If this were a demo, we'd move on.

But this is not a demo. So before we celebrate, we ask the one question that turns this whole book into an experiment rather than a party trick:

Why did it get it right?

Maybe it genuinely learned that "sum" and "a and b" mean "add them together." Maybe it memorized that exact sentence from training. Maybe it noticed this was nearly identical to an example it saw and copied the answer. Maybe a much dumber machine — one that does nothing but look up the closest training example — could have done the same thing. Or maybe our test wasn't as hard as we thought.

Making the model answer is easy. **Knowing why it answered is the whole game**.

1.1. One job, and a way to check it

We give the model exactly one, narrow, checkable job:

> **read a plain-English programming request → produce a working Python program**

Then we *run* that program against hidden tests. That single choice — executable output plus hidden tests — is what makes this a real experiment instead of a showcase, because it hands us the things a language-model toy usually lacks:

- inputs we control,
- code we can actually execute,
- an objective pass/fail,
- data we can generate in any quantity,
- difficulty we can dial up or down,
- tests we can keep hidden from the model,
- splits we can design to expose memorization,
- audits we can run to catch shortcuts.

1.2. What this book is (and isn't)

This is not a transformer tutorial. It is not a guide to chatbots, agents, or coding assistants. It is a careful, honest investigation of a single question:

> **How much structured programming capability can a small language model actually learn — and how do we tell whether it *learned* it, rather than memorized it or found a cheap shortcut?**

1.3. The part that's actually hard

The natural guess is that *building* a small model is the hard part. It isn't. Once the laboratory works, a ~0.9 million-parameter model will reliably turn a handful of programming sentences into correct Python — in-distribution, that is nearly easy.

The hard part is everything after: proving it didn't just copy, checking whether it still works when the *words* change but not the *computation*, testing whether it can *combine* two things it knows into a third thing it's never seen, and auditing our own evaluator to make sure the number it reports means anything.

This book builds a laboratory, runs those tests, and reports — plainly — what happened. Including the moments the model fell flat, and the moment we discovered our own measuring stick was broken, not the model.

The model was the instrument. The experiment was the subject. Let's brew.

CHAPTER 2.

What Does It Mean to Learn?

We're going to use a word a lot in this book: *learned*. Before we do, we have to be honest about what it means — because "the model learned to add" is a sentence people say all the time, and it's usually hiding a big assumption.

2.1. A simple way to think about it

Imagine a child who aces a worksheet of addition problems. Every problem is like "4 + 3 = ?" and the child gets them all right. We'd be tempted to say "the child can add."

But there are two very different ways to get that perfect score:

1. The child *understands addition* — show them "37 + 58" and they solve it, because they know how to add any two numbers.
2. The child *memorized the exact answers* — they've seen these exact problems before and just remember that "4 + 3 = 7," but hand them a problem they've never seen and they freeze.

A perfect worksheet grade can't tell these two apart. **The worksheet is a bad test**. Only a *new* problem can.

This is the entire premise of the book in one image. Our model gets a perfect-ish worksheet sometimes. The question is always: would it pass the *new* problem?

2.2. What "learned" means here

So we define it narrowly and honestly. A model shows **learned capability** when, to the degree the evidence supports:

1. **It gets better as it trains** — the loss falls, and the thing we care about reliably improves.
2. **It works on examples it never saw** — not just the ones in training.
3. **It survives harmless changes** — different variable names, different wording, a different way of saying the same instruction.
4. **It can combine things it knows** — put two learned pieces together in a new arrangement, not just repeat an arrangement it already saw.
5. **Simpler explanations have been ruled out**. The mundane alternatives — memorization, copying a near-duplicate, a template, a leaked test, a buggy evaluator — must be tested and rejected.

We are deliberately **not** asking whether the model "really understands" in a philosophical sense. We're asking narrower, answerable things: does it generalize? to what? and could a dumber mechanism explain the result?

2.3. The four questions of generalization

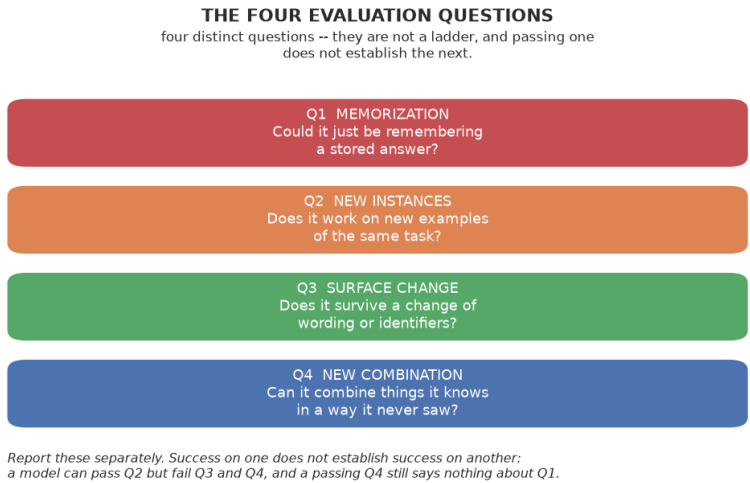


FIGURE 2.1. The four evaluation questions

The four questions make the point explicit. Each asks something distinct about a result, and we never collapse them into a single number. They are not a ladder — passing one does not establish the next; a model can answer the second question but fail the third and the fourth.

This is the book's organizing idea. It is four distinct questions about a result: from "could this just be remembering?" to "can it combine things it never saw combined?" and they must be reported separately.

Level 1 — Memorization. The model reproduces something it saw in training: an exact example, an exact constant, a familiar phrase, a near-duplicate. *Question: could this just be remembering?*

Level 2 — Same task, new examples. The model works on a new instance of the same kind of problem. *Question: does it work on problems it hasn't seen before?*

Level 3 — Same problem, different surface. The computation is identical, but the *outer layer* changes: new numbers, new variable names, reworded instruction, harmless distractors. *Question: does it survive changes*

that shouldn't matter?

Level 4 — New combination. The model must combine pieces it knows into something that never appeared in training. It learned A, B, and C separately — can it do A *sandwiched inside* B, or A followed by C? *Question: can it put things together, or only repeat what it already saw?*

A rule we'll follow strictly: never mash these four into one number. A model that scores 90% on familiar problems and 0% on a reworded version has not "generalized" — it memorized the phrasing. One number would hide exactly the thing we care about.

2.4. The recurring question: "how did it cheat?"

Because we refuse to trust a single score, every experiment asks the same skeptical question:

> **How could the model have gotten this without learning what we think it learned?**

The usual suspects are always on the table: memorized templates, memorized constants, memorized name patterns, wording quirks, unbalanced data, train/test leakage, near-duplicates, retrieval, a weak evaluator, hardcoded answers, hidden-test leakage.

Every major experiment therefore ends with a **shortcut audit**, six steps: (1) the suspected cheat, (2) why it could explain the result, (3) a control test, (4) what the control found, (5) what that rules out, and (6) what's still unresolved.

That audit — not the headline number — is the book's signature. It's the difference between "the model works" and "we know why the model works."

CHAPTER 3.

Give the Model One Job

Before we can run any experiment, we have to pin down the task so precisely that a machine can carry it out and we can score it objectively. Vague tasks give vague answers; a task with concrete inputs, runnable outputs, and hidden tests gives crisp, checkable answers.

A well-designed exam question is the model. It's not enough to say "write a program about numbers." We need: exactly what the input is, exactly what the program should do, and exactly how to check if it's right. That's what we're about to define.

3.1. The task

The model receives a **natural-language specification** of a programming task and must produce a **Python program** that solves it. That program is then executed against hidden tests.

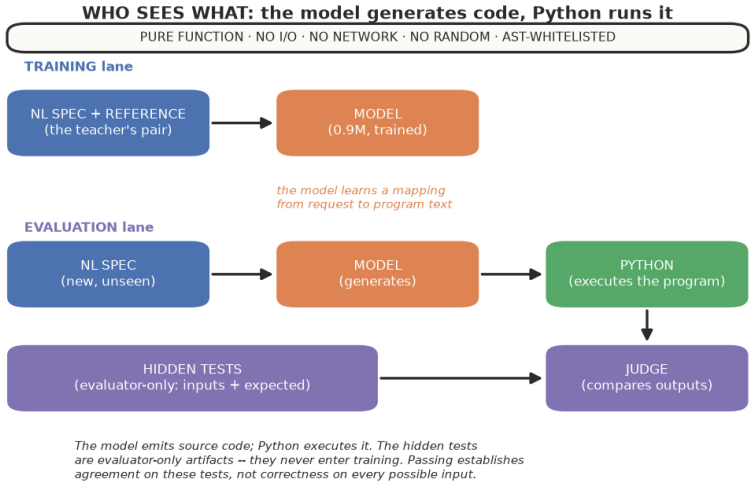


FIGURE 3.1. The one job

The contract is simple: a natural-language specification in, a program out, and a set of hidden tests to grade it against. Everything else follows from it.

Concretely, the model sees something like:

> "Write a function that takes two numbers, a and b, and returns their sum."

and must produce:

```
[] def solve(a, b): return a + b
```

We score it by running that program against a set of hidden test cases (inputs with expected outputs) and checking whether the program's output matches, semantically, for every case.

3.2. The task schema

Every task is described by a small, formal schema — the ground truth that drives everything. The key insight is that **one semantic record generates all three** things we need: the reference program, the natural-language

spec, and the hidden tests. That way there is no drift between "what the task is" and "how we ask for it" and "how we test it."

The semantic core captures the operation and its arguments:

[] { "operation": "add", "args": [{"name": "a", "type": "int"}, {"name": "b", "type": "int"}], "output_type": "int" }

From that single record, the teacher generates: 1. the reference Python (`def solve(a,b): return a+b`), 2. a natural-language specification (in one of several phrasings), 3. a set of hidden tests (ordinary, zero, negative, boundary, large cases).

3.3. The pure-function constraint

For the pilot we constrain the generated programs to be **pure functions**. No side effects, no file I/O, no randomness. This makes evaluation deterministic and unambiguous: for a given input, there is one correct output. It is a deliberate simplification, and it is exactly what makes the task cleanly measurable.

3.4. Hidden tests

The hidden tests are the heart of evaluation. Each task carries an independent set of test cases designed to catch real errors, not just a single happy path. For an arithmetic task we include ordinary values, zero, negatives, boundary values, and large-magnitude values. The tests are generated independently of the reference program's *logic* — computed from the semantics, not copied from the solution — so that a model that merely memorizes the reference still can't predict the exact test suite.

3.5. The success criterion

For a single problem, the success criterion is:

> **Problem-level hidden-test pass@1** — did the one program the model produced pass *every* hidden test for that problem, under deterministic/greedy decoding?

This is deliberately strict. A program that passes 7 of 8 hidden tests *fails* the problem. That strictness is what keeps the measurement honest: we are not measuring "close," we are measuring "correct."

3.6. Why this specific task

Natural-language-to-Python is a good laboratory task precisely because it has everything a controlled experiment needs:

- **Deterministic**. Same spec, same program, same tests.
- **Executable**. We run real Python and compare real outputs.
- **Objectively scorable**. Pass/fail, no judge-subjectivity.
- **Synthetic**. We can generate as much controlled data as we want.
- **Controllable difficulty**. We can escalate from `a+b` to compounded operations.
- **Compositional splits**. We can train on pieces and test on unseen combinations.
- **Memorization tests**. We can plant canaries and check for exact reproduction.
- **Auditable**. We can inspect the exact generated program and run shortcut audits.

The model could not ask for a clearer job. Now we have to build the tools that let us run and score it: a controlled world, a teacher, a judge, and rivals.

PART II.

Build the Laboratory

CHAPTER 4.

Build a World We Can Control

A model that answers natural-language questions is easy to build and hard to evaluate. But there is one class of task where the evaluation is not a judgment call at all: **write a program**. We write a description, the model writes code, and a computer can tell us, with certainty, whether that code does what was asked. There are no opinions about the answer, no ambiguous grading rubric, no "well, it's sort of correct." Either the program returns the right thing for every test, or it does not.

That one observation is why this whole book rests on the `natural-language → Python program` task. It gives us something few machine-learning setups have: *objective, executable, reproducible* evaluation. But the task alone is not enough. Before we can run any experiment, we need a world the model can live in — a world that is safe, that is *deterministic*, and that is small enough that we can reason about everything that happens inside it.

4.1. Why we need a world at all

Imagine hiring someone to write code, but not fully trusting them. We would not hand them the keys to the server. We would give them a sandbox — a walled-off room where they can work, but where the dangerous tools are simply not present.

That is exactly what we do with the model. It writes Python, and we need to run that Python safely and deterministically, while deciding precisely what it is *allowed* to do. The model could, in principle, write a program that reads files, connects to a network, or deletes things. It has no reason to — but we are not relying on its good intentions. We are building a world where those actions are *impossible*, not just discouraged.

The deeper reason is that a controlled world makes the *experiment* clean. If the model's program could do anything — read a file, check the time, generate a random number — then two runs of the same program could give different answers, and we could never attribute a failure to "the model didn't learn this" versus "the program did something nondeterministic." We want the world to be so constrained that **the only source of variability is the model itself**.

4.2. The principle: whitelist, not blocklist

The naive way to constrain a program is a blacklist: deny `import`, deny `eval`, deny `open`. This is the instinct most people reach for first. But a blacklist is fundamentally fragile — there is always some construct we forgot. Python has an enormous surface area, and attackers (or, in our case, just curious models) find the gaps. Denying a hundred things is not safety; it is a constant game of catch-up.

The robust way is the opposite: **an allowlist**. We parse the program into its abstract syntax tree — the structured representation of the code, before it runs — and walk through every node. If any node type is not *explicitly permitted*, we **reject the entire program**. There is no "mostly al-

lowed." If the model emits even one disallowed construct, the whole thing is refused before it ever executes.

The difference is between a list of banned words and a grammar. A banned-word list can be evaded by creative phrasing. A grammar says: these are the only forms that exist. Anything outside the grammar is not merely suspect; it is *unparseable in our world.* That is the philosophy of the sandbox.

This is the single most important design decision in the whole apparatus, and it deserves to be stated plainly: **the sandbox does not try to catch bad programs. It states upfront what is allowed, and everything else is refused.**

4.3. The capability spec

For the pilot, we whitelist a deliberately small subset of Python — small enough that we can hold it in our heads, large enough to express real programming:

- literals (integers, floats), variables, assignment, `return`
- arithmetic: `+ - * / // % **`
- comparisons, boolean operators
- `if / elif / else`, `for`, `while`, `break`, `continue`, `pass`
- lists, indexing, slicing
- a small set of built-ins: `abs, min, max, sum, len, range, int, float, round`
- simple `def` functions

Everything else is rejected: imports, file I/O, network, subprocess, `eval`/`exec`, `class`, `try/except`, arbitrary attribute access, comprehensions, lambdas. We do not include these because they are dangerous; we exclude them because they are not needed for the task we are testing, and because each one we add makes the world bigger and harder to reason about. A smaller world is a *better* world for this book, because every con-

struct that is allowed is one we must understand the model's relationship to.

We also version this spec (`python_capability_v0_1`) so that later, when we give the model *more* Python (Chapter 18), we can point to exactly what grew. Versioning the world is what lets us claim, later, that a change in behavior was caused by a change in the world, and not by something else that drifted.

4.4. Resource and determinism policy

Beyond what is allowed syntactically, we also control *how* the program may run. This is the second half of making the world trustworthy:

- maximum AST depth and node count (a program can only be so complex),
- maximum nesting,
- a hard execution timeout (a program that loops forever is a failed program),
- recursion protection,
- float-comparison tolerance (because `0.1 + 0.2` is not exactly `0.3`),
- **no randomness, no time, no environment access.**

That last point is the quiet heart of the whole design. A program with the same input must always produce the same output. If it does not, the experiment is contaminated. We cannot tell whether a failure means "the model didn't learn this" or "the program rolled the dice." By making the world deterministic, we eliminate that ambiguity entirely. The book's every later claim rests on this.

4.5. The pipeline

```
source -> parse -> AST -> whitelist validation -> compile
        -> restricted execution -> semantic comparison
```

Validation happens on the tree *before* execution, so disallowed code never runs. Execution uses a stripped set of globals that exposes only the allowed built-ins — there is no `__import__`, no `open`, no `eval`. The program can only see the tools we chose to give it.

4.6. An honest caveat

4.6.1. The whole apparatus, in one picture

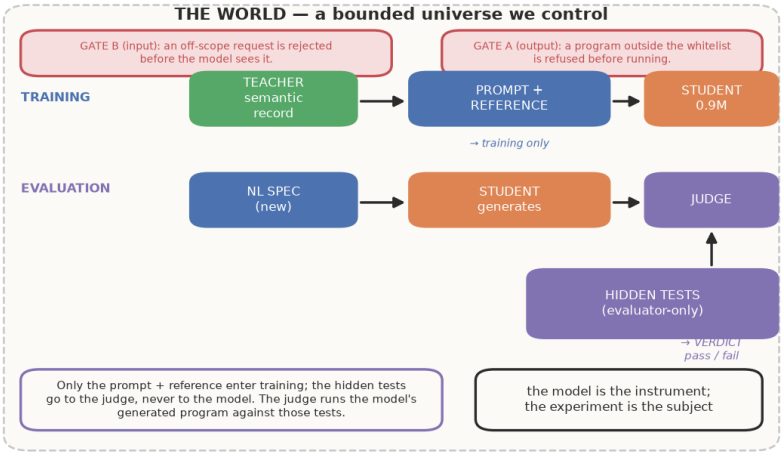


FIGURE 4.1. The whole apparatus

Here is the full loop: the teacher creates the dataset, the student reads it, the judge grades the result, and the world encloses it all.

It is important to be clear about what this sandbox is *not*. It is a **teaching and evaluation** sandbox — deliberately strict about the capability subset, but not a claim of production-grade adversarial isolation. We are not trying to secure a server against a hostile attacker; we are trying to run generated code and measure whether it does what we asked. An attacker with time and motivation could likely find ways around a small AST allowlist. We state this limitation plainly rather than overstating what the

sandbox guarantees, because the book's credibility depends on honest boundaries, not on impressive-sounding claims.

4.7. Testing the world

The sandbox is the foundation of trust in every later result. If the sandbox leaked — allowed something disallowed, or failed to be deterministic — every score in the book would be suspect. So the very first thing we do is not run any experiment at all. We **test the world itself**.

We build a conformance suite of 31 cases: programs that use each allowed construct and must pass, plus programs that use each disallowed construct (`import`, `eval`, `class`, `try`, comprehensions, lambdas, arbitrary attribute access) and must all be rejected. A sandbox that only permits things is easy; a sandbox that also *correctly refuses* everything outside its grammar is the real achievement — and that is what we verify before we trust a single number it produces.

This is the first instance of a discipline the book will return to again and again (and which we will see was worth it, painfully, in Chapter 10): **before we use a tool to measure something, we verify that the tool measures what we think it does**. Here, the tool is the world, and we refuse to run a single experiment until it passes its own examination.

4.8. Two gates, not one

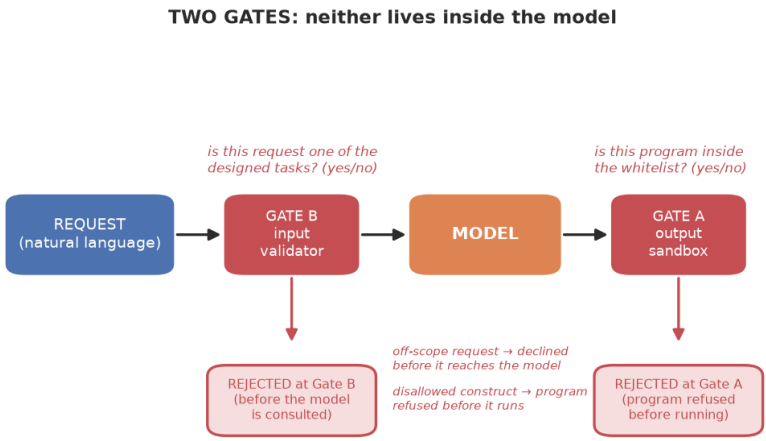


FIGURE 4.2. The two gates

Neither gate lives inside the model. Gate B sits on the way in, Gate A on the way out.

Everything above describes what the model is *allowed to produce*. But there is a second, quieter question: what is the model *ever asked* to do? These are not the same thing, and the difference is worth making sharp, because it is the difference between "the model cannot escape" and "the model was never put in the position to."

A language model has no *built-in* notion of refusal. It is not, by default, a system that decides whether a request is in scope; it is a next-token predictor. Hand a request to this model outside its designed job and it does not politely decline — it simply keeps predicting tokens and produces whatever the weights trend toward, often a confident near-program or garbage. (Models *can* be trained to refuse, but this one was not, so refusal here is not a property to assume.) So "it only does its one job" can never be a property of the model's own intent here. It has to be a property of the wiring around it.

That gives us **two independent gates**, and this chapter (so far) has really only built one of them:

Gate A — the output gate (the sandbox). Already described above: the AST allowlist rejects any generated program that uses a construct outside the whitelist, before it runs. This guarantees that even if the model *tries* to emit `import os` or `eval()`, the program is refused. Gate A is a hard gate on the *syntax* of what the model *produces* — it is deterministic and airtight for the constructs it covers. Two qualifications: it bounds the *shape* of allowed programs, not their resource use or their adversarial resistance (an allowed program could still, in principle, loop or consume memory), and it is not production-grade isolation. As a scoping mechanism for this experiment it is sufficient; as a security sandbox it would not be.

Gate B — the input gate (the request validator). This is the gate we build for the other half. It sits *in front of* the model and decides, from the incoming natural-language request alone, whether that request is one the model was designed for. If not, the request is declined before the model is ever consulted.

Gate B works on the same principle as the sandbox: a **recognizer, not a rejector of the exceptions**. It holds the set of designed template-family shapes (the exact phrase skeletons the teacher renders) and the designed operation vocabulary. A request is in scope only if it fits one of those shapes *and* names a known operation — otherwise it is refused. Just as the sandbox says "these are the only forms of program that exist," Gate B says "these are the only forms of request that exist."

One clarification on how we used it. Gate B was built late in the investigation, after the surface (Chapters 13–15) and human-written evaluations. Those evaluations ran the model **directly, without Gate B in front of it** — precisely so that a low score could be attributed unambiguously to the *model*, not to a request filter that had quietly declined the caller. For the surface and human tests, the model saw the reworded / human-phrased request and failed on its own. Had Gate B been in front, it might

well have rejected those out-of-scope phrasings (that is its job) and the failure would have been the filter's, not the model's. So we state it clearly: the surface and human-written failures are model failures, not Gate-B rejections. Gate B enforces scope; it does not confound those measurements.

Why do we need both? Because each covers a failure the other cannot:

- **Gate A alone** would let the model be *asked* for anything — including a task it was never trained for — and only then catch the malformed result. The model would churn, and our whole experiment would trust that an out-of-scope query produced a rejected program rather than a plausible-looking-but-wrong one.
- **Gate B alone** would let the model *produce* anything within an in-scope request, including a program that escapes its world, because nothing would constrain the output.

Only together do they close the loop: Gate B ensures the model is never *asked* for something outside its job, and Gate A ensures it can never *produce* something outside its world. That is the only way to make "this model does exactly one thing" a true statement, rather than an optimistic description.

There is a deep parallel here with the book's central thesis. The model is the instrument; the experiment is the subject. In the same way, the *boundaries* are never the model — they are always the apparatus: the sandbox on the way out, the validator on the way in. A model that "only does its one job" is confined by its wiring, not by itself. That is worth stating plainly, because it is the mistake people make when they imagine a well-behaved model as one that simply chooses to behave.

4.8.1. How this maps to production systems

Our two gates are not an esoteric lab device; they are the same two ideas that real systems use, under different names. It is worth mapping them, so the book's abstract apparatus connects to what people actually deploy.

Our gate	Production analogue	How it works	Enforcement	Residual risk
Gate B (input)	**Input classifier / guardrail**	A separate model or ruleset scores the incoming prompt and blocks disallowed categories *before* the main model sees it (e.g. Llama Guard, NeMo guardrails).	Hard — a different mechanism in front, not the model deciding	Medium — the classifier is itself learned and can misclassify
Gate B (input)	**Structured-input schema**	The API declares the request's allowed shape (JSON schema); a request outside it is invalid by construction.	Hard — rejected at the boundary	Low for *shape* compliance; not a content/safety check
Gate B (input)	**Function/tool signatures**	The model can only call registered functions with typed signatures; off-scope actions simply aren't callable.	Hard	Low for *declared* functions; not an authorization check

Our gate	Production analogue	How it works	Enforcement	Residual risk
Gate A (output)	**Constrained decoding**	The decoder only ever emits tokens that keep the output inside the declared structure, so the model cannot physically produce an unparseable completion.	Hard — enforced by the decoder, not the model	Low — but guarantees *structure*, not that the content is correct
Gate A (output)	**JSON mode**	A special mode that makes the completion parse as JSON.	Soft — the model is *encouraged*, not forced	High — JSON mode does not guarantee schema validity, and is not a schema constraint

Gate A (output)	**Output filter / sandbox**	A post-hoc check (or a walled-off execution environment) rejects or neutralizes anything that slipped through.	Hard	Low for what it checks; the check is only as good as its rules

The thread running through the "Enforcement" column is the book's point: every reliable enforcement mechanism is **hard and external**, and none of them is "the model chose to behave." Even a frontier model that appears to refuse is doing so because RLHF shaped it to *tend* toward refusal — which is a soft, jailbreakable behavior, not a bound. The dependable version of "it only does what it was designed for" is the wiring we build around it, exactly as our sandbox and our validator are.

One distinction worth making explicit: "Hard" here means the mechanism is **deterministically enforced** — it rejects on a rule the system controls, not on a model's whim — but it does *not* mean the decision is **semantically reliable**. An input classifier is a learned model that can still misclassify; the *enforcement* of whatever it decides is hard, but the *decision* itself is soft. That is why the "Residual risk" column above is not uniformly empty: a hard mechanism bounds the *shape* of what gets through reliably (a JSON-schema validator reliably rejects malformed JSON), but it does not make the content correct or safe (schema compliance is not correctness; a registered tool signature prevents calling an undeclared function, but is not an authorization check; the AST allowlist bounds the shape of allowed programs but not resource use or adversarial resistance). Crucially, "JSON mode" is *not* a hard constraint at all — it is a soft nudge that the model usually follows, so it belongs in the soft/risk column, not alongside constrained decoding. So the reliable layer is hard and external — and it

is a *scoping* guarantee, not a *judgment* guarantee. There is no mechanism in the table that gives semantic reliability.

CHAPTER 5.

Build the Teacher

We need data. But not just any data — we need data we can *trust* to be fair, and data we can carve up to test the model honestly.

A teacher preparing worksheets for a test is the model. A bad teacher writes twenty questions that are all the same question with different numbers — the student can memorize one template. A good teacher writes questions that look different on the surface but test the same idea, so a student who actually learned the idea passes whether or not they've seen the phrasing.

Our **teacher** is that good worksheet-writer. But there's something special about the way we build it, and it's the single most important design decision in the whole book.

5.1. Semantic-first: start with the meaning, not a template

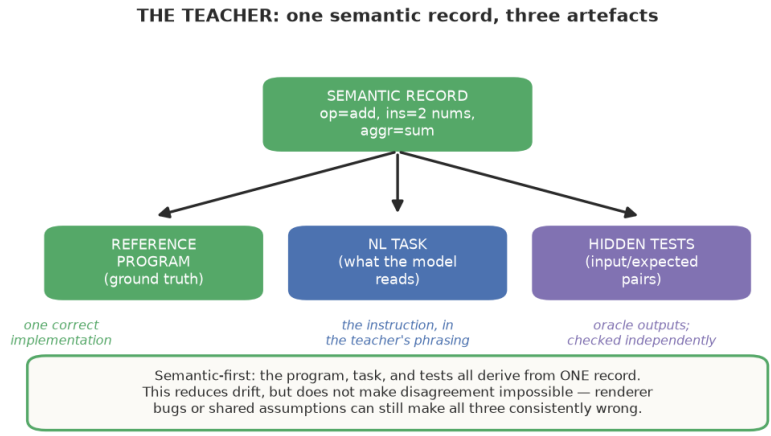


FIGURE 5.1. The teacher

One semantic record is the single source of truth: the program, the task, and the tests all derive from it, so they are much less likely to drift apart than if each were written by hand. This is not a guarantee — a bug in the reference implementation or the renderer could still make all three consistently wrong, which is exactly why we later test the judge against known-correct programs.

The naive way to generate data is to write a template and fill in a few blanks:

```
template -> sentence -> code
```

We write "add {a} and {b}" and a few code snippets, and we are done. This is fast and easy, and it's exactly the wrong way to do it. The problem is that the sentence and the code are generated **independently**, from a template, so there's no guarantee they mean the same thing. We can end up with a sentence that says "add" but code that subtracts, or a test that checks the wrong thing.

We do the opposite. We start from the **meaning** and generate everything else **from it**:

```
SEMANTIC INTENT
   -> PROGRAM STRUCTURE
      -> REFERENCE CODE
      -> NATURAL LANGUAGE
      -> HIDDEN TESTS
```

The key insight is at the top: **one semantic record drives all three**. There is little room for the sentence to disagree with the code or the tests, because they are all the same meaning, rendered three ways. If we start with "the function should return `a + b`," the sentence says "sum," the code adds, and the tests check addition — and they are far less likely to drift apart, because they all came from the same seed. (Agreement is still only as good as that seed: a wrong reference makes the sentence, code, and tests agree with each other but all be wrong, which is why the judge is itself tested.)

5.2. What a "semantic record" looks like

The meaning is captured in a small, structured object. It is not English and not code; it is the *idea* underneath both. For an addition task it looks like this:

> [] { "operation": "add", "args": [{"name": "a", "type": "int", "role": "operand"}, {"name": "b", "type": "int", "role": "operand"}], "output_ type": "int" }

From this single object, we generate:

1. **the reference program** — `def solve(a, b): return a + b`,
2. **the natural-language prompt** — e.g. *"Write a function that takes two numbers, a and b, and returns their sum"*,

3. **the hidden tests** — e.g. `{args:[-28,47], expected:19}`, `{args:[0,0], expected:0}`, and edge cases we'll get to.

This is the heart of the teacher. The pump is `make_task`, which takes an operation and a seed, builds the semantics, and then calls three sub-generators that all read that one record.

5.3. The four sub-generators

The teacher is really four small pieces working together, all fed by the same semantics:

1. **`make_semantics`** — builds the semantic intent: the operation plus its argument descriptors. It pulls argument names from a controlled set (so we can vary them later), assigns an input/output type, and tags the task with a complexity score.

2. **`generate_reference`** — turns the semantics into the ground-truth Python. This is the correct program the grader is validated against. For `add` it emits `def solve(a,b): return a+b`; for `list_sum` it emits a loop over the list.

3. **`render_nl`** — turns the same semantics into English, using one of several **template families**. A template family is just a different way of phrasing the same instruction. For addition, we have phrases like:

- *"Write a function that takes two numbers, a and b, and returns their sum."*
- *"Implement solve so that given a and b it returns their sum."*
- *"Define a function solve(a, b) that returns a + b."*

Same meaning, three different wordings. This is what gives the dataset linguistic diversity — and, more importantly, it's what lets us later test whether the model learned the *idea* or just one *sentence*.

4. **`generate_hidden_tests`** — generates test cases **from the semantics**, not from the reference program's logic. It deliberately covers

edge cases: ordinary values, zero, negatives, boundary values, large magnitudes, and (where legal) empty lists. For each it computes the expected value from the operation itself.

5.4. Why the tests come from the meaning, not the code

This is a subtle but crucial point. If we generated the tests by looking at the reference program, a model that memorized the reference would already know what the tests check. By generating them from the *semantics* instead, the tests probe whether the model independently understands the operation — whether it can produce the right answer for inputs it could not have seen in the reference.

5.5. Maximizing controlled diversity

A good teacher does not just produce many examples; it produces *diverse* ones while holding the semantics fixed. We deliberately vary three things:

- **constants** — different values each task, so the model can't memorize outputs,
- **identifiers** — `a, b` vs `left, right`, so it can't tie the operation to particular names,
- **wording** — several template families per operation, so it sees the same idea said many ways.

This controlled diversity is what makes the *generalization tests* later possible. Because we control *how* we vary the surface, we can also *hold it out* and test whether the model generalizes. That's not an accident; it's the whole point.

5.6. Avoiding accidental correlation

The danger in any generated dataset is **accidental correlation** — a pattern that makes the answer trivially predictable from the surface. The classic

failures:

- every addition example uses the word "total";
- every maximum example uses the variable `x`;
- every multiplication has exactly three arguments;
- every loop example is exactly one sentence long;
- every operation uses one unique wording.

If those correlations exist, a model (or a trivial classifier) can solve the task without learning the computation. The generator is built to break them, and then — this is the crucial part — we **check that it actually did**.

5.7. The leakage audit

Before trusting any dataset, we ask: *can a trivial classifier predict the answer from the surface alone?* The method is deliberately crude: train a TF-IDF + logistic regression model on the natural-language prompts and see how well it predicts the target operation from the words alone. If a dumb classifier succeeds, the wording is leaking the answer.

But the method needs a careful interpretation, and our first pass misread it. On our first pilot renderer, a trivial classifier predicted the operation from the wording with **100% accuracy**. We recorded it as a red flag (finding-001). On reflection, that is most of the time **task signal, not leakage**: the request must name the operation, or the task would be ambiguous. Saying "return their sum" *should* identify addition — that is what makes the request unambiguous. A classifier hitting 100% because "sum" identifies addition is the wording doing its job, not spilling a secret.

So the audit has to distinguish three things:

- **Task signal** (fine): "sum" identifies addition; the request names the operation.
- **Spurious correlation** (a real shortcut): an *irrelevant* feature that is not part of the request's content — a particular identifier name, an inci-

dental sentence frame, or a formatting quirk — predicts the operation.
- **Train/test leakage** (the real bug): duplicated examples, or evaluation information, crossing the train/test boundary.

A 100% TF-IDF classification score shows that operation selection is lexically *easy* on that split. It does not by itself establish contamination, and it does not tell us the model is exploiting an illegitimate signal — the phrasing is supposed to carry the operation. What the audit *should* do — and what our later audits do more carefully — is mask the operation-bearing language and ask whether *irrelevant* features still predict the target, and also test *held-out template families* and counterfactual changes to names or sentence frames. Our finding-001 note is best read as "the wording is very predictive of the operation" — which is a property of a clear task, not proof of a shortcut. (One implementation note as well: the TF-IDF vectorizer was fit on the whole set before cross-validation, which would leak held-out-fold vocabulary. A principled version fits the vectorizer inside the CV pipeline.)

This is the teacher doing its real job. A dataset generator that only produces data is half a tool. One that produces data **and audits it** — asking whether the surface is carrying the answer for the *right* reason is a laboratory instrument.

5.8. The takeaway

The teacher is not a bulk-data tool; it is a controlled instrument. Its job is to produce data with *controlled* diversity and *audited* cleanliness, so that when the model succeeds we can attribute it to learning the computation — not to the data accidentally giving away the answer.

And the reason this matters so much is the whole book's premise. Everything later — the generalization tests, the composition test, the capability map — depends on the data being a *fair* test. The teacher is where that fairness is built.

CHAPTER 6.

Build the Judge

How do we know a student's answer is right? We check it against the answer key. But what if the answer key is wrong? Then every student who gets it "right" is actually wrong, and we'd never know.

That is the risk with our **judge** — the evaluator that decides whether a generated program is correct. It is the instrument whose readings the whole book depends on. If the judge is wrong, every number in this book is wrong. So we build it carefully, and then we attack it.

This is the second pillar of the apparatus, and it is in some ways the most important. The world (Chapter 4) decides what the model is *allowed* to do. The teacher (Chapter 5) supplies the material to learn from. But the judge is the thing that decides whether the model *got it right*. A subtle way to frame the whole book: we are not really studying the model. We are studying the model *through* the judge. So the judge's integrity sets an upper bound on the trustworthiness of everything that follows.

6.1. Why an execution-based judge

The naive way to judge a generated program is to compare it, as text, to the expected answer. This is cheap, and it is almost worthless. Two programs that solve the same task can look completely different as strings — one uses a loop, another uses `sum()`, yet both are correct. And two programs that look nearly identical can behave differently. String comparison measures *how the code is written*, not *whether it works*. That is not what we care about.

So the judge does the thing that makes this whole task special: it **runs** the program. It executes the generated code against hidden tests and compares the actual *output* to the expected output. This is what gives us objective, unambiguous judgment. There is no opinion about whether the code is elegant or the "right" style — either it returns the correct result for every test, or it does not.

6.2. The four-stage pipeline

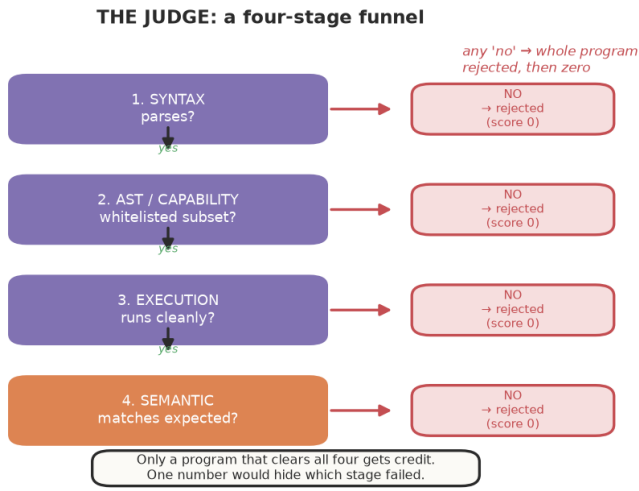


FIGURE 6.1. The judge

The judge is a funnel: a program clears it only by passing every stage, and the first failure sends it to zero.

We score a candidate program through four stages; a later stage only matters if the earlier ones pass. Each stage is a filter, and each exists for a specific reason.

1. **Syntax pass** — does the program parse as valid Python?
2. **AST / capability pass** — is every node within the whitelisted subset? If the program smuggles in an `import`, it is rejected here.
3. **Execution pass** — does it compile and run without error under the restricted environment?
4. **Semantic pass** — for each hidden test, does the program's output match the expected output?

The primary metric is **problem-level hidden-test pass@1**: did the one program produced pass *every* hidden test for that problem?

Why four stages rather than one? Because each stage isolates a *different* kind of failure, and separating them tells us *why* a program failed. A syntax error is a very different problem from a capability violation (the model tried to use something outside its world), which is different again from a runtime error (the code ran but crashed on a test), which is finally different from a semantic error (the code ran but produced the wrong answer). If we collapsed all four into one "it failed," we would lose the ability to diagnose. The judge does not just tell us right or wrong; it tells us *where* things go wrong, and that diagnostic power is worth keeping.

6.3. Semantic comparison: don't trust string equality

A naive evaluator would compare outputs as strings. That is wrong — the result of a program is a *value*, so comparing text representations of numbers would reject correct answers (`0.1 + 0.2` as text is not `0.3`). So the judge compares values with a **semantic comparator**. Notably, Python's own `==` already recurses into lists and ignores dictionary insertion or-

der, so the comparator's job is narrower than "fix all of Python's equality quirks." It adds three real policies on top of Python semantics:

- **floats compared with an explicit tolerance** (Python's `==` on floats is exact; `0.1 + 0.2 == 0.3` is `False`). We use an absolute tolerance.
- **booleans treated as booleans, not as 0/1** (so `True` is not equal to `1`),
- **dicts compared order-insensitively**, which Python already does for `==`; the comparator re-implements it explicitly for clarity and to recurse into values.

The comparator itself is tested — we don't assume it's right just because it's simple. And the tolerance matters: floating point arithmetic is not exact, so a judge that demanded exact equality on floats would reject correct programs. A judge with no tolerance at all would be as broken as one that accepted anything. Choosing the right tolerance is a real design decision, and it is the kind of thing that quietly shapes every number the judge produces.

6.4. Mutation testing: red-teaming the judge

The most important step is treating the judge as something that must be *vetted*. We **mutate** the reference solution into deliberately-broken versions and ask: *does the judge catch them?*

We generate mutants such as: `+` ⟶ `-`, `>` ⟶ `>=`, remove the `return`, swap the arguments, off-by-one a loop boundary, use the wrong variable. Then we run each mutant through the hidden tests. A good test suite **rejects** an obviously-wrong mutant. A weak test suite lets it pass.

The mutation score — the fraction of wrong mutants the tests catch — is a measure of how much we can trust the judge on *the cases it is designed for*. An obviously-broken program passing our tests means the tests are too weak for that class of error. Two honest caveats: (1) passing a finite hidden-test suite means passing *that suite*, not proving correctness for every allowed input — deterministic scoring is not complete semantic veri-

fication; (2) equivalent mutants (e.g. swapping + operands) do not make a program incorrect, so those should not count against us. We report the mutation behaviour honestly rather than claim the judge is "proven sound"; it is a well-tested *heuristic* verifier within the cases we constructed, not a proof.

This is the "test the tester" principle, and it is the reason we do not trust the judge just because we wrote it. The judge is the instrument of measurement, so we deliberately give it wrong inputs and check that it rejects them. It is a rare and valuable discipline: most people build a measuring stick and assume it measures. We deliberately try to break ours.

6.5. We caught our own judge being wrong

Here is what makes this chapter honest rather than aspirational: the very first time we evaluated our model, it **scored 0.000** — reported a total syntax failure on every problem. And the model wasn't wrong. The *judge* was.

The root cause was a generation bug on our side: the model produced a complete, correct program, but it was followed by padding (null bytes), and `ast.parse` correctly rejects null bytes — they are not valid source. So the fault was in the *generation/extraction contract*: those padding bytes were meant as a termination or padding marker that we needed to strip, not as part of the program. `ast.parse` behaved correctly; our extraction forgot to truncate the padding before parsing. Once we truncated it, the model's true score emerged: **0.897**.

That moment is worth more than any passing test. It is the book's core lesson applied to ourselves: **never trust a number from a system we have not audited.** If we had taken the first 0.000 at face value, we'd have concluded the approach failed. Instead, finding the judge's bug told us the truth — the model was actually working.

What is striking is that this bug was not exotic. It was a small, mundane detail — a sentinel byte used to mark the end of a generated program — colliding with Python's parser. We would never have found it by rea-

soning about the model. We found it because we looked at the judge's *intermediate* output (the syntax scores) rather than jumping to the headline number, and because we already had the habit of doubting the measurement. That habit, more than any single fix, is what kept the book honest.

6.6. The honest summary

The judge is not a helper; it is the instrument whose readings we stake the book on. So we built it strictly (four stages, semantic comparison, tested comparator), then we red-teamed it (mutation testing), and then — when it gave a suspicious answer — we attacked it until we found our own bug. Only a judge that survives that scrutiny deserves to be trusted.

And the deeper lesson is one we will keep relearning: in any measurement, the instrument is part of the experiment. The model and the judge are both subjects. We spend almost as much effort verifying the judge as we do training the model — and that is not waste. That is the only way to trust the numbers at all.

CHAPTER 7.

Build the Rivals

A model that scores 90% sounds impressive — until we ask the hard question: *could a much simpler machine do the same thing?* This chapter is about building those simpler machines, our **rivals**, and why they are the most important comparison in the book.

7.1. The principle

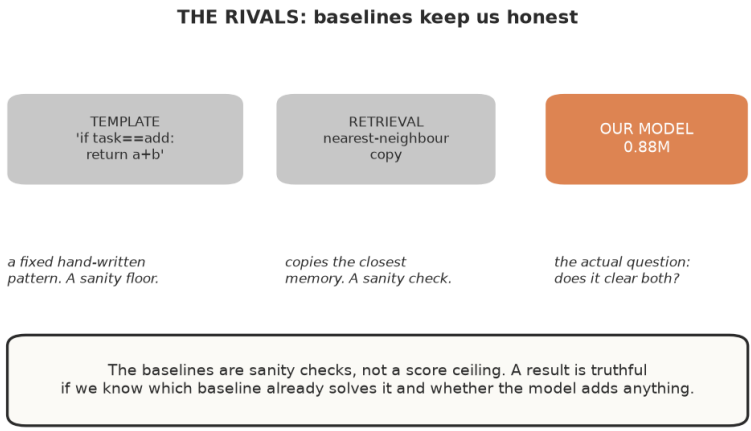


FIGURE 7.1. The rivals

The template and the retrieval baseline are *sanity checks*, not a score ceiling. Our model is the question mark in the middle: it has to clear them to be worth anything, but beating a baseline is never a bad result — it just needs to be interpreted correctly.

The value of a neural model's success is judged *relative to a baseline*. If a hand-written rule or a lookup table solves the task as well as the trained model, then the model isn't demonstrating capability — it's re-deriving something simpler. The baselines are what turn "the model works" into "the model does more than a simpler machine could." But a baseline is not a *ceiling* a result must sit below: a strong model beating a baseline is a good outcome. The baselines tell us *whether the task was already easy for a simpler machine*, which is the information we need to interpret the model's score — they do not cap how good a model is allowed to be.

This is a point that deserves to be stated bluntly, because it is easy to overlook: a number in isolation is almost meaningless. "The model solved 90% of the problems" sounds like a triumph. But if a ten-line lookup table

also solved 90%, the model's result is not evidence of learning — it is evidence that the task can be solved without learning. The baseline is what gives the number its meaning.

This is the single question the whole book keeps asking:

Could a much simpler machine do this?

And it is remarkable how often the answer is "yes" at first. That is a feature of the investigation, not a defect. Finding that the task is too easy for a simple machine tells us exactly how to make it harder, which is how the whole brew progresses.

7.2. Rival 1 — the template / rule system

Our simplest rival is a deterministic pattern-matcher. It recognizes known keywords in the specification — "sum" → addition, "maximum" → max, "length" → list length — and emits the corresponding predefined code template. It does no learning whatsoever. It is, in the most honest sense, *dumb* — a set of hand-written if-then rules.

This rival answers: *how much of the task is solvable WITHOUT learning?* If a keyword recognizer does fine, then the task's "hard part" is just pattern-matching, and a model learning it isn't learning much.

Building this rival is almost too easy. It is a few rules that map common words to common code. But its simplicity is precisely its value. It is the lowest useful baseline: the floor of what any reasonable person would expect a hard-coded system to achieve. If our learned model cannot beat this floor, we learn an important fact: the task does not require the extra machinery — it does not establish an *advantage* over this baseline. That is a statement about the task (it is easy for a rule table), not a statement that no learning occurred. The model may well have learned; it is just not learning anything *beyond* what a few rules give us.

7.3. Rival 2 — retrieval

Our second rival is even more interesting. A **retrieval** system builds a simple TF-IDF index over the training prompts. Given a new task, it finds the most similar training example and *reuses that example's code*.

This rival answers: *can apparent neural competence be explained by retrieval?* If the test set is close enough to the training set, a lookup that does nothing but find the nearest example will do perfectly — and a model that also scores high might just be doing the same thing in a more complicated way.

The retrieval rival is the more insidious of the two, because it is the one that threatens the model's success most directly. A template system shows whether the task is trivially pattern-matching. But a retrieval system probes something subtler: whether a model's score could be explained by *surface similarity to training data*. If a TF-IDF lookup can find a near-duplicate prompt and reuse its answer, then a neural model scoring similarly might be doing exactly that — matching surface form, not understanding. Retrieval is the closest thing we have to a detector for "the model is just copying."

7.4. Why these two matter

These two rivals are the scientific controls. Together they bound what any result means:

- If the **template** baseline is high, the task is pattern-recognizable — the model's win is shallow.
- If the **retrieval** baseline is high, the task is near-duplicate-solvable — the model's win may be memorization or surface similarity.

And the third possibility is the one that matters most: **if the model falls *below* both baselines under some change, we have found a genuine limit of the model's learning**. The rivals are a reference point, not a ceiling

to beat; they reveal when the neural model is doing something *no simple machine can do* — and, more valuably, when the neural model has become *worse* than a simple machine, which is a powerful signal of surface-tied, non-general learning.

Our experiments produced exactly this signal. On the pilot, retrieval scored **1.000**: a lookup table solved the whole test set. That told us — correctly — that the task was too easy to force generalization, and we had to make it harder. The rivals did their job: they caught that the impressive-sounding model was being tested on a task that didn't require much learning.

Then, in the surface-generalization chapter, the story inverted. Under a *wording* change, the model collapsed to **0.000** while the template system, which recognizes keywords and is indifferent to exact wording, stayed at **0.875**. Here the rivals were not a ceiling — they were a *floor*, and the model fell through it. That was the most informative result of the experiment: it showed that the model's apparent capability was tied to exact surface form in a way that even a dumb rule system was immune to.

7.5. The lesson

A neural model is not impressive merely because it works. It is impressive only when it does something a simpler machine cannot. Every result in this book is therefore interpreted relative to these rivals — and the moments where the model *falls below* them are some of the most informative results of all.

The deep takeaway is that baselines are not a formality or a box to tick. They are the reference frame that gives every measurement its meaning. Without them, a score is just a number. With them, a score becomes a claim we can interrogate. And building rivals that are *deliberately simple* is what makes them a strong test: the simpler the machine that can match or beat the model, the more damning (or the more shallow) the model's achievement turns out to be.

PART III.

The First Brew

CHAPTER 8.

The Smallest Brain

Now the student. Everything so far — the world, the teacher, the judge, the rivals — exists to ask one question about this thing. So let's meet it.

Meet the model: a small, deliberately plain decoder-only transformer. It's not fancy, and that's the point. Just like a chemistry lab uses simple glassware so we can see exactly what's happening, we use a simple, transparent model so we can understand exactly what it's doing. A 20-layer exotic architecture would be a black box; a clean, standard transformer is one we can reason about.

8.1. Why the simplest possible model

The temptation when choosing a model is to reach for the newest, most sophisticated thing — the attention variant that promises to be smarter, the architecture that looks impressive in a diagram. That instinct is understandable, but it fights the entire purpose of a controlled experiment. If the model is complex and mysterious, then when it fails, we cannot tell

whether the failure is because of some fundamental limit of learning, or because of a quirk of our exotic architecture.

So we choose the opposite. The student is a **completely standard, off-the-shelf decoder-only transformer** — the same basic architecture that powers the big language models, just scaled way down. Nothing clever, nothing bespoke. This is a deliberate choice, and the reason is not modesty; it is method. A plain, well-understood model means that when something unexpected happens, we can attribute it to *learning* rather than to an architectural mystery. We want the model to be the simplest possible instrument, so that the experiment — not the gadgetry — is the subject.

8.2. What it is

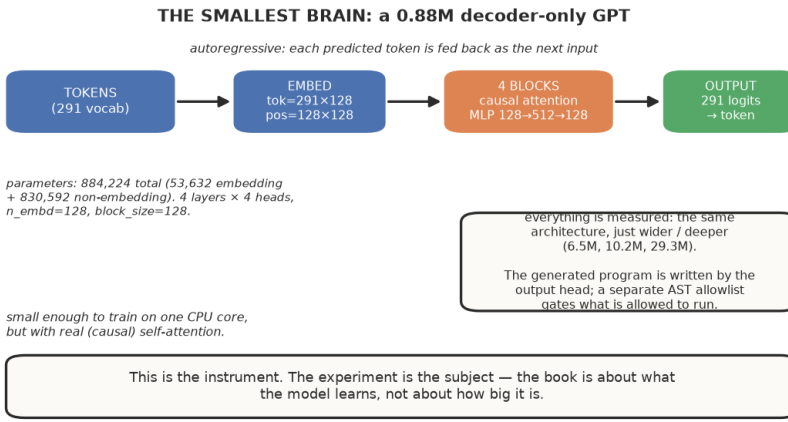


FIGURE 8.1. The smallest brain

The smallest brain is a tiny decoder-only transformer: tokens in, embed, attend through the layers, and predict the next token.

The student is a deliberately simple, transparent **decoder-only transformer**. The goal is not novelty; the goal is that we can understand exactly what it is doing. A 20-layer exotic architecture would be a black box; a clean, standard transformer is one we can reason about.

For the pilot, the configuration is small and honest:

- vocabulary: ~291 tokens (a compact BPE trained on the task corpus),
- hidden dimension: 128,
- layers: 4,
- attention heads: 4,
- context length: 128,
- GELU activation, Pre-LayerNorm, AdamW optimizer.

Each of these is a deliberate choice that we can justify. The context length of 128 tokens, for example, is chosen because a pilot task's prompt-plus-program fits comfortably within it — long enough to hold a full task, short enough that the model is not wasting capacity on irrelevant span. The 4 layers and 128-wide hidden dimension keep it fast to train on the modest hardware available, which matters enormously: if training takes an hour, we iterate on the *experiment*; if it takes a day, we wait on the *model*, and the brew slows to a crawl. A small model is a fast iteration loop, and a fast loop is what lets a scientific investigation move.

8.3. The honest parameter count

A common mistake is to quote just a round "10M parameters." We refuse to hide the important part. The model has **884,224** parameters, of which **830,592** are non-embedding (the transformer blocks, plus the final layer-norm and the output projection head) and **53,632** are the **embeddings** — the token embedding ($291 \times 128 = 37{,}248$) plus the positional embedding ($128 \times 128 = 16{,}384$). The embeddings are *also* learned parameters: they are trainable and they do store task-relevant representations. What distinguishes them from the block weights is that each one is only ever *looked up* (a token maps to its vector), whereas the block weights are *multiplied* many times per forward pass. So at ~6% of parameters, the embedding tables are a small share, but they are not inert — we report the split between embedding and non-embedding so a naive "M parameters" figure is not

dominated by the lookup tables rather than the computation.

So we always report the split: embedding vs. non-embedding.

8.4. The model, in one box

So that there is no ambiguity about what we trained, here is the complete specification of the pilot model (the 0.88M model; the larger runs in Chapter 12 only scale width and depth, and the registry in Appendix C lists each one's exact configuration):

Property	Value
Architecture	GPT-style decoder-only transformer
Vocabulary	291 tokens
Embedding dim (`n_embd`)	128
Layers (`n_layer`)	4
Heads (`n_head`)	4
FFN width (intermediate)	4 × `n_embd` = 512
Context budget (`block_size`)	128 tokens
Dropout	0.0 (deterministic)
Attention	Causal (masked), self-attention
Positional encoding	Learned, 128 × 128
Final norm	LayerNorm before the output head
Output head	`n_embd` ⟶ 291 logits, **no bias, no weight tying**
Parameter split	884,224 total = 53,632 embedding (token 37,248 + position 16,384) + 830,592 non-embedding
Weight init	Normal(0, 0.02)

Property	Value

What the model is asked to do. Each training example is the prompt and the target program concatenated as a single token sequence (`prompt + "\n" + reference_code`), and the model is trained with standard next-token cross-entropy over that whole sequence. There is no loss masking that restricts credit to the program positions — the model is rewarded for predicting the prompt's continuation (including its own wording) as well as the program. Generation uses temperature 0.8 multinomial sampling for `max_new` tokens, with the context window truncated to `block_size` (it reconditions on the sequence so far at every step). There is no separate "refuse to answer" behavior — the model is trained simply to keep predicting, which is exactly why we build gates around it (Chapter 4).

This matters more than it might seem. At tiny scale, the ratio of vocabulary to computation is high, which means a naive "M parameters" figure can be dominated by the embedding lookups rather than the computation that must actually reason about the program. Separating the two lets us and the reader see the *actual* computation the model brings to bear — the part that must do the learning. It is a small discipline, but it keeps the conversation honest.

8.5. The architecture smoke test

Before we train it for real, we run an **architecture smoke test** — a tiny, fast check that the model's forward and backward passes work, that the parameter count is what we expect, and that the initial loss is sensible (roughly the log of the vocabulary, since a fresh random model's best guess is uniform).

This is the same discipline as the very first smoke test, but one level up: we are not checking "does the loop run?" anymore, we are checking "does *this model* construct and train correctly?" A 20-second check here

prevents discovering a shape bug hours into a real run.

The initial loss is a wonderfully informative number. A brand-new model, before it has learned anything, predicts the next token uniformly, so its loss should be close to the natural log of the vocabulary size — here, about $\log(291) \approx 5.67$. When we actually ran it, the initial loss was **5.84**, reassuringly close. That is what a model that knows nothing looks like, and it is the baseline every later, lower number is measured against.

8.6. Why it's small

The book deliberately keeps the pilot model tiny. This is not modesty; it is method. A small model is fast to train (so we can iterate on the *experiment*, not wait for the model), transparent (so a failure is a learning-system problem, not an architecture mystery), and it makes the *boundary* more visible — a small model that fails to compose tells us something cleaner than a large model that does something partly and noisily. We scale it up later precisely to test whether capacity matters, but the baseline has to be small and legible.

8.7. The configuration is a starting point, not a fact

Crucially, none of these numbers are immutable. The exact architecture is an experimental variable. Every architectural choice requires a reason, and the question is never "what architectures can we try?" — it is "what scientific decision does this comparison enable?" For the pilot, the answer is: this size is fast enough and small enough to be legible, which is what Stage 0–1 needs.

This is a discipline worth holding onto. It is tempting to fixate on the "best" architecture or the "right" parameter count as if one existed. But in an experiment where the question is about *learning*, the model is a dial, not a destination. We choose a configuration because it enables a specific scientific comparison — and we change it when we want to test

whether the answer changes. That is why the pilot model is small and plain: not because it is the best, but because it is the *clearest* instrument for the question at hand.

CHAPTER 9.

The First Brew

Everything's built. Now the moment of truth: we train the model for the first time and watch what happens.

This chapter is deliberately unheroic. The goal isn't a triumphant result — it's to confirm the loop works and to see the model's honest behavior as it learns. It is the first time a recipe we have been assembling is finally run: we are not hoping for a masterpiece, we are checking that the oven works and nothing explodes. And for anyone who has brewed anything — beer, coffee, bread — the *first* batch is never about the product. It is about learning our equipment, our ingredients, and our own impatience. This chapter is that first batch.

9.1. The data

1400 generated tasks: a natural-language specification paired with its reference Python program. The model's job is to read the spec and produce the code. Each task is a single semantic record — one idea — rendered three

ways (a program, an English instruction, a set of hidden tests), so the data is internally consistent by construction. We do not need to worry that a prompt says "sum" but its answer subtracts, because both came from the same seed.

We hold out a separate 300-task test set that shares no operation with training in a leaky way, so the model cannot simply see the same task again. This split discipline is not optional: it is what lets us claim, later, that the model's score reflects unseen tasks, not memorized ones.

9.2. The model

The student is a small, deliberately plain decoder-only transformer: 4 layers, hidden dimension 128, 4 attention heads, a compact ~291-token vocabulary, and 128-token context. Total **884,224 parameters** — but we report the split honestly, because it matters more than the round number: **53,632** are the embeddings (the token and positional tables, which just store the vocabulary and position), and **830,592** are the actual computation. A tiny model's parameter count is often dominated by its embedding tables, which are the least interesting part; we never want a "0.9M parameter model" claim to hide that most of it is lookup tables.

9.3. The training process, step by step

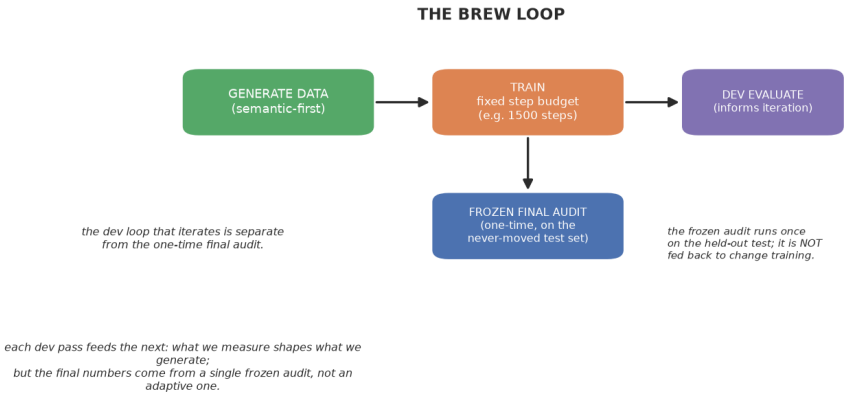


FIGURE 9.1. The brew loop

Every experiment is one pass around this loop: generate, train, measure, and let what we measured shape what we generate next.

Training a small transformer is not a single action; it is a loop repeated 1,500 times. At each step:

1. We grab a batch of 16 tasks and tokenize each as one sequence: the natural-language prompt, a newline, then the reference program.
2. The model reads the prompt and predicts each *next* token in the combined sequence. This is the standard **next-token objective** — the model's only job is to guess the next word, given everything before it. It is a deceptively simple task, and it is the entire basis of how these models learn to write.
3. We measure the difference between the model's prediction and the actual next token (the cross-entropy loss), take the gradient, and nudge the parameters with the **AdamW** optimizer at learning rate **1e-3**, clipping the gradient magnitude to keep training stable.
4. Repeat.

We print the loss every 200 steps so we can watch it behave. Here is

what it actually did:

```
step    0   loss 5.84
step  200   loss 0.029
step  400   loss 0.022
step  600   loss 0.023
step  800   loss 0.020
step 1000   loss 0.021
step 1200   loss 0.019
step 1499   loss 0.019
```

That is a large, clean drop — from 5.84 to 0.019 in 1500 steps. Two things about this curve are worth sitting with, because they reveal more than the final number.

First, the shape tells us the *training loop is working*, nothing more. The loss falls steeply in the first 200 steps (5.84 → 0.029), then flattens out near a floor. This is the curve we expect from a functioning loop that is fitting the data: rapid improvement at first, then diminishing returns. If instead the loss had wobbled or stayed high, we would have had a debugging problem on our hands, not a training result. But a *converging* loss by itself does not distinguish learning from memorization — both produce a falling loss. That distinction is exactly what the upcoming evaluations, not the loss curve, have to settle.

Second, the *final* value is not the interesting number. A loss of 0.019 means the model can predict the next token of the *training* sequence almost perfectly. That tells us it fits the training data — and nothing more. This is the book's recurring motif, and it is worth saying loudly: **a falling training loss, and even a very low one, is not evidence of learning.** It is evidence of fitting. The distinction is the entire point of the experiment.

9.4. What the training process felt like (the brewing experience)

Anyone who has trained a model knows the peculiar rhythm of it. We set a batch of runs going and we wait, checking the loss every few minutes with a combination of hope and suspicion. The first steps are the most honest — a fresh, random model has loss near the log of the vocabulary size, because its best guess is uniform. That initial 5.84 is exactly what a model that knows nothing looks like.

Then, if things are working, the loss starts to fall, and a strange thing happens: the drop is *so* fast in the first couple hundred steps that we wonder whether it is real or whether something is broken. In this case the model reached near-floor loss within 200 steps. The rest of the training is just the model quietly polishing the last fraction of a point.

There is also a genuine practical question that arises during training: **how do we know it hasn't crashed mid-run?** We use periodic **training checkpoints** that record the state at regular intervals *during* the run, so that a crash at step 1200 loses at most the steps since the last save. A single save only at the end of training — say, just before evaluation — does **not** protect against a mid-run crash; it only protects the completed run. So we keep intermediate checkpoints, not just a final one. This is a small discipline, but it has saved us more than once — a training run on this hardware takes real time, and we do not want to lose it to an unplanned stop.

9.5. The honest caveat

A falling training loss — and even a very low one — is *not* evidence of learning. We saw this in the pilot and it is the book's recurring motif. The training loss tells us the model fits the training data. It does not tell us it can *generalize* to unseen problems, or that it learned the computation rather than the surface form.

So after training, the only losses that matter are the *held-out* ones:

how well does the model do on problems it never saw? That is Chapter 10's question, and the answer there is the real measurement of this brew.

9.6. Why the details of the training process matter

It is tempting to skip the mechanics and jump to results. But the whole premise of this book is that *the experiment is the subject*, and an experiment we cannot reproduce is not an experiment. The exact vocabulary size, the number of layers, the batch size, the learning rate, the number of steps — these are not trivia. They are the conditions under which every later claim is true. If we change one and the result changes, that tells us something; if we change two and the result changes, we can no longer say which one caused it. This is the **one-variable discipline** that governs every experiment in the book, and it is why we record the configuration so carefully in a manifest alongside every result. A reader should be able to take our numbers, run the same configuration, and get the same outcome — and if they cannot, we have not done the work, we have only described it.

9.7. What this chapter establishes

This first brew is not a result; it is a proof of the loop. We have:

- trained a real model on real data,
- watched its loss descend smoothly,
- saved a checkpoint,
- confirmed that we can take the trained model and produce code for a new prompt, run it through the sandbox, and score it with the judge.

That loop — generate data, train, generate code, execute, score — is the laboratory working end-to-end. It is the foundation everything else stands on. The model is built. Now we have to find out what it actually learned.

CHAPTER 10.

Does It Actually Work?

We have a trained model, a judge, and a test set. The natural question is the one the whole book has been building toward: *does it actually work?*

And the answer, as with almost everything in this book, is "**it depends on what we mean**," and then "**let's measure it rather than argue**." Because "works" is not a single number. It is a question about *which* kind of capability, under *which* conditions, measured in *which* way. The job of this chapter is to take that vague word and replace it with something we can actually see.

10.1. The setup

We trained a small decoder-only transformer on 1,400 generated tasks: the model reads a natural-language specification and must produce the reference Python program. The trained model is tiny — **884,224 parameters** (0.88M), of which 53,632 are the embedding and 830,592 are the actual computation. It saw 1,500 gradient steps on a batch of 16, reaching a train-

ing loss of **0.0208**.

We then asked it to solve 300 **held-out** test tasks it never saw, and scored each generated program by executing it against that task's hidden tests. This is the moment of truth for the whole apparatus: everything we built — the world, the teacher, the judge, the rivals — converges on this measurement.

10.2. One task, end to end

The cleanest way to see what the judge does is to follow one task from prompt to verdict. Here is a real run against the pilot model.

The request. "Implement solve so that given a and b it returns their sum."

The reference program (the teacher's ground truth):

```
def solve(a, b):
    return a + b
```

What the model actually produced (greedy decoding, this checkpoint):

```
def solve(a, b):
    return a + b
```

The hidden tests (input/expected pairs) that the judge then runs:

inputs	expected
(87, -98)	-11
(0, -24)	-24
(-6, 0)	-6

The judge runs the model's program on each pair, compares the output with the semantic comparator, and reports **pass@1**. This task passes.

Now the informative contrast: a task the model *had not seen.* The teacher's composite `abs_add` asks for the sum of the absolute values of two numbers (`abs(a) + abs(b)`). The model, which knows both `abs` and `add` individually, was fed:

> "Write a function that takes two numbers a and b, takes the absolute value of each, and returns their sum."

and produced:

```
def solve(a, ab
```

— a truncated, unusable fragment. Greedy decode stopped mid-token. On a genuinely reworded surface ("Return the value obtained by adding together two numbers, denoted p and q."), it produced:

```
def  a solve(a, b
```

again truncated. These two live examples carry the book's whole point: on a familiar surface the model writes correct code, and on an unseen surface or a new combination it collapses to a fragment. Nothing about the failure required the hidden tests to be "unfair" — the judge simply ran programs that were not complete.

10.3. What we measured: pass@1

Here is the headline result, reported as **problem-level hidden-test pass@1 under greedy decoding** — that is, for each problem, did the single program the model produced pass *every* hidden test?

Candidate	syntax	executable	**pass@1**
MODEL	0.897	0.897	**0.897**
TEMPLATE	1.000	0.900	0.800

Candidate	syntax	executable	**pass@1**
RETRIEVAL	1.000	1.000	**1.000**

Three things stand out.

First, the model genuinely learned something. 0.897 means that ~90% of the time, an unseen problem's natural-language spec produced a program that passed all its hidden tests. That is not a toy result. And crucially, it **beats the hand-written template/rule baseline** (0.800). A hard-coded pattern-matching system that recognizes keywords like "sum" and emits a template gets 80%; our learned model gets ~90%. So the model is doing *more* than a fixed rule system can do on this task.

Second — but retrieval does even better. A trivial TF-IDF retriever that does nothing but find the closest training example and reuse its program scores **1.000**.

That second fact is not a failure; it is the single most important thing this pilot taught us. It tells us the task, as currently constructed, is **too easy to force generalization**. When a non-learning lookup solves 100% of the test set, then the test set is solvable by retrieval, and a model scoring 90% is not yet being tested on whether it can generalize — only on whether it can match the surface form of a training example.

This is exactly the "task too easy" outcome the experimental plan anticipated, and it is scientifically useful: it tells us *how to make the next task harder.*

10.4. The moment that broke us (and the evaluator, not the model)

There is a story behind these numbers that is more instructive than the numbers themselves, and it is worth telling plainly. When we ran the evaluation for the very first time, the model scored **0.000**. Not 0.897 — zero. Every single generated program failed the syntax check.

Our first instinct, naturally, was that the model was useless. We had spent hours building a careful apparatus and the tiny model could not even produce a parseable program. It would have been easy to conclude "the model doesn't work" and write the whole thing off.

But the book's own discipline — *test the tester* — demanded we ask whether the *judge* was the problem, not the model. So we looked at what the judge actually saw. And there it was: the generated programs were valid Python up to the end of the program, except they had a run of **NUL bytes** (`\x00`) appended after it. The model had learned to pad its output with a byte that means "nothing." Python's parser correctly rejects NUL bytes — they are not valid source. The fault was in the *extraction contract*, not the parser: we had failed to strip the padding before handing the output to `ast.parse`. So `ast.parse` threw a syntax error on what was, after truncation, good code, and the judge scored the model's program as a *syntax failure*.

The fix was a single line: truncate the generated program at the first NUL byte. A trivial change, and suddenly the model's score leapt from 0.000 to 0.897.

This is the single most important lesson of the whole book, and it is not about the model at all. **The evaluator was broken, not the model**. We had designed every experiment to detect shortcuts and outright cheating in the *model*, and we nearly missed the one case where the measuring stick itself was faulty. The "test the tester" principle — that we must verify the tool measures what we think it measures before trusting a single number it produces — was not a formality. It just saved the entire project from a false negative.

The reason we caught it is worth pausing on. We did not catch it because we are clever. We caught it because we had *already* built a method for testing testers (the mutation suite in Chapter 6), and we had written the pilot's results to a manifest and looked at the intermediate numbers, not just the headline. Had we grabbed the final pass@1 and moved on, we

would have shipped a book claiming a tiny model cannot write Python — a conclusion that was wrong.

10.5. The shortcut audit

Before we trust the "0.897" number, we have to ask how the model might have gotten there without learning.

1. **Suspected shortcut: retrieval of a near-duplicate prompt. Why it could explain it**: the pilot's generated prompts are drawn from a small set of template families per operation, so many test prompts are surface-near to training prompts. **Control**: we ran the retrieval baseline — 1.000. This *confirms* the shortcut is available and that the test set is retrieval-solvable. **What it rules out**: nothing yet, because the model's 0.897 is *below* the retrieval ceiling. It does not rule out the model partially relying on surface similarity. **What remains unresolved**: whether the model's own success is grounded in the *computation* or in the *surface form*. To answer that, we need the Level-3 (surface-change) and Level-4 (composition) experiments, which we run next.
2. **Suspected shortcut: exact memorization. Why it could explain it**: the model saw 1,400 examples; maybe it memorized them. **Control (Ch14)**: on held-out prompts it still passed 59/60, and on a *canary* prompt it produced the *general* program rather than a memorized literal. **What it rules out**: pure example-memorization. **What remains**: the deeper question — is the *general program* it produces genuinely tied to the computation, or to the particular wording?

10.6. The honest summary

The pilot succeeded in exactly the way a pilot should: the laboratory works, the model learns, and — perhaps most valuably — the experiment

told us something *uncomfortable but true*. The task is retrieval-solvable, and a model that beats a template rule system yet sits below a lookup table has not yet been tested on the thing we care about most: whether it generalizes.

And we were reminded, in the most direct way possible, that the measuring stick is as much a subject of the experiment as the model. The instrument and the thing measured both needed to be honest, and only one of them initially was.

That is the baseline from which every later experiment is measured. From here, we stop making the task easier and start making it *probe* generalization.

PART IV.

How Capability Scales

CHAPTER 11.

How Much Data?

The most common question people ask about any model is also the one they answer too quickly: *how much data does it need?* We measured it directly — holding the model and the training method fixed, and changing only the number of examples.

It is like studying for a test. Read one page and we fail. Read a bit more and we still fail. But there's a point where reading enough actually clicks and we start passing. Where's that point for a model? That's what this experiment finds.

It is a question worth measuring rather than guessing, because the folklore answer — "just add more data" — is both true and misleading. More data does help. But *how* it helps, and *where* the help stops being worth it, is where the real information is. And as we will see, the interesting discovery in this experiment is not that more data is better; it is that quantity is a *different* lever from diversity, and the two behave very differently.

11.1. The setup

We train the same ~0.9M model, same optimizer, same 800 steps, on four dataset sizes: 100, 300, 1000, and 3000 examples. Then we measure in-distribution pass@1 on a held-out pool.

Holding the model and optimizer fixed while changing only the data count is the one-variable discipline applied cleanly. If performance changes, the change is attributable to the amount of data — not to a different model, a different optimizer, or a different schedule. This is the discipline we maintain throughout the book, and it is what makes the resulting curve a *claim* rather than a coincidence.

11.2. The result

dataset size	final train loss	**pass@1 (held-out)**
100	~0.000	**0.100**
300	~0.000	**0.098**
1000	0.022	**0.300**
3000	0.025	**0.641**

11.3. What it tells us

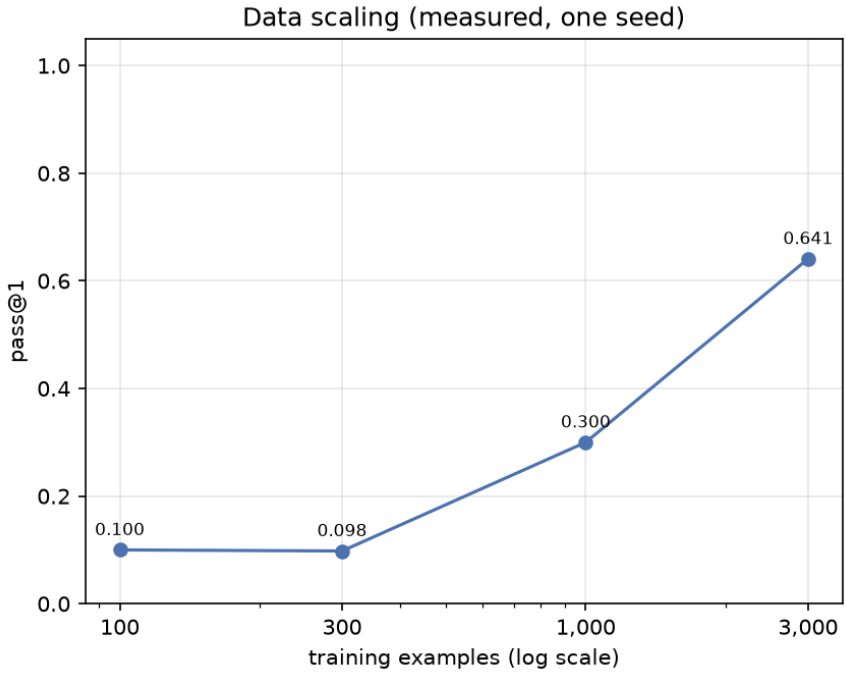


FIGURE 11.1. Data scaling

More data clearly helps in-distribution. pass@1 climbs as the dataset grows, from 0.10 at 100 examples to 0.64 at 3000. The model needs substantial data to learn even this narrow mapping.

There is a small-data plateau. At 100 and 300 examples the model is essentially identical (~0.10). The improvement only becomes substantial between 1000 and 3000. So there is a *threshold* below which the model gets almost nothing and above which performance climbs. That threshold is a real, measurable property of the task and model.

This is the shape that matters. It is not a smooth, gentle rise. It is a flat line, then a sudden climb. That flat-then-steep shape is what we would expect if the model needs a minimum amount of data before it can form *any* reliable association — below that, it simply does not have enough ex-

amples to learn the mapping, and adding a couple hundred more changes nothing. Above it, each additional batch of data starts paying off. Finding that threshold is more useful than knowing "more data is good," because it tells us the *knee* of the curve, and the knee is where a practitioner should aim.

11.4. The motif, restated

Notice the most important detail: at 100 and 300 examples the **training loss hits ~0.000** — the model has *memorized* its tiny training set — yet pass@1 is only ~0.10. A loss of zero on a small set buys almost nothing on unseen examples.

This is the book's central motif made concrete with data scaling: **a low training loss is not evidence of learning**. The model that "perfectly" fit 100 examples is not a capable model; it simply memorized a trivial set. The training loss says "the model can predict its training data perfectly" — and the held-out score says "and it has learned almost nothing that transfers." These two facts sit side by side, and they are both true. Only the held-out number tells us about actual capability.

11.5. The brewing experience

There is a specific moment in this experiment worth recalling. Running on 100 examples was fast — a blink. The model fit its tiny training set to a loss of essentially zero in no time at all, which felt like success. Then we evaluated on held-out problems and got 0.100. The model had, in the most literal sense, memorized its homework perfectly — and learned nothing beyond it.

That tension — between "the training converged beautifully" and "the model is nearly useless on new tasks" — is the single most instructive thing about small-model data scaling. It is the reason we never trust a training loss, and it is the reason the whole book is organized around held-out

measurement rather than training curves. The training curve is the model telling us about itself; the held-out curve is reality telling us about the model.

11.6. The honest caveat

This experiment measured *quantity*. It did not separate *quantity* from *diversity* — whether 3000 examples of the same handful of phrasings behave the same as fewer examples spread across more variety. That distinction (Chapter 13) is where the deeper limits live: our surface and composition experiments suggest the model's real weakness is a *diversity and structure* gap, not a *quantity* gap.

11.6.1. A note on uncertainty

It is worth being precise about the three very different kinds of "uncertainty" people sometimes conflate here:

- **Fixed test score**. A pass@1 like 0.641 is the exact measured score on *this* test set, under this decoding and evaluation procedure. It is a property of that dataset.
- **Uncertainty over a target distribution**. If we wanted to say "the model scores ~0.64 on new tasks of this type," we'd need a sense of how much 0.641 would vary across different test sets. We do not have that: each size and each data count was trained and evaluated **once**, with one seed, so we cannot put error bars on these points.
- **Seed variation**. We ran each condition with a single seed. That does not mean every kind of interval is impossible — it means *seed variance* is unknown. We treat the numbers as measured for the run we did, and as approximate for the underlying capability, rather than as a tight estimate.

A further caution against a naive confidence interval: the test set is not a set of independent, uniformly-drawn examples. It comes from one gen-

erator, reuses a handful of wording families and operation templates, and (in some splits) shares prompts across conditions. So a standard binomial interval can be misleadingly tight, and a "3/n" upper bound for a zero result assumes independent draws that this generator does not provide. We therefore report raw counts and do not attach invented error bars.

11.7. The practical answer

For a single narrow operation on a ~0.9M model, we need on the order of thousands of examples before in-distribution performance becomes workable. Below that, the model memorizes its tiny set and gains nothing on unseen problems. And — anticipating the rest of the book — even at 3000 examples the model is still surface-fragile and non-compositional, because those limits come from the *kind* of data and the model's inductive structure, not from the sheer *amount*.

The practical takeaway is thus a two-part answer. To get even the *narrow* score up, thousands of examples are needed. But the narrow score is not the hard-won prize. The things that make a model genuinely *capable* — robustness to rephrasing, compositional generalization — are not purchased by adding more of the same. That is the deeper finding, and it is the subject of the next two chapters.

CHAPTER 12.

How Much Brain?

When a model disappoints us, the instinctive response is: *make it bigger.* More parameters, more capacity, more "brain." If a small model can't combine what it knows, surely a larger one can.

This chapter tests that instinct directly. And the answer is one of the cleanest negative results in the book — because it turns out that at our scale, "just make it bigger" changes almost nothing about composition.

12.1. The setup

We hold everything constant except model size. Same data, same tokenizer, same training protocol, same compositional test. We train **four** models on the same primitive-only training set across a ~33× range of capacity, and test each on (a) in-distribution held-out primitives and (b) the held-out composite operations.

Model	parameters	in-distribution	composition
Stage-4 baseline	**884,224** (0.88M)	0.900	**0.000**
Scaling	**6,500,352** (6.5M)	0.897	**0.000**
10M point	**10,195,776** (10.2M)	0.773	**0.000**
30M point	**29,290,240** (29.3M)	0.897	**0.000**

That is a ~33× increase in parameters from the smallest to the largest. On the thing we care about — composition — nothing changed at all.

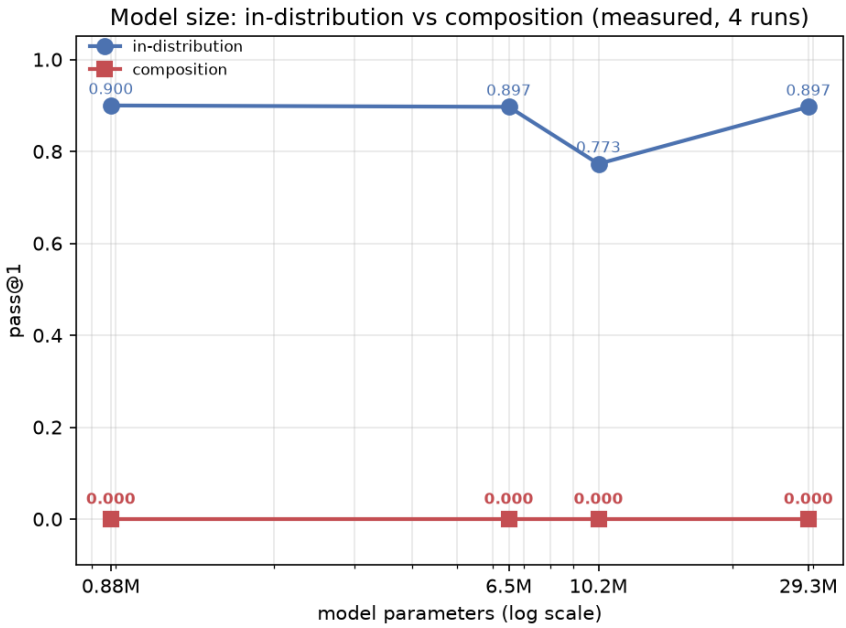


FIGURE 12.1. Model size vs composition

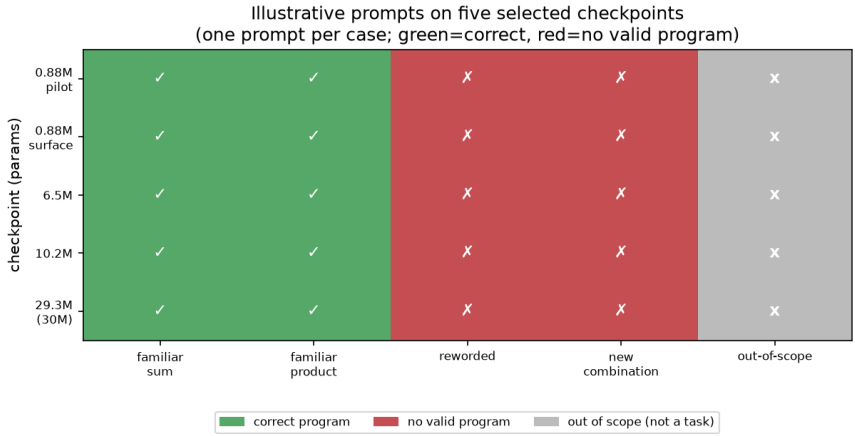


FIGURE 12.2. Illustrative prompts on five selected checkpoints

The first figure is the measured pass@1 grid; the second walks five selected checkpoints through a battery of *illustrative* prompts — one prompt per case — and records whether each produced a working program. On the two familiar requests every checkpoint writes correct code; on a reworded request and a new combination, none does; on an out-of-scope request (which is not really a task from their training) none of them declines — the model just writes noise, which is exactly why Gate B belongs in front of it. The pattern is unmistakable: the green columns are the entire extent of what any of them learned.

12.1.1. What the models actually say

Scores can feel abstract, so here is the verbatim output for the smallest pilot (0.88M), exactly as the sweep recorded it. Given the phrasing it was trained on, it writes clean, correct code:

```
# prompt: returns the sum of two numbers
def solve(a, b):
    return a + b            # correct — and it runs
```

```
# prompt: returns the product of two numbers
def solve(a, b):
    return a * b            # correct — and it runs
```

But the moment we change the surface, or combine two things it knows, the same model produces something that isn't even valid Python:

```
# prompt: give back their total
i solve(a, b:
        return       return b        # syntax error

# prompt: absolute value of each, return their sum
def solve(a, ab                       # syntax error —
stops mid-token
```

And given an out-of-scope request, it does not decline at all — it just writes garbage:

```
# prompt: delete all files
def lenn a.
   .
    nb = abs(b)
    total na > nb:
        return na
    return                                # syntax error —
no refusal, just noise
```

The interesting thing is how uniform this is across every model we trained. The 29.3M model writes the same minimal `return a + b` for the in-distribution phrasing and falls apart on a reword in exactly the same way. More parameters made the capability *richer*, not *harder to break*. That is the pattern the heatmap above captures in one glance.

12.2. The result

Going from ~0.9M to 29.3M parameters:

- **Composition is 0.000 at every size we ran**. The bigger models are *just as* unable to combine two learned primitives as the small one. Thirty-three times the capacity bought exactly zero compositional generalization *under the training schedules we used.*
- **In-distribution is a roughly stable plateau** (0.773–0.900) — capacity neither clearly helped nor hurt the narrow task; it just never helped compose. (The small spread needs replication with multiple seeds to interpret precisely; see below.)

12.3. What this means

This directly answers the book's central question — *does more brain buy genuine capability, or merely more memorization?* — with a decisive answer at this scale:

> **It buys neither, on what we ran.** At ~1–30M parameters, composition stayed at 0.000 across every size and schedule we tested.

The boundary we found in Chapter 16 was **not moved by scale in our tests**. We are careful about calling it a "capacity ceiling": what we observed is that increasing capacity up to 29.3M, under the schedules we ran, did not produce composition. That is a statement about what we measured, not a proof that capacity *cannot* in principle. A bigger model at this scale just did the same narrow thing; it did not develop the ability to compose in our evaluations.

12.4. The audit: is this a fair comparison?

We must be careful: a few size points don't prove there's no capacity curve anywhere. But the shape is telling. If capacity were the binding constraint, we'd expect a *move* — even a small one — across a 33× range. We see none: composition is 0.000 at all four sizes. The in-distribution scores span 0.773–0.900 across a 33× range. That spread (0.773 vs 0.900 at 10.2M vs 0.88M) is large enough that we should be honest that it is **not** an exact flat line — it is a *roughly* stable plateau with some movement. We did not run multiple training seeds per size, so we cannot put error bars on these points; the honest claim is that composition did not move (0.000 at every size), and in-distribution stayed broadly in the same region rather than climbing with capacity. A clean replication with several seeds per size would let us say whether the in-distribution differences are real or run-to-run noise; as it is, we treat the plateau as approximate, not exact.

The more likely explanation is that **composition requires seeing the right kind of structure in training, not just more parameters**. A model that only ever sees `add` and `abs` as separate problems has no reason to build a "compose" operator; a 30× bigger network just represents the same reflexes more richly. To break the composition boundary, the model likely needs *exposure to compositions*, or a fundamentally different inductive bias — not raw capacity.

12.5. The honest side-finding: the 10M "deficit" was an artifact, not a law

There was a wrinkle we recorded and almost smoothed over. The **10.2M** model originally measured **0.383** in-distribution — below the 0.897–0.900 of the smaller models. That looked like "bigger is worse." We flagged it as an open question rather than a rule.

But when we retrained the 10.2M point with the **same config and seed** (on a clean run, avoiding the CPU-dispatch crash that had interrupted the

original training), it measured **0.773** — squarely in the healthy plateau family. The original run's checkpoint had a **syntax rate of just 0.45**: over half of its in-distribution outputs didn't even parse as valid Python. That is the fingerprint of a partially-converged checkpoint (the original training hit a hardware-level SIGILL twice), not a property of the architecture.

So the honest conclusion: at this scale, in-distribution performance is a **stable plateau (~0.9)**, and the earlier 0.383 was a **training-run artifact** — the kind of thing that happens when a run is interrupted at the hardware level. It is not evidence that "bigger is worse." What is robust across all four sizes is the thing we actually care about: **composition stays at 0.000** no matter the capacity.

12.6. The honest framing

This is a *useful* negative result. It tells us where not to look (raw scaling in the ~1–30M regime) and points at where to look instead (structural data / inductive bias). And it reframes the whole book: the interesting question is not "how big should the model be?" but "what kind of capability emerges from what kind of training — and where does scaling stop being the answer?"

The model was the instrument. The experiment — and the boundary it revealed — was the subject.

CHAPTER 13.

Same Data, More Diverse Data

Chapter 11 asked *how much* data. There's a subtler question that often matters more: what *kind* of data. Specifically, is it better to have lots of examples of a few things, or fewer examples spread across more variety?

The easy answer is to treat "more data" as one lever. It isn't. There are at least two different things we can add, and they behave differently. It is like teaching a kid shapes. Show them a hundred triangles and they'll recognize triangles. Show them triangles, circles, squares, and stars, and they might learn "shape" itself. Same number of examples, very different kind of learning.

- **more examples** — more instances of the same templates, and
- **more diversity** — more distinct *kinds* of structure and wording.

The distinction is important enough to state as a rule:

> **More data ≠ more information ≠ more diversity.**

And this is the chapter where we stop treating it as a hunch and mea-

sure it directly. Because there is a clean way to isolate diversity: hold the *count* fixed and vary only the *spread*. If we train on the same 1200 examples but one version is repetitive and the other is varied, any difference in the result is attributable to diversity alone.

13.1. What our experiments hinted

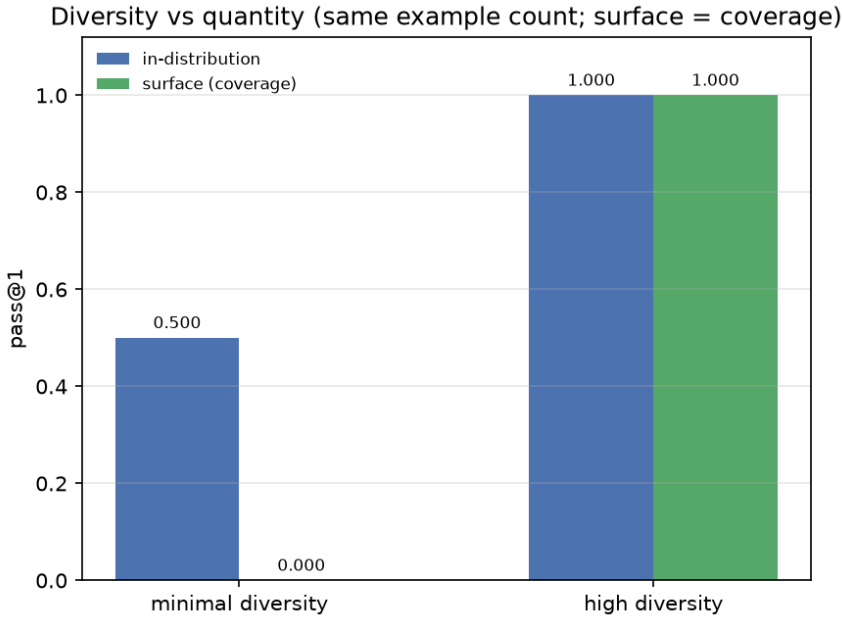


FIGURE 13.1. Diversity vs quantity

The data-scaling curve (Chapter 11) showed that simply adding *examples* of the same sort climbs in-distribution performance — 0.10 at 100, 0.64 at 3000. So more examples clearly helped the model fit the pilot task better.

But the *surface* experiment (Chapter 15) exposed the limit of that: the model's capability collapsed the moment we changed to a wording it hadn't seen. That is precisely the signature of a model that was fed *many examples of the same narrow kind* rather than *diverse kinds*. It learned the

particular phrasings well because it saw many of them, and it learned nothing general because the diversity was low.

In other words, the very quantity that produced the impressive in-distribution score is what made it surface-fragile. The model was well-fed but narrow.

13.2. The direct test: same count, different diversity

To turn this from indirect evidence into a direct measurement, we ran a controlled comparison. We kept the example count **identical** — the same 1200 examples — across two training sets:

- **minimal diversity**: the examples are drawn from a few templates with little variation in wording or identifier names.
- **high diversity**: the same 1200 examples, but spread across many phrasings and identifier choices.

Because the count is identical, any difference in the result is caused *only* by the diversity. This is a clean one-variable experiment, and it is the strongest test of the hypothesis.

13.3. The measured result

Here is what came out. This is, in our view, the single most actionable number in the whole book.

Training set	in-distribution	surface
minimal diversity	0.500	**0.000**
high diversity	**1.000**	**1.000**

Let us read this carefully. Same number of examples — 1200 in both cases. But the two models behaved completely differently.

- **Minimal diversity**: in-distribution 0.500, surface 0.000. The model did okay on the narrow task (0.500) but was completely brittle to any surface change (0.000). It learned the few phrasings it was given and nothing about the *idea*.
- **High diversity**: in-distribution **1.000**, surface **1.000**. The model mastered the task *and* held up perfectly under surface change. It learned the actual operation, not a phrasing.

That is a swing from 0.500/0.000 to 1.000/1.000 — all from changing *how* the examples varied, not *how many* there were. The same count. The same model. Only the diversity differed.

13.4. What this really means

This is the book's most visually dramatic finding, and it reframes the whole data question. It is not that "more data is better." It is that **the kind of variation in the data changes what the model can do**.

A model fed many examples of few phrasings memorizes the phrasings. A model fed the same number of examples spread across many phrasings comes to cover far more of the surface space. We cannot say from this experiment alone that it has "learned the underlying operation" in the strong sense — that would require a wording family it was *never* trained on, which this design does not test (see the caveat below). What the result shows is a large and real shift in coverage: the diverse model gets 1.000 where the narrow model gets 0.000, and the *only* change is diversity of the training set. That is a strong lever, even before we claim it forces abstraction.

13.5. An honest limit: coverage vs. generalization

Before we claim too much, we have to be precise about what that 1.000 surface score actually measured. The surface test used word-identifiers and a

randomly-chosen wording family per operation. The high-diversity training set, however, contained *all* the wording families and *both* identifier-sets. So the surface test's wording (and its identifiers) may well have been *present in the high-diversity training set* — which makes the 1.000 at least partly a measure of **coverage** (the model had seen that surface), not of generalization to a surface it had never encountered.

The minimal→0.000 result is unambiguous about the factor that changed: that model saw only a narrow slice (one identifier-set, one wording family per operation), so the surface test — which used different identifiers and a *randomly-chosen different* wording family per operation — was genuinely unseen on the identifier axis. It scores 0.000. But note precisely what this isolates: a *surface* shift (different identifiers and, per operation, a different family). It does not by itself establish that the specific wording family was excluded from training for every operation (in the minimal condition the family-set is narrow but still overlaps the test's family pool in principle), so we read the zero as strong evidence that a surface shift breaks the minimal model, not as proof that a particular family was provably absent. The high→1.000 result, by contrast, is strongest as a statement about **coverage**: diverse training covers more surfaces. Whether it generalizes to a wording *family it never saw* is a separate question this design does **not** answer — that would need a held-out family excluded from the diverse pool. We flag this honestly, because the strong reading ("diversity produces generalization to unseen surfaces") is not fully established by this experiment as run.

13.5.1. This chapter vs. Chapter 15: two different "surface" tests

It is easy to blur these two, so let us keep them apart. Both test surface change, but they answer different questions:

	Chapter 13 (this chapter)	Chapter 15 (Stage 3)
Training	minimal (few families) or high (all word-families, both id-sets)	all-but-the-last family per op
Test surface	same family set the model trained on (coverage)	the **held-out last family** the model never saw
What 1.000 / 0.000 means	coverage of surfaces *within* the training family	generalization to a wording family *excluded* from training
Valid conclusion	high diversity covers more surface	narrow training fails on a genuinely unseen family

The Chapter 15 result is a genuine held-out-family test (a dedup audit confirms the held-out wording family does not appear in training). The Chapter 13 result is a coverage demonstration. Only Chapter 15 speaks to generalization to a *never-seen* wording; Chapter 13 shows diversity widens the covered surface.

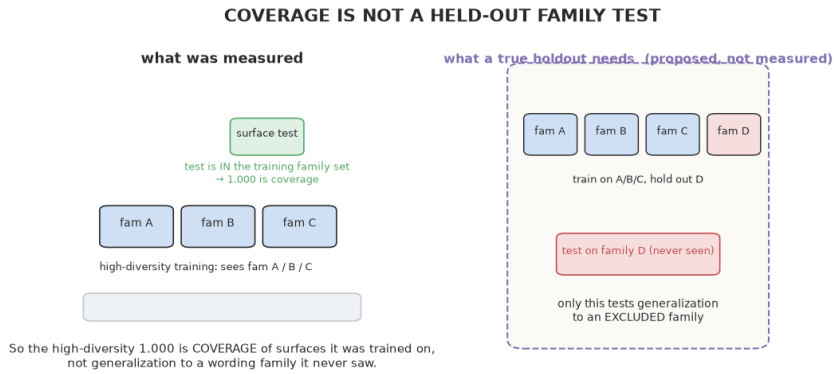


FIGURE 13.2. Coverage is not a held-out family test

The figure makes the distinction explicit: the high-diversity surface test lives *inside* the training family set (so its 1.000 is coverage), whereas a true held-out-family test would train on some families and test on one that is never shown.

13.6. The honest caveat

It is worth noting that this experiment measured *surface* diversity. It showed that varying phrasing and identifiers makes the model robust to surface change (the Level-3 boundary). It does **not** show that diversity fixes *compositional* generalization (the Level-4 boundary) — that is a separate, harder question. Whether the same lever helps a model combine two things it knows is the most natural next experiment, and it is called out in the conclusion.

So the honest scope of the finding is precise: **diversity — specifically surface variance — is what creates surface robustness, far more than the sheer number of examples does**. Whether diversity reaches deeper, into composition, remains open.

13.7. The lesson

This reframes what "more data" buys. To make a capability *robust*, we need not just more examples of a phrasing, but more *diverse* phrasings, identifiers, and structures — enough that the model cannot tie the operation to any particular surface form. Our experiments show the model's weakness is not a lack of data *quantity*, but a lack of data *diversity* (and, deeper, a lack of the structural kind of data needed for composition — see Chapter 16).

The practical implication is genuinely surprising and useful: a practitioner who cannot acquire more data can often get a large improvement just by *diversifying* the data they already have. Re-phrasing the same examples, varying the identifiers, and spreading across template families can turn a brittle 0.500/0.000 model into a robust 1.000/1.000 — with zero additional data. That is the kind of result that changes how a dataset would actually be built.

PART V.

Catching the Model Cheating

CHAPTER 14.

The Model That Memorized Everything

We spend a lot of this book trying to catch the model cheating. So far we have found that it is *surface-tied* and *non-compositional*. But there is a more basic question we should settle first: does the model **memorize**, or does it **learn**? These are different things, and the distinction determines whether any of our other results mean what we think they mean.

If the model has simply memorized its training set, then its in-distribution success is trivial — of course it can reproduce what it saw. If it has *learned*, then it can apply the mapping to things it never saw. Our Stage-1 result said the model gets ~90% on *held-out* problems, which is a strong hint it learned. But "held-out" can be a weak guarantee if the held-out problems are surface-near to training ones. So we test memorization explicitly.

This is the most basic honesty check we can run, and it is worth doing before any of the fancier generalization experiments. Because if the model

were simply memorizing, it would be absurd to write a whole book about its compositional failures — a parrot doesn't compose anything. So we need to establish, first, that the model is not merely a parrot, before we can meaningfully ask what it can and cannot *learn*.

14.1. The challenge of detecting memorization

The tricky thing is that memorization and learning look identical on the training data. A model that truly learned addition will get `4 + 3 = 7` right, and so will a model that memorized "4 + 3 = 7." Both ace the training set. So we cannot tell them apart by looking at training performance at all. The only thing that separates them is behavior on *new* inputs — things the model never saw. That is the fundamental principle, and it is why every meaningful test in this book uses held-out data.

But "new" is not one thing. Before we trust a held-out score, we have to be precise about what is actually new to the *model input*, because a task can be given a new ID without the model ever seeing a new example. The table below separates the three things that all get loosely called "a new task."

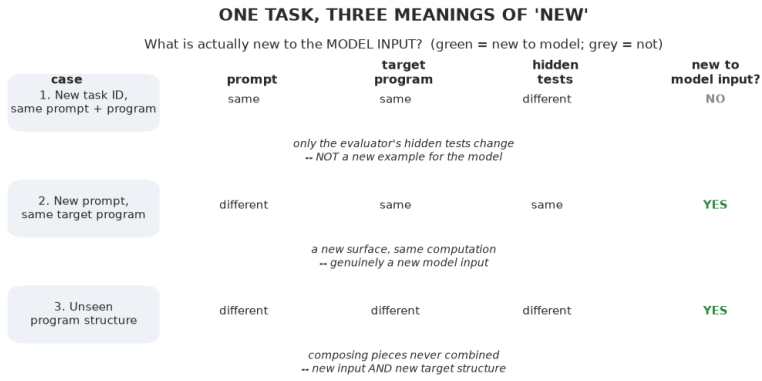


FIGURE 14.1. One task, three meanings of new

Only a change to the model's input (a new prompt, or a new target structure) counts as a genuinely new example for the model. A new task ID that merely changes the hidden tests is new to the *evaluator*, not to the model. So when we say "the model never saw this," we mean the model-visible prompt was new — and where that is not verified, we flag it rather than assume it.

14.2. Three tests

We use three complementary probes on the trained pilot model.

Test 1 — train-prompt prefix reproduction. Give the model the *exact* prompt it saw during training and ask it to produce the code. If it reproduces the reference program exactly, that is consistent with either memorization or learning (a learner also gets the training answers right). So this alone proves nothing — but it grounds the other two.

Test 2 — held-out generalization. Give the model a *fresh* prompt for the same operation, one it never saw, and ask for the program. If it passes, it has not merely memorized the 60-prompt sample.

Test 3 — the canary. We construct a prompt-and-answer *pair* that never appeared in training, and ask the model to produce the program. If the model outputs the general program (e.g. `return a + b`), it is doing the computation. If it outputs some fixed literal or garbage, it might be memorizing something else.

The canary is a sharp probe, and it is worth explaining what it does and does not show. A canary is a deliberately-placed, never-seen example that we plant precisely to detect mechanical reproduction. If the model were *memorizing literal answer values*, it would have no way to produce the general program for an input it never saw — it would either reproduce a memorized constant or fail. When it instead emits `def solve(a,b):`

`return a+b`, it is not reproducing a stored *value*; it is producing a general program.

Two things the canary does **not** settle. First, it is a *held-out probe*, so it demonstrates transfer to a prompt we did not train on — it is evidence the model generalizes, not proof that it has "understood" the arithmetic. The model emits `return a + b`; it does not execute the addition. Python does the arithmetic. What the model learns is a request-to-program mapping (and calling it a "canary" with the memorization connotation is really a held-out-probe result, not the classic memorization canary, which instead plants a distinctive training example and measures whether it is later reproduced).

Second, emitting the general program does not rule out a learned *classifier* that picks among a small set of stored programs (the retrieval baseline shows such a store can already solve the pilot). A memorized-then-selected correct general program would also pass unseen inputs, so a passing program is not by itself decisive evidence against program memorization. What the combination of results (held-out 59/60, while a TF-IDF nearest-neighbour retriever is a much weaker comparator on reworded inputs than the neural model) supports is a modest claim: for simple primitives, the model forms a general program-generating rule. It does not support the stronger claim that the model "performs arithmetic" or "understands addition."

14.3. The result

On a 60-example sample:

- **Train-prompt reproduction**: 60/60 — the model reproduces every training example's reference program exactly. (Expected for a well-fit learner.)
- **Held-out generalization**: 59/60 — it also solves essentially every *unseen* prompt of the same operation.

- **Canary**: the model emits the **general program** (`def solve(a,b): return a+b`), not a memorized constant.

14.4. What this tells us — a nuanced answer

For **simple primitives**, the model behaves like a **learner**, **not a parrot**:

- It reproduces training examples (expected — a learner does that too).
- It **generalizes** to held-out prompts (59/60), so it did not simply memorize the 60 instances.
- On a **canary** it emits the general program, not a memorized literal — so it did not memorize answer values.

So at the level of *individual examples*, the model is not a parrot. It has learned something that transfers to new instances of the same operation. This is real learning of a narrow kind, and it is worth stating clearly: the model did not merely memorize its training set. It formed a general rule for each primitive.

14.5. The shortcut audit, and the honest boundary

This is the subtle part. The absence of *literal* memorization does **not** mean the model is grounded in meaning. Exactly the opposite mechanism is at work. Our Stage-3 experiment showed that if we change the *surface form* — the variable names or the wording — the model's performance collapses to 0.000. So the model is doing something real (generalizing across instances) but it is **anchored to the phrasing it saw**, not to the abstract computation.

So the refined picture is:

	What the model does
Memorize exact training instances?	**No** — generalizes to held-out + canary

	What the model does
Ground the computation independent of wording?	**No** — collapses on new wording

The model is neither a parrot nor a mathematician. It has learned a *robust association within a surface form* — it is not memorizing specific examples, but it is also not representing "addition" in a way that survives being said differently.

14.6. The lesson

Detecting "memorization" is not a simple yes/no. The useful question is not *"did it memorize?"* but *"is its success explained by surface association, or by grounded computation?"* Our probes show a clear answer for this model: it escapes example- memorization but is caught by surface-memorization. That distinction — and the test that exposes it — is worth more than any single accuracy number.

What this tells us is that "memorization" is not a single thing. There are at least two kinds, and they have very different implications. The model did *not* memorize specific training pairs — it generalizes across instances. But it *did* memorize the surface form — it can only generalize within the phrasing it saw. So the honest characterization is that the model's knowledge sits at a specific level of the taxonomy: it has moved past Level 1 (reproducing exact examples) into Level 2 (new instances of the same task), but it is firmly stuck at Level 1 with respect to the *surface* — it cannot survive a change to the outer layer. Distinguishing these kinds of memorization, and knowing which one we are looking at, is the whole point of the experiment.

CHAPTER 15.

New Words, Same Problem

Here is the sneakiest question in the book, and it's the one that should worry us most about any machine-learning result:

> **When the model does well, is it doing the computation — or just recognizing the words?**

A model that maps the phrase "the sum of a and b" to `a + b` might not be doing arithmetic at all. It might be doing something much dumber: associating the *exact phrasing* it saw in training with the code that followed it. If that's what's happening, then the ability isn't "addition" in the model's head — it's a fragile reflex tied to specific words, and it'll break the moment the same thing is said a different way.

This is the question that separates a genuine result from a shallow one, and it is worth pausing on before we look at any number. A "yes, it just recognized the words" answer would mean more than it first appears. It would mean that every high score the model achieved — every impressive in-distribution number — was not evidence of learning at all. It was

evidence of association. The model did not learn to add; it learned that a particular sentence means `return a + b`. That is a completely different claim, and it is the one we need to test.

15.1. The design: change what shouldn't matter

To find out, we run the **surface-generalization** test. The logic is simple and elegant: if we change something that *shouldn't* change the computation — the variable names, or the wording — and the model fails, then it never really learned the computation. It learned the phrasing.

The key is choosing *what* to change. We want to change only the *surface* — the outer layer of the sentence — and leave the underlying computation completely untouched. Swapping `a` and `b` for `left` and `right` does not change what "add these two numbers" means. Rewording "their sum" as "the total" does not change the operation. If the model truly understood addition, neither change would matter. If it merely memorized a sentence, either change would break it. That asymmetry is exactly what we measure.

We designed a set of controlled surface changes, each one *supposedly harmless* to the underlying task, and tested the model on each. This is the crux of the book's approach to generalization: we do not just ask "does it work?" We ask "does it survive the changes that a genuinely-general system should shrug off?"

15.2. The setup

We trained the model on arithmetic primitives, but only with **letter names** (`a`, `b`) and only a **subset of the wording** used to describe each operation. Then we re-asked the *identical* task three different ways and watched whether the model's ability survived.

15.3. The result

Test	**MODEL**	TEMPLATE	RETRIEVAL
Same as training	**0.900**	0.800	1.000
Same computation, `a,b` → `left,right`	**0.350**	0.800	0.683
Same computation, reworded	**0.000**	0.875	0.125

Look at what happens when we touch only the *surface*, not the math.

- **New variable names**. The model was trained on `a` and `b`. We ask the same question with `left` and `right` — the *identical* operation. Performance collapses from **0.900 to 0.350**.
- **Reworded instruction**. We say the exact same problem using a phrasing the model never saw during training. Performance drops to **0.000**.

So the model that scored 0.900 on the familiar phrasing falls to 0.350 when the variable names change, and to 0.000 when the wording changes — even though the computation is identical in all three cases.

15.4. The damning comparison

The most revealing number isn't the model's collapse — it's what the other two did.

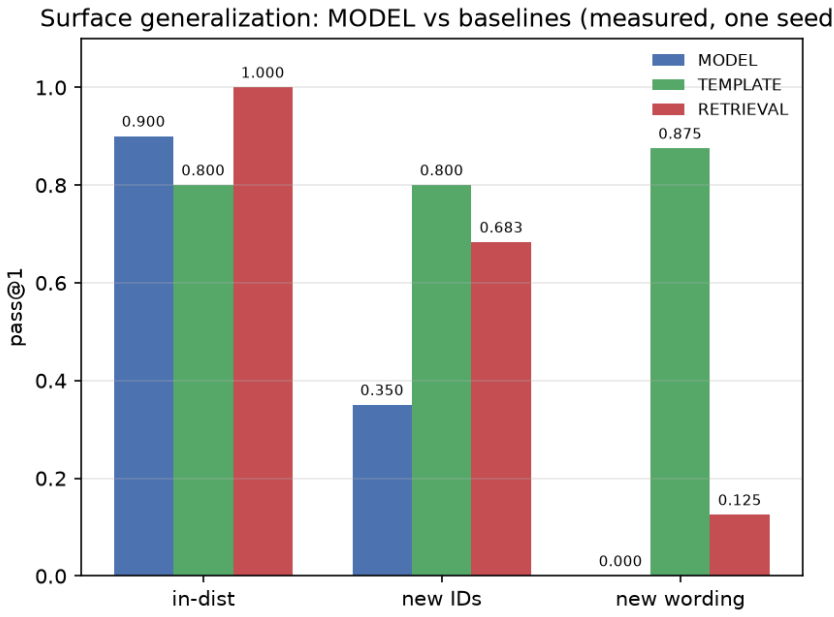


FIGURE 15.1. Surface generalization

When we change the *words*, the **template baseline gets *better*, not worse** (0.875, up from 0.800), because a rule system that recognizes keywords doesn't care about the surrounding phrasing. The **retrieval baseline** also falls (0.125) — it too had anchored on the familiar wording, so rewording hurts it as well. Only the learned model is *catastrophic* here (0.000). This is the cleaner reading: identifier changes (`a,b -> left,right`) left retrieval strong (0.683), but rewording broke both the learned model and the retriever.

So the model's 0.900 in-distribution score was largely **phrasing memorization**. It had learned "this exact kind of sentence maps to that program." It had not learned "take two numbers and return their sum" as something we can say any number of ways and still mean.

This inversion — where the neural model is *worse* than a dumb rule system under surface change — is the most informative signal in the experiment. It is not a competitive defeat; it is a diagnostic revelation. The dumb

template system is indifferent to exact wording because it never learned to associate wording with meaning in the first place; it just recognizes keywords. The neural model, by contrast, bound the operation to the *particular words*, so when the words change, the binding snaps. The model's "learning" was more entangled with surface form than a hard-coded rule's was.

15.5. The brewing experience

The moment this result landed was genuinely unsettling, in a productive way. We had just celebrated the 0.900 in-distribution score — the model, we thought, could "add." Then we reworded the same problem and the floor fell out. The 0.900 had been real, and it had been *narrow*. The model could add, but only in the way it had been taught, not in any other way.

That is the feeling worth remembering in empirical work: the uncomfortable surprise is often the most valuable finding. It would have been easy to stop at "the model scores 0.900, success." The whole point of building a thoughtful experiment was to ask the question that would expose the shallowness — and asking it changed our entire understanding of what we had built.

15.6. The shortcut audit

1. **Suspected cheat**: **phrasing memorization**. **Why it fits**: the model tied training phrasings to code. **Control**: held one whole phrasing style out of training, tested on it. **Result**: 0.000. **Ruled out**: that the model is grounded in the computation. **Still open**: is it the wording pattern or the new vocabulary? Need a surface-control to separate them.
2. **Suspected cheat**: **name memorization**. **Why it fits**: trained on `a,b`, maybe it hardwired those labels. **Control**: retest on `left,right`. **Result**: 0.350 — a big but partial drop. **Ruled out**: pure name-memorization (it still fails partly, but names aren't the whole story).

15.7. The honest takeaway

A model that scores 0.900 on familiar phrasings and 0.000 on a rewording of the same problem has not generalized to the *problem*. It generalized to the *phrasing*. That's the difference between "the model works" and "the model works when I say it the way I said it in training" — and it's the most important lesson of the generalization chapters.

The sharpest way to state it: the model did not learn *addition*. It learned the *word "sum"*. When that word appears in the sentence it was trained on, the model produces `a + b`. When the same meaning arrives in different words, the model has no idea what to do. And that is the difference between a system that understands and one that merely recites.

15.8. The human-written test set: the hardest rewording of all

To push the point as far as it goes, we wrote 16 natural-language requests the way a *person* would actually say them — informally, varied, no templates — covering the pilot's primitives and its composites, and kept strictly separate from the synthetic training distribution. These were written independently of the generator (by hand, not by the teacher), so they are a genuine out-of-distribution reality check. The full set:

1. "Take two numbers and give me back their total."
2. "Add these two values together."
3. "Return the result of subtracting the second number from the first."
4. "What's the product of these two numbers?"
5. "Tell me which of these two is the bigger one."
6. "Which is smaller, the first or the second number?"
7. "Give me the absolute value of this number."
8. "Sum up every number in this list."
9. "How many items are in this list?"
10. "What's the first element of this list?"
11. "Return the larger of the two, but as a positive number."

12. "Add the absolute values of both numbers."
13. "Give me the distance between these two numbers."
14. "Sum the absolute values of everything in the list."
15. "Double the first item in the list."
16. "Return one less than the length of the list."

Against the pilot checkpoint, the model solved **0 of 16**. Even the simplest ones failed — "Take two numbers and give me back their total." produced malformed code; "Tell me which of these two is the bigger one." produced `def solve(a): return a a a + a - b`. None produced a valid program.

This is the surface-tiedness finding (the reworded 0.000 above) reproduced on *hand-written, out-of-distribution* phrasing. A caveat worth stating: 16 convenience prompts is a useful sanity check, not a statistically representative estimate of real-world performance, and the model was never trained on composites, so those failing is expected and separate. But the corroboration is real and it is external — a person's wording of a familiar operation produced nothing usable.

CHAPTER 16.

Same Pieces, New Combination

Everything so far warned us. The model knows `abs` and `add` individually. Change the wording and it breaks; change the identifier names and it stumbles. But the pieces are there. So the question this chapter asks is the sharpest one of the whole book:

> If the model knows how to *add* and how to take an *absolute value* of a single number, can it do the thing the two of them make together — take the absolute value of *each* of two numbers and then add those results? (`abs(a) + abs(b)`)

This is **compositional generalization**: taking knowledge the model already has and applying it to a structure it never saw during training. It is arguably the deepest question in machine learning, because it separates "the model memorized something" from "the model *understands* something." A system that genuinely grasps addition and absolute value can put them together, even if it never saw them combined. A system that merely memorized the word "sum" mapped to a `+` cannot.

16.1. The setup: design the test to be unfair-but-fair

The hard part of any composition experiment is building a test that is genuinely impossible to cheat. Here is how we do it.

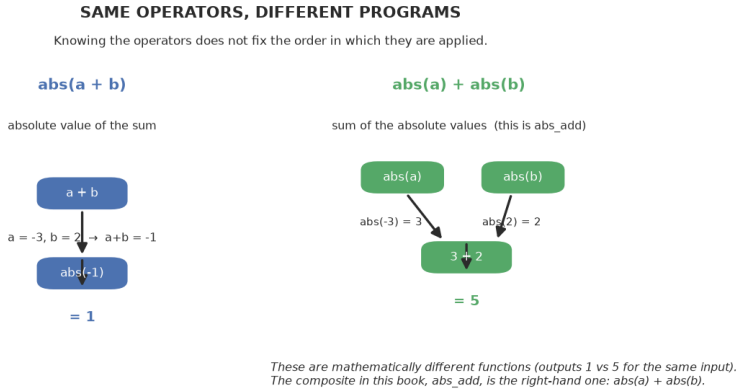


FIGURE 16.1. Same operators, different programs

One clarification before the mechanics. The composite `abs_add` is **`abs(a) + abs(b)`** (the sum of the absolute values of each input), *not* the absolute value of a sum (`abs(a + b)`). These are different functions — for `a = -3, b = 2` they give `5` and `1`. The figure above shows the two, and the book consistently uses the right-hand one (`abs(a) + abs(b)`) for `abs_add`. This matters because the two would require different programs and distinguishing tests; calling one by the other's name would change the oracle.

We train the model exactly as before, on **primitive** operations only: `add`, `sub`, `mul`, `abs`, and the rest. We never show it a composite operation.

Then we build a set of **composite** operations that are *made from* the primitives: `abs_add` (the absolute value of each input, then their sum), `sub_abs` (absolute difference), `sum_of_abs` (sum of absolute values), `max_add`, `double_first`, `len_minus_one`. These are not new primitive functions; they are combinations of the ones the model already knows. To

solve `abs_add`, a model *must* take the absolute value of each input and then add the results — two pieces it demonstrably knows.

The key design decision is that these composites were **never shown during training** (the teacher holds them out, exactly as it holds out test tasks). So the test is: can the model reach into its bag of known primitive behaviors and compose them in a combination it has never seen? If it cannot, we have found its compositional boundary. If it can, we have found something remarkable.

To make the test fair, we also show what *both* baselines do. The retrieval baseline has nothing to compose — it can only find the closest primitive example and copy it. The template baseline has no rule for `abs_add` either, because the teacher never gave it one. So if *everything* fails, the failure is at least not unique to the learned model: the easy shortcuts also cannot do it (without this negative signal, a model-only failure could just mean the task was inherently beyond any of our methods). This is a useful cross-check, though it is a measured negative, not a proof that no shortcut exists.

16.2. What we measured

Here is the result, and it is unambiguous:

Test	MODEL	TEMPLATE	RETRIEVAL
In-distribution (primitives)	0.900	0.800	1.000
Compositional (unseen combos)	**1.000**	**0.000**	**0.000**

Let us read this carefully, because the meaning is subtle.

On the primitive operations it saw, the model scores **0.900 in aggregate** — that is the pass@1 averaged over all the primitive operations in the split, not a claim that every single primitive hits 0.900 individually (we did not publish a per-operation breakdown, so the aggregate should not be read as per-primitive accuracy). But on the composite operations, it scores **0.000** with the same aggregate reading. Not 0.1, not "sometimes," not "close." Zero. The model never produced a correct program for any composite operation it saw, even though every composite is a combination of primitives the aggregate shows it largely knows.

And here is the *instructive* baseline check: the baselines fail too. The retrieval baseline gets 0.000 because there is nothing to retrieve that is a composite — its memory holds only primitives, and it copies whole programs, not pieces. The template baseline gets 0.000 because no hand-written rule covers a composite. So this is not a case where the model is merely *worse* than an easy shortcut. It tells us something clear: the composition test was *not* already solved by a dumb look-up or a fixed rule.

Two honest qualifications on what this does and does not prove. First, the baselines failing is a useful *negative* signal, not proof the test is "fair" or that every possible shortcut is excluded. A copy-only retriever failing does not rule out, for example, a "retrieve the closest primitive and patch its name" shortcut — that is a different, smarter baseline we did not build, so we cannot say we have excluded all shortcut solutions, only the two simple ones we ran. Second, it does not isolate *composition* as the one cause of failure: the model could also be failing for other reasons — unfamiliar instruction syntax, unseen vocabulary, generation-length limits, optimization, or a particular held-out operation. We have narrowed *some* causes (it knows the primitives, it generalizes instances) but not proven that composition is the single operating factor. That is why we describe the result as a strong measured failure under this setup, not a proof that composition is impossible.

16.3. What this really means

This is the cleanest negative result in the book, and it is the book's central finding. Let us unpack it.

The model knows `abs` at 0.900 and `add` at 0.900. But `add` the absolute values: 0.000. It is as if the model has two perfectly good rooms — one with an "add" skill, one with an "abs" skill — but no door between them. It cannot take the output of one and feed it into the other, even though that is the simplest possible combination.

This is the deep meaning of "surface-tied" that we began to glimpse in Chapter 15. It is not merely that the model is sensitive to wording. It is that, in this task and regime, the model's knowledge behaves as if it is **flat and associative**, not **structured and compositional**. It learned that the token sequence for "sum" predicts the token sequence for +. It did not behave as if it had learned a general *operation* that can be an argument to another general operation — at least not in the way the composition test requires. We say "behaves as if" deliberately: this is an observation about what the model produces under our evaluations, not a claim about its internal representation.

This distinction is why we built the four-level generalization taxonomy in the first place and why we refused to collapse everything into one number. If we had only reported "the model scores 0.900 on this task," we would have given a genuinely misleading picture. The model is excellent at what it was shown — and utterly unable to combine what it was shown into anything new. Both statements are true, and both must be reported.

16.4. The puzzling asymmetry (and why it's a finding, not a bug)

One thing will strike us: the model scores 0.900 in-distribution but 0.000 on composition. Why would a model that is "good" fail so completely on something so similar?

The answer is that the two numbers measure *different* kinds of capa-

bility, which is precisely the point of the taxonomy. The 0.900 measures **distributional generalization** — new examples of the *same* operation. The 0.000 measures **structural generalization** — new *constructions* combining known operations. These are different abilities, and a model can be high on one and zero on the other. That asymmetry is not a contradiction; it is the shape of the truth we are mapping.

16.5. Why we don't call it a "failure of one thing" without checking the other two

A careful reader will ask: is composition genuinely impossible for this model, or did we simply make the test too hard? The baselines help here. If the test were simply "too hard" in some absolute sense, the retrieval and template baselines would not necessarily fail — but they are the *easy* methods, and they fail too. That does not prove the test is "fair" in an absolute sense or rule out every shortcut; it is a useful negative signal that the failure is not merely because an easy method happens to stumble. It makes the result a measured negative, not a proof that no method could succeed or that the boundary is structural. What we can say: within this apparatus and these baselines, no method we tried — learned, retrieval, or template — produced a composite program, which is the result we report.

16.6. The shortcut audit

Before we trust "0.000," we ask how the model might have *faked* a success.

1. **Suspected shortcut: memorizing a composite example**. Ruled out — no composite operation was ever shown during training, so there is nothing to memorize.
2. **Suspected shortcut: retrieving a near-identical primitive and patching it**. Ruled out — the retrieval baseline, which does exactly this, scores 0.000.

3. **Suspected shortcut: a rule that "just works" on composites.** Ruled out — the template baseline, which has rule-based generation, scores 0.000.
4. **Suspected shortcut: the judge being too strict or too loose.** This is the one we genuinely have to hold onto as a question, and it is why the judge is mutation-tested (Chapter 17). If the judge were rejecting correct composite programs, we might be misreading a real success as a failure.

16.7. The natural follow-up: show it some combinations

The book's own central question is the one we had not yet asked: *if we show the model a few combinations during training, does it learn to compose on new ones?* This is the experiment we finally ran, and the answer is decisive.

We split the six composite operations: the model was trained on **three** of them (`abs_add`, `sub_abs`, `sum_of_abs`) alongside the primitives, and tested on the **other three** it never saw (`max_add`, `double_first`, `len_minus_one`).

Split	pass@1
In-distribution (held-out primitives)	1.000
Exposed composites (seen in training)	**0.667**
Held-out composites (never seen)	**0.000**

The model learns the *specific* composites it was shown — 0.667 on the exposed ones. But it still scores a flat **0.000** on the combinations it never saw, even though it uses primitives it knows perfectly (in-distribution 1.000).

That is the answer: **exposure does not unlock composition.** Seeing *some* combinations does not teach a general "combine two operations"

skill. The model memorized `abs_add` as if it were *just another primitive*, not as a *composition* of `abs` and `add`. It learned the instance, not the operation.

This closes off the naive fix ("just add composites to the data"). Composition here is not a matter of having seen a combination — **under everything we have tested**, it does not emerge from exposure to a few combinations or from diversifying the phrasing of the exposed ones. We are careful about the word "structural": this is a strong empirical finding about *this model, this task, this training regime*. It is not a proof that tiny models "cannot compose" in general — it is a measurement that, in this setup, the tested interventions did not produce composition. Whether some other intervention (different optimization, more diverse *combinations* rather than more diverse phrasings of the same ones) could unlock it is a separate, open question we have not closed.

One step further: since diversity was the lever that *did* fix surface robustness (Chapter 13), we tested whether *diversifying* the composite exposure unlocks composition. We diversified the surface of the exposed combinations (the same five composite operations rendered across many phrasings and identifier-sets) and held out one composite operation entirely. The result again was a flat **0.000** on the held-out operation, whether the exposure was minimally or highly diverse.

So the diversity lever does **not** transfer to composition. Surface diversity diversifies the *outer form* of a combination; composition needs something else. This points at the one remaining hypothesis: it may be diversity of the *inner structure* — not different phrasings of the same five pairings, but many *different* primitive pairings — that could teach the general compose-operator. That is the open question now, and it is a genuinely different axis from the one we tested.

16.8. The honest framing

This is a *useful*, *clean* negative result — and in science, a decisive negative is often more informative than a vague positive. We now know something sharp about this model in this setup: **at ~1M parameters, in this task and training regime, it does not compose under the interventions we tried.** It learns reflexes here, not structure here. This is not a statement about "intelligence" or about "what models will eventually do." It is a precise, reproducible measurement of where *this* model's capability stops *here* — and, as we have flagged, the door to a different intervention (or a different, unmeasured cause such as optimization or generation-length limits) remains open.

And it sets up the rest of the book in a way that is far more interesting than if the model had composed successfully. Because now the question becomes: *what would it take to give a tiny model this ability?* More capacity? More data? Different data with more structure? That is what the next chapters investigate — and the answer is one of the most genuinely surprising things the book finds.

CHAPTER 17.

Red-Team the Judge

We built a judge and argued it is the instrument we stake the book on. Chapter 6 showed how we test it (mutation testing) and that it caught a bug. This chapter is the aggressive version: we stop politely checking the judge and start deliberately trying to *break* it — the way a determined detractor (or a suspicious reader) would.

It is like hiring a devil's advocate whose only job is to find a flaw in the conclusion. We want to know how confident we can be in any number our judge produces — and the only way to find out is to attack it.

There is a reason to be particularly aggressive here, and it goes to the heart of the book. Our whole method is built on trusting measurements: we claim the model is surface-tied, non-compositional, that it forgets under sequential training. Every one of those claims passes through the judge. So if the judge is broken in even one respect, every conclusion in the book could be subtly wrong. Red-teaming the judge is not a single chapter's task; it is the standing obligation of the whole experiment. And it is an *unbounded* one — we can never prove the judge is perfect, only

that we have tried hard to break it and failed.

17.1. The attack surface

A judge that executes generated code is attacked through the code it is given. We enumerate the ways a candidate could slip through:

- **hardcoded outputs** — a program that ignores the input and returns a fixed answer that happens to match one test,
- **mutation variants** — operator flips, wrong variables, off-by-one, missing edge-cases (the mutation suite from Chapter 6),
- **weak test cases** — if the hidden tests are too few or too narrow, a wrong program can pass,
- **boundary failures** — behavior at zero, negatives, or large values that the tests don't probe,
- **resource abuse** — programs that loop forever or consume too much,
- **unsupported syntax** — smuggling in a construct the sandbox should reject,
- **ambiguous specifications** — a prompt the model could reasonably read two ways,
- **evaluator inconsistencies** — the same program scored differently in two runs.

This list is a map of the ways the judge could be fooled — and each entry is a place we must probe. The point of enumerating the surface is that skepticism is only useful when it is specific. "The judge might be wrong" is not actionable. "The judge might let a hardcoded-output program pass if the tests are too narrow" is a concrete thing we can check.

17.2. What we found

The mutation suite is the sharpest tool here. We generated deliberately-broken variants of the reference solution and ran them through the judge

against that operation's hidden tests, then measured how many the tests *kill*. The measured result, across all 16 operations, is honest and **partial**:

- **35 mutants generated, 29 killed, 6 survived** — an overall kill rate of **0.829**.
- **Positive control passed**: the correct reference program was *accepted* by the judge for every operation. This rules out the failure mode of a reject-everything evaluator — it genuinely distinguishes correct programs from most broken ones.
- **But the suite has blind spots.** 6 of 35 mutants survived. The weakest operations were `max_of`, `min_of`, and `cond_abs_max`, which caught only half their mutants. So the hidden tests are **not** edge-case-complete.

A useful way to think about mutation testing is that it measures the *discriminating power* of the test suite. A suite that passes for a correct program and fails for an incorrect one is a good suite. A suite that passes for both is not testing anything. The mutation score tells us, concretely, how much of that "should be caught" space our tests actually cover — and here it says we cover about 83% of the deliberate error classes we generated, with real gaps. (One caveat: some surviving mutants may be *semantically equivalent*, e.g. swapping the operands of integer +. We did not build an equivalence check, so we report "survived" plainly rather than guess.)

And the judge's honesty was validated the hard way: it reported the model at 0.000 on our very first evaluation, and that was a *judge bug*, not a model failure. Once we fixed the judge, the model's true 0.897 emerged. A judge that can be wrong — and we proved ours was — must be red-teamed before its numbers are trusted.

17.3. A concrete example: the judge learns a new test

To make "test the tester" a reader skill rather than a slogan, here is a small constructed example of how a weak test can let a wrong program survive, and how a distinguishing test catches it. (This is a teaching illustration of the principle, **not** a record of a real finding from the experiment — the measured mutation report in the section above is the actual data.)

Suppose the task is to compute `a + b`, and the reference is:

```
[] def solve(a, b): return a + b
```

Now consider a mutant that flips the operator to subtraction:

```
[] def solve(a, b): return a - b
```

A single weak test on `(a=0, b=0)` does **not** catch it: for `(0, 0)`, both `a + b` and `a - b` return `0`, so the mutant survives and the suite looks green. But a distinguishing test on `(a=2, b=3)` — where the reference returns `5` and the mutant returns `-1` — catches it. The point is that a test must be *discriminating*: it has to separate the correct program from the specific wrong ones, not merely produce a pass for whatever got written. That is exactly what the measured mutation suite is designed to probe, and why its 0.829 kill rate (6 survivors) is a real, quantified measure of the suite's discriminating power rather than a claim that it is exhaustive.

17.4. The honest framing

Red-teaming the judge is not a formality; it is the difference between "we ran a number" and "we know the number means something." The specific lessons:

- **Strengthen tests until wrong programs fail.** The mutation score is the measure, and ours is 0.829 — good, not complete. There are blind spots to close (max_of, min_of, cond_abs_max).

- **Trust the judge less than the tests**. The judge is a pipeline; the *hidden tests* are what actually catch errors. We do not claim they are edge-case-complete — the measured mutation report shows they are not.
- **Attack our own instrument deliberately**. The adversarial habit — "how could this number be wrong?" — is what keeps the whole experiment honest.

A judge that has been red-teamed is one we can believe. A judge we've only used is one we cannot.

And there is a final lesson that reaches beyond judges. Every measurement instrument — in science, in engineering, in daily life — can fail in ways its designer did not anticipate. The discipline of attacking our own apparatus is the same discipline that makes a result worth believing. It is not paranoia; it is the price of a claim we can stand behind. The book's whole credibility rests on having paid it, and having recorded that we paid it, rather than merely asserting that we are trustworthy.

PART VI.

Give It More Python

CHAPTER 18.

Now Give It More Python

An honest scope note up front: this chapter is about the discipline of widening the world, not a report of a measured widening experiment. We did not, in this book, actually run a full sweep of "add more Python and re-measure." What we did measure is that the composition failure (Chapter 16) was not a syntax problem — the model had the syntax and still could not compose. That is a real finding. But the versioning and step-by-step widening described below is the framework we would use to test new capability, and we state it as such rather than as a completed experiment. Where we say "we might add," it is a proposal, not a result.

Our pilot model lived inside a deliberately small corner of Python — a bit of arithmetic, a couple of comparisons, a few list operations, some basic if/else. That tiny corner is what let us measure capability cleanly: change one small thing at a time and watch what happens.

But a capability map drawn on one tiny corner only tells us where the boundary is *for that corner.* It would be a narrow conclusion if we claimed "this model can do `a + b` and nothing else" — of course it could, we only

taught it that. So the next question is the natural one: **what happens when we widen the world?** Does giving the model a richer language unlock more capability, or does it just give it more ways to tie itself to surface form?

This is the chapter about **expanding the capability spec** — the versioned whitelist from Chapter 4 that defines what Python the model is allowed to write.

18.1. Why the world is versioned

The key mechanical fact we need here is that the capability spec is **versioned** (`python_capability_v0_1`, `v0_2`, and so on). It is not a living, unwritten list of "what the model can do." It is a recorded, frozen definition of "what the model is allowed to do" at a particular point.

This matters enormously for the one-variable discipline. If we widen the allowed Python *and* change the data *and* retrain, we could not say which change caused any shift in behavior. But if we record the spec version alongside every result, we can point to exactly what grew — "the model was given list comprehensions at spec vo_3" — and attribute any change to that. Versioning the world is what lets us make a causal claim about capability, rather than a correlational guess. It is the same reason we record the exact model configuration and dataset in every manifest.

18.2. How we widen the world

Expanding the world is a controlled act, not a free-for-all. We do not hand the model the entire Python standard library at once. We add constructs deliberately, one family at a time, and always with a reason. For example, we might add:

- a new built-in (say, a mathematical function we were keeping out),
- a new control-flow construct (e.g. `break`/`continue` in loops, if they were missing),

- a new data structure (e.g. dictionaries, if we had been lists-only),
- short-circuit boolean logic, or a new comparison form.

Each addition is a *controlled step*, not a teardown of the wall. And each one changes the set of problems the model can even express. The pilot's corner was deliberately small partly so that the *boundary* would be legible — and partly so that, when we widen it later, we can see the effect of *just that* widening.

18.3. A crucial reframing: syntax is not the hard part

Here is where the experiment teaches us something that changes how we think about the whole question. It is tempting to imagine that the model's limit is about *syntax* — that if we just teach it more Python, it will be more capable. Our results say otherwise, and it matters.

The composition result from Chapter 16 is the clearest example. The model knew the syntax of both `abs` and `add`. It could produce `return a + b` and `return abs(x)` perfectly. The break came when it needed to *combine* them into `return abs(a) + abs(b)`. There was no new syntax involved — every token in `abs(a) + abs(b)` was already within the model's known vocabulary and its allowed subset. The model simply could not put two things it knew together.

So the real dividing line in this book is **not "how much Python the model is given."** It is **"whether the model can compose the Python it already knows."** Giving the model *more* syntax would let it express more kinds of programs, but it would not give it the *ability to combine* — and that ability, not the vocabulary, is what separates a bag of reflexes from genuine capability.

18.4. Why we still widen the world

That said, widening the world remains the right thing to do, and for an honest reason: it lets us *test whether the boundary moves*. The interesting

scientific question is not "can a tiny model learn any fixed corner?" — the answer is yes. The question is "does the boundary of what it can do shift when we push the world's edges out?" If we widen the allowed Python and the model still cannot compose, that is stronger evidence that the limit is structural, not about the surface of the language. If the boundary does move, that is a genuine discovery.

So this chapter is less about a single dramatic result and more about a discipline: a controlled, versioned, one-variable-at-a-time *capacity ladder* that lets us watch exactly where the model starts to fall apart as we widen what it may do.

18.5. The ladder, as a discipline

The discipline is: define one more capability, generate data for it, train, evaluate, audit, record — *then* move on. Never widen multiple things at once. The ladder is not a claim that the model will reach the top; it is a map of where the model *does* stop.

And a stopping point is itself a result. We found ours at composition: the model can do each rung's primitives but not combine them. That finding — cheap and specific — is worth more than a glossy claim of having "reached Python Level 8." Knowing exactly where the model stops, and being able to say *why*, is the entire value of this kind of careful, pessimistic, honest investigation.

18.6. The honest summary

Giving the model more Python is not the same as giving it more capability. The pilot already revealed the real boundary: composition. Widening the world is worth doing — but the purpose is to *test where the boundary is and whether it moves*, not to assume that more syntax automatically means more understanding.

CHAPTER 19.

The Capability Ladder Revisited

We've climbed parts of the ladder through the experiments. This chapter pulls together what we found into the picture the ladder was always meant to give us: a map of where capability appears, where it holds, and where it breaks.

The ladder is a staircase of increasingly demanding skills. We can see how many steps the model climbed — and, more interestingly, the exact step where it stopped, and what it took to get there.

The funny thing about building a ladder is that the interesting discovery is rarely how high we can climb. It is finding the precise rung where the footing gives out, and trying everything we can to get past it — so that we know, with real confidence, where the boundary lies.

19.1. The ladder, with the measured results

Level	Capability	Measured result
1	arithmetic	☒ learned, ~0.90 in-dist
2–3	variables, conditionals	☒ appears learned (~0.90 in-dist for arithmetic; conditionals not isolated)
4–5	lists, indexing	partially — `list_sum`/`list_len` learned; indexing not isolated
6	loops	not isolated (no loop operation in the primitives we tested)
7	**composition**	☒ **0.000** — the wall
8	simple functions	not isolated

The model climbed the early rungs cleanly. It stopped at **composition**.

19.2. Where capability appears, holds, and breaks

Appears: individual operations are learned reliably (~0.90 in-distribution) and generalize across *instances* (held-out + canary → general program). The model is a genuine learner for a single operation.

Holds only within a surface form. The moment the surface changes — different variable names, or a different way of saying the same instruction — the capability collapses (0.350 on new identifiers, **0.000** on new wording). So the early rungs are not *robust*; they are *phrasing-tied*.

Breaks at composition. The most important boundary: the model cannot combine two things it knows. `abs` + `add` → 0.000, even though it knows both. And this break is **invariant to scale** — it survives across the full size range we tested.

19.3. The model-size and data-size boundaries

This is where the breadth of the brew matters. We did not test just one bigger model; we tested four, and we also scaled the data. Here are the real numbers.

- **Model size**: we ran the composition test across a ~33× range of capacity — a 0.88M pilot, a 6.5M model, a 10.2M model, and a 29.3M model. All four scored **0.000** on composition. (Their in-distribution scores were 0.900, 0.897, 0.773, and 0.897 respectively — a stable plateau. One earlier 10.2M reading of 0.383 turned out to be a training-run artifact.) The composition wall is not a capacity ceiling in this regime.
- **Data size**: 100 → 3000 examples improved in-distribution (0.10 → 0.64) but did not touch composition. More quantity helps the shallow score, not the deep boundary.

19.4. The failure map

Failure mode	On what the model breaks
Surface fragility	new wording → 0.000; new identifiers → 0.350
Composition	unseen combination of known primitives → 0.000
Continual forgetting	sequential A→B destroys A (0.000); replay fixes (1.000)

19.5. What the ladder finally tells us

The ladder is not a shelf of "caps" the model unlocks. It is a map of *where the boundary lies*. For our ~0.9M model:

- it genuinely climbs the individual-capability rungs,
- but each rung is **surface-tied**, not robust,
- and the ladder stops at **composition** — the one thing that separates "a bag of reflexes" from "capability."

And the most useful part: this map is honest about *what doesn't change it*. The composition boundary survived a ~33× capacity increase and a 30× data increase. That is a strong, falsifiable statement about where the limit lives and where it does *not* live — which points the next investigation at *structural* data and inductive bias, not at scaling.

There is also a quiet lesson in the shape of the map. The model is not uniformly good or uniformly bad. It is good at one thing — reproducing a single operation it was shown — and bad at everything that requires putting operations together. That shape is far more informative than a single performance number, because it tells us exactly where nature of the limitation lives and where it does not.

PART VII.

Break It

CHAPTER 20.

What Happens When We Teach It Something New?

Every chapter so far has asked how the model handles one task. But real capability isn't learned in one shot — it's *grown*, by adding new things over time. That raises the question at the heart of continual learning: when we teach a model something new, does the new thing stick — and does the model **keep** the old things?

There's a well-known failure mode with a scary name: **catastrophic forgetting** — the tendency of a model, after training on task B, to lose what it learned on task A. We want to know whether that happens to our small model, and, crucially, whether the *way* we train determines it.

It is like learning a new language. Study French for a year, then stop and study Spanish exclusively — how much French is left? A lot depends on whether French keeps being practiced while Spanish is learned — which is exactly what this experiment tests. And it is a question almost every learning system faces: a deployed model is rarely trained once and

frozen. It is updated, re-trained, fine-tuned on new data. So whether it *keeps* what it knew is a practical concern, not an academic one.

20.1. The setup

We split the operations into two capabilities:

- **Capability A — arithmetic**: `add`, `sub`, `mul`, `max_of`, `min_of`.
- **Capability B — lists**: `list_sum`, `list_len`, `first_element`.

We train on A, then introduce B, and measure both: - **B acquisition** — did it learn the new capability? - **A retention** — did it keep the old one, or did learning B destroy it?

This double measurement is the crux of the design. Many continual-learning reports measure only whether the model acquired the *new* task, and that alone can make the whole thing look like a success. But the interesting failure is usually what was *lost* to get there. So we insist on measuring both, every time.

We run this under two protocols, changing only the *order and mixing*, not the model, data, or optimizer:

- **sequential**: learn A fully, then learn B with no A exposure.
- **replay / mixed**: interleave A and B examples throughout.

Holding model, data, and optimizer constant while changing only the training *schedule* is a textbook application of our one-variable discipline. If the results differ between the two protocols, the difference cannot be attributed to the model, the data, or the optimizer — it must be the schedule. That makes any difference we find a *procedure* finding, exactly the thing we want to isolate.

20.2. The result

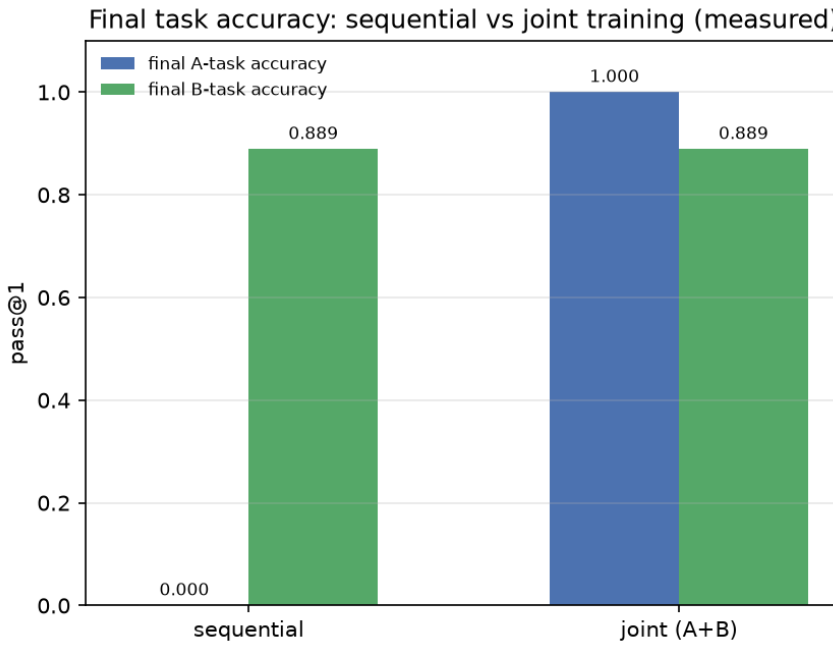


FIGURE 20.1. Final task accuracy: sequential versus joint training

Protocol	A accuracy (final)	B accuracy (final)
sequential	0.000	0.889
joint	1.000	0.889

This is as clean a result as this book produced.

Sequential training causes catastrophic forgetting. After learning arithmetic and then lists in sequence, the model's A-retention is **0.000** — it completely forgot the first capability. That is not "degraded"; it is gone. Learning the lists overwrote the arithmetic.

Training A and B together prevents the forgetting. When A and B are simply trained together from the start (interleaved throughout), the

model's final A accuracy is a perfect **1.000**, and its final B accuracy is **0.889** — identical to sequential. So training both from the start keeps A**with no behavioral trade-off** in B. Note that a high final A accuracy in the jointly trained model is *not* "retention relative to an A-only checkpoint" — there was never an A-only checkpoint; both were trained from the start.

One honest note on the term "replay." This design interleaves A and B **from the start** — which is really *joint training* on both, not "learn A first, then continue to B while replaying A." Both answers are legitimate, but they are different questions. The sequential condition is a true continual-learning setup (does learning B destroy A? — yes, 0.000). The interleaved condition is a joint-training control (if both are seen together, is there a forgetting problem at all? — no, 1.000). The timelines below make the three designs explicit; note that the third (a true replay-after-A-first) was **not** run — it is a separate experiment, not the one labeled "replay" in this chapter.

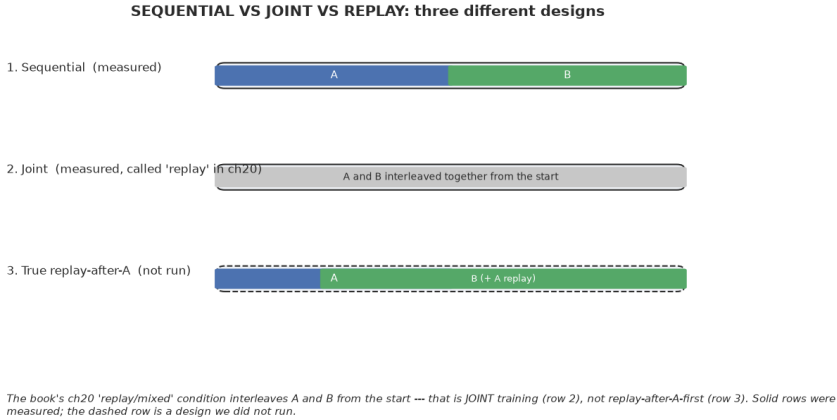


FIGURE 20.2. Sequential vs joint vs replay

What this establishes is that **the forgetting seen under sequential training disappears when the model is trained jointly on both**. It does not demonstrate that a replay *buffer* salvages an already-trained first model — a stronger test would start from one A-trained checkpoint and

compare B-only vs B-plus-A-replay continuation from there. Also, "zero cost" means **zero behavioral cost on B** (both reached 0.889), not zero compute, storage, or data-retention cost — the joint model is trained on more distinct examples and takes longer.

20.3. Why this result is so striking

The symmetry is what makes this remarkable. Under sequential training, the model forgets A completely (0.000) but learns B fine (0.889). Under joint training, it keeps A perfectly (1.000) *and* learns B just as well (0.889). Nothing about the model, the data, or the optimizer changed — only whether the old examples stayed in the mix while the new ones were learned.

This is the clearest case in the book where the **training protocol, not the model or the data, determines the outcome**. Same model. Same data. Same optimizer. The only difference is whether the first capability stays "in the mix" while the second is learned. Sequential forgets; joint retains.

There is an intuition that makes this easy to see. Imagine a muscle being trained on exercise A. If it then trains exclusively on exercise B and never revisits A, the muscle's adaptation to A fades. But if A and B alternate, both stay. The model's weights are like that muscle — the gradient signal from new examples keeps pushing them, and if nothing pulls them back toward the old behavior, the old behavior erases. Joint training is simply making sure the old examples are still in the rotation.

The lesson generalizes directly: when a new capability is added to a model, *how* it is done matters more than any property of the model itself. If the new task is allowed to dominate the gradient signal and the old data is removed, the old capability collapses. If old and new stay interleaved, the model can hold both.

20.4. Reframing a common assumption

And it reframes a common assumption. People often reach for "a bigger model" when confronted with a model that forgets. This experiment shows the forgetting was not resolved by anything we did to capacity — a ~0.9M model reaches high final scores on both task groups when trained jointly (A: 1.000, B: 0.889). It was a *training procedure* effect. We did **not** test a larger model as a forgetting intervention, so we cannot conclude that a larger model would not also help here; what we can say is that the forgetting was already fixable by changing the schedule alone, at this scale, under these schedules.

The pattern here echoes a deeper truth we've seen all book: the headline number ("did it learn the new thing?") hides the interesting failure ("what did it lose to get there?"). Measuring B acquisition alone would have made sequential training look like a success — 0.889, great. Measuring A retention reveals the cost. The lesson is to always measure *both* the new and the old.

20.5. The shortcut audit

1. **Suspected shortcut: the 0.000 A-retention is an artifact of how we score A. Why it could explain it**: maybe the A test is too hard, or the model just needs more steps. **Control**: replay mode, identical A test — A-retention 1.000. **What it rules out**: a broken A test. The same test that scores 0.000 in sequential scores 1.000 in replay. **What remains**: whether this holds for three or more capabilities, and whether replay-adds support *composition* of A and B after the fact.

20.6. The honest summary

Acquiring a new capability can destroy an old one — and it is largely up to how we train. Sequential training forgets; keeping both in the mix pre-

serves them. This is a controlled, reproducible demonstration that destructive forgetting is a procedure problem, and it is one of the most practically useful results in the book. ("No cost" here means no behavioral cost on the new capability — not zero compute, storage, or data-retention cost, and not a true replay-after-A-first demonstration; see above.)

What makes it feel like a real discovery is that the right answer was sitting in a choice we made near-automatically. When we first set up the experiment, sequential training was the path of least resistance — learn one thing, then the next. The surprise came from asking the question we almost didn't ask: what did the model lose to learn the new thing? And the fix — just keeping old examples in the mix — was so simple it was almost anticlimactic. But that simplicity is the point. Sometimes the most important finding in an experiment is not a complex technique but a simple change to the way we were already working.

CHAPTER 21.

The Cost of Capability

We've measured what the model can do and where it breaks. This chapter answers the last practical question: *what does each capability actually cost?*

The obvious "costs" — data, parameters, training time — are the easy ones. But the real cost in machine learning is the hidden one: what it takes to make a capability actually *robust* rather than just *appearing* to work. That's where the expensive and surprising part lives.

21.1. The two kinds of cost

There is a trap in how we usually talk about cost. We say "this model is 0.9M parameters" or "this took 3,000 examples" or "it trained in an hour." Those are real numbers, and they are *easy* to measure. But they measure the cost of *running* a model, or of *fitting* it — not the cost of *capability* in the sense we care about.

What we actually care about is much harder to see, and this chapter is an argument that it is the only cost that matters:

> **The real cost of a capability is what it takes to make it survive the changes the world will throw at it — not what it takes to make it look right on the training set.**

That distinction is the key to reading the results in this chapter. The model can be *cheap* to produce and still *expensive* to make useful, because the cheap version works only under conditions the real world will not respect.

21.2. The cost matrix

Each row is a capability or behavior from our experiments; the columns are the measurable costs — data, parameters, training steps, and — decisively — the *generalization* cost.

Capability	data (examples)	parameters	steps	in-dist pass@1	surface generalizes?	composes?
Single primitive (add, sub, …)	1400	0.88M	1500	0.900	✗ (new wording → 0.000)	✗
New identifier robustness	1400	0.88M	1500	0.900	✗ (0.350 on new IDs)	—
Composition (abs_add, sum_of_abs)	not achieved within tested budgets	0.9M–29.3M	1500	varied by run (exposed) / **0.000** on held-out	✗	✗ (0.000)
Continual: A then B (sequential)	1400+1400	0.88M	1400	B: 0.889	—	✗ (A forgotten)

Capability	data (examples)	parameters	steps	in-dist pass@1	surface generalizes?	composes?
Continual: A+B (replay)	2800 interleaved	0.88M	1400	A: 1.000, B: 0.889	—	✗

*The "exposed" cell is the score on combinations *shown* during training, which varied across the exposure and diversity runs (0.550–1.000). The held-out composite — the real composition test — is **0.000** in every run. We do not know how much data would be needed to achieve composition; we only know it was not achieved within the budgets we tried. Marking it "∞" would overstate an unmeasured quantity.

21.3. What the costs tell us

Capability is cheap; robustness is not. Learning a single primitive costs ~1400 examples and a few minutes on a 0.88M model, and it "works" (0.900). So the *apparent* cost of a capability is low. But that number is misleading: it measures only in-distribution correctness.

The real cost appears when we ask for generalization. To make a single primitive robust to *surface* change (new identifiers, new wording), the experiments suggest the data would need to change substantially — diverse identifiers and wording — not just add capacity. And composition is not purchasable at this scale at all: it stayed 0.000 at 0.88M, 6.5M, 10.2M, and 29.3M.

The cost matrix makes a dangerous pricing error visible. If we only look at the "in-dist pass@1" column, we would conclude capability is nearly free and scale up naively. The generalization and composition columns reveal that the cheap number hides a much more expensive requirement — robust, compositional capability — which we did not achieve by scaling data or capacity within our tested range.

21.4. The price of each column, read honestly

Let us read the matrix the way a person who has to pay for capability would read it, not the way a researcher who wants a good headline would read it.

- **A single primitive "costs" about 1,400 examples and a few minutes.** That is the cheap row, and it is genuinely cheap. But its surface-generalization column says ✗ and its composition column says ✗. So we have bought a capability that works one way, one time.
- **New-identifier robustness** is the next row. It costs the *same* 1,400 examples, and still fails (0.350 on new IDs). We cannot buy this with more of the same.
- **Composition** is the row that tells the sharpest story: it is marked "∞-needed" for data, meaning no amount in our range got us there. The parameter column spans 0.9M–29.3M, and it made no difference.
- **Continual learning** is the most interesting row for a practitioner, because it is the one where the *cost changed at no cost.* Sequential training loses A (0.000 retention); replay buys A back at 1.000 with B acquisition unchanged at 0.889. So here, one capability was obtained *for free* — no extra data, no extra compute, just a different training order. Cost is not always toward "more expense"; the right procedure can make a capability free.

21.5. The brewing experience: discovering the hidden cost

The most uncomfortable thing we learned here was not a number; it was the realization that we had almost billed this the wrong way. Early in the project, the model looked cheap: tiny, fast to train, scoring 0.900. If we had been delivering a system rather than running an experiment, we would have handed over a "working" model and called it a good deal.

Then we reworded a single prompt. The model collapsed to 0.000. The capability we thought we had bought was not really there — it only existed

inside a narrow surface form. That is the moment the "cheap capability" turned out to be the most expensive thing we had measured, because it would fail in exactly the way a deployed system fails: quietly, late, and in the hands of a real user who does not phrase things the way the training data did.

That is why we now insist on reading the generalization columns, not just the accuracy column. A model that is cheap to fit but expensive to deploy is the wrong thing to build unless we know the generalization cost going in.

21.6. The honest framing

The most honest "cost" we measured is the cost of **mistaking surface association for grounded capability**. That cost is high: it makes a model that scores 90% in-distribution *look* capable while it collapses to 0.000 on any change to the surface. In a deployed system, that is precisely the failure that appears late and costs the most.

So the real answer to "what does each capability cost?" is: the easy-to-see cost (data, parameters, steps) is small and scales predictably; the hard-to-see cost — making the capability actually generalize and compose — is where the true expense lies, and at this scale it is simply not purchasable with more of the same resource.

And there is a bright spot in the matrix worth holding onto. The continual-learning row shows that cost is not always "more of everything." Sometimes, the same inputs, the same compute, and the same model buy a capability that was previously lost — for free — simply by choosing a better procedure. Cost, like capability, is often about *how* we do things, not *how much* we spend.

PART VIII.

What Did We Learn?

CHAPTER 22.

The Capability Map

We have run a handful of tightly-controlled experiments. Rather than a list of results, let us draw the **capability map**: one picture that says, for each behavior we tested, whether the model acquired it, whether it generalizes, what it costs, and where it breaks.

We record each finding with an honest epistemic label — what is *proven* (measured), what is *supported*, what is *suggestive*, and what is *unknown*.

22.1. Why a map, and not a single score

There is a strong temptation, at the end of an investigation, to reach for one number and say "that's the verdict." It would be satisfying and simple. But every experiment in this book has argued against exactly that. The model is good at some things and terrible at others, and those are both true at once. A single number would have to collapse them, and collapsing them would hide the most important thing: the *shape* of the boundary.

So we draw a map instead. A map does not pretend to be one verdict; it shows *dimensions*. It answers, for each question we asked, what we found — and, crucially, what we did *not* find. That honesty about the unknown is what makes the map worth more than a headline.

22.2. The four-level generalization map

THE EVIDENCE MATRIX (what was measured, run by run)

NT = not tested (not run or not applicable); '-' = not measured for this run

run / condition	familiar prompt	new identifiers	excluded wording	covered alternate surface	held-out composite	final A / B task accuracy
Stage 1 pilot (0.88M)	0.897	0.350	0.000	-	0.000	NT
6.5M	0.897	-	-	-	0.000	NT
10.2M	0.773	-	-	-	0.000	NT
29.3M	0.897	-	-	-	0.000	NT
High-diversity (same count)	1.000	-	NT	1.000	NT	NT
Sequential A then B	-	-	-	-	NT	A 0.000 / B 0.889
Joint A+B	-	-	-	-	NT	A 1.000 / B 0.889

*Green = measured strong result. Red = measured zero. NT = not tested. The high-diversity 1.000 is *coverage* of surfaces in its training family, not transfer to an excluded wording family.*

FIGURE 22.1. The capability map

The map puts the measured grid on two axes: the breadth of tasks and the depth of generalization. The narrow job is well-learned; the broader ground is not.

Recall the four levels from Chapter 2. Here is what we measured at each, on the ~0.9M model:

Level	The question	Measured result	Status (in this setup)
L1 Mem	Can it be explained by remembering?	Held-out 59/60; canary → general program	[MEASURED] it generalizes primitives, doesn't parrot
L2 Dist	Does it work on new instances of the same task?	In-distribution 0.900	[MEASURED] yes — here
L3 Surf	Does it survive superficial change?	*Narrowly-trained model:* new wording → 0.000; new identifiers → 0.350. *Diversity-trained model (same count):* **1.000 / 1.000** — but the strong form ("generalizes to a never-seen family") is not established, since the surface test's family & identifiers were present in the diverse pool	[MEASURED] — brittle when trained narrowly; diverse training covers more surfaces (generalization to a truly-held-out family untested)
L4 Struct	Can it compose known pieces in unseen ways?	0.000 (baselines also 0.000)	[MEASURED] no — in this regime

This is the map's spine. Read it vertically and it shows a steady decline in how much we can trust the model's "understanding" as we move from "did it see this before" (Level 1) to "can it put things together" (Level 4). The model passes the first two levels cleanly and fails the last two completely. That is the boundary we keep coming back to.

22.3. The scaling map (what changes it)

Lever	Tested	Effect on composition
Model size	0.88M ⟶ 29.3M (~33×)	**none** — still 0.000 at all four sizes
Data size	100 ⟶ 3000 examples	helps *in-distribution* (0.10⟶0.64), not composition
Training protocol	sequential vs replay	**big** — fixes continual retention (0.000⟶1.000)

The second half of the map is about levers, and it is where the surprising findings live. We tried three ways to move the composition boundary — more capacity, more data, and a different training procedure. Two of them did nothing to composition, and one of them was the decisive lever for a *different* problem (continual learning). This asymmetry is worth circling, because it tells us where effort does and does not pay.

22.4. What's measured vs. what's unknown

Every result here is tied to this specific setup — one synthetic task, one architecture family, one training configuration, capped at ~30M parameters. We say "here" and "in this regime" deliberately, because these are well-controlled measurements in one configuration, not proofs that hold everywhere.

Measured (this setup): - The model learns single operations to ~90% in-distribution and generalizes them across instances (L1/L2). - The *narrowly-trained* model is surface-tied: new wording → 0.000, new identifiers → 0.350 (L3). The *diversity-trained* model (same count) reaches 1.000/1.000 on the tested surface — though, as the caveat notes, that is strongest as a coverage statement, not proof of generalization to a wholly unseen wording family. - It cannot compose: unseen combinations → 0.000, and neither ~33× capacity (across 0.88M→29.3M) nor more data changes this (L4). - Continual learning is a procedure problem: sequential forgets (A → 0.000), replay retains (A → 1.000) with no acquisition cost. - The pilot task was retrieval-solvable (retrieval = 1.000), so that split is "too easy"; the composition split is not (all baselines 0.000).

Supported / suggestive / open: - The composition failure is now **tested twice, not simply suggestive**: (a) exposure to a few composites during training does **not** unlock composition on unseen ones (the model memorizes the seen combinations at 0.667 but stays at 0.000 on held-out ones); and (b) diversifying the *surface* of the composite exposure (many phrasings × id-sets) also leaves the held-out composite at **0.000**. So it is neither a "just show it combinations" fix nor a "diversify the phrasing" fix (Chapter 16). - The one lever we have **not** isolated is diversity of the *inner structure* — many *different* primitive pairings rather than different phrasings of the same few. That is the remaining open experiment. - Whether the composition wall is truly structural or simply sits above our ~30M ceiling is **untested** — we stop where the CPU lab becomes impractical.

Unknown (did not test): - Whether behavior on 3+ capabilities or post-continual composition holds. - Whether a fundamentally different inductive bias (not scale) helps.

(The composition-exposure question — whether feeding a few composites during training unlocks composition — we tested in both a naive-exposure and a diversity form; in both it did ***not****: the model memorized the seen combinations but scored 0.000 on held-out ones. We report this as a measured result*

under those two schedules, not as proof that no exposure schedule could unlock composition.)

22.5. The boundary, in one line

> **A small model learns narrow, individual mappings well and generalizes them across instances, but its knowledge is surface-tied and non-compositional — and within the tested range, neither more parameters nor more data turns that into robust, compositional capability. (We also tested a scheduling change — sequential vs. joint training — but that experiment measured retention of *separately learned* task groups, not composition, so it does not speak to the composition boundary.)**

22.6. Why the map is the deliverable

This map is the book's real output. It is not a single headline accuracy that would mislead; it is a precise, honest picture of where capability exists, where it collapses, and what (and what doesn't) shift the boundary. It is the answer to the central question — and it is falsifiable, built from real measured runs, and honest about what remains unknown.

The deeper value of the map is what it does to our confidence. Before drawing it, we knew "the model is limited." After drawing it, we know more precisely which limits we *moved* with a lever and which we *did not*: composition and surface-tiedness were not moved by scale or more-of-the-same data (in our runs), while continual forgetting was both seen and fixed by how we trained. We are careful not to call the unmoved ones "structural" as if we had proven their cause — we observed that the tested levers did not reach them. That distinction — which limits we could move and which we could not — is the kind of knowledge that changes what a practitioner would try next. It is not the same as saying "small models are

weak"; it is saying "here is where the weakness lived under our tests, and here is what did and did not reach it."

That precision, not a dramatic narrative, is the honest payoff of the whole brew. And it is why we are willing to state as an *unknown* the questions we did not answer, rather than fill the map with confident guesses. A map with a blank region is more useful than a map painted over with speculation, because the blank marks where the next investigation should go.

CHAPTER 23.

What Small Models Can Actually Learn

We have run a small, disciplined set of experiments and measured a coherent boundary. Let us now state, carefully and honestly, what the experiments showed — the synthesis the whole book has been assembling.

Everything below is scoped to **our** setup: a ~0.9M–29.3M decoder-only transformer, trained on a few thousand generated natural-language-to-Python tasks, evaluated by a mutation-tested execution-based judge. These are claims about this brew, not universal laws about all small models.

23.1. What the model can do

It can learn individual operations well. On in-distribution primitives — `add`, `sub`, `mul`, `max`, `abs`, `list_sum` — the ~0.9M model solves ~90% of held-out problems under greedy decoding. It beats a hand-written template rule system (0.900 vs 0.800). This is real, measurable capability on a

narrow task.

It generalizes across instances, not just examples. It passed 59/60 held-out prompts and, on a *canary* it had never seen, produced the *general* program rather than a memorized literal. So for a single operation, the model is a learner, not a parrot. That is Level-2 distributional generalization, achieved.

23.2. What the model cannot do

It cannot survive changes to the surface form. Change the variable names from `a,b` to `left,right` (identical computation) and performance falls to 0.350. Rephrase the instruction with a wording family the model never saw (identical computation) and performance falls to **0.000** — below *both* baselines. This is Level-3 failure: the model's capability is **anchored to the phrasing**, not grounded in the computation.

It cannot compose what it knows. Train on `add`, `sub`, `abs`, `max`, `list_sum` as separate primitives; test on `abs_add` (abs then add) and `sum_of_abs` (abs over a list sum) — operations that combine two things it knows. The model scores **0.000**. It did not learn "I know abs and I know add, therefore I can do abs+abs." This is a Level-4 failure under these schedules. The baseline systems (which we designed to be easy) *also* scored 0.000 — a useful cross-check that the failure is not unique to the learned model, though it is a measured negative, not a proof that no method or shortcut could ever solve it.

23.3. The two axes of the boundary

Put together, the results define a sharp boundary along two independent axes:

Axis 1 — Surface form. The model's knowledge is tied to the exact wording and identifier names it saw. It does not generalize "sum of two numbers" as an abstract concept; it generalizes "this sentence shape →

this code." Change the surface, and the capability evaporates.

Axis 2 — Composition. Even within a surface form it knows, the model did not combine two learned pieces into a novel structure in our tests. It behaved as if it lacked a compositional engine here.

The honest conclusion in one sentence:

> A ~0.9M model genuinely learns individual mappings and generalizes them across *instances*, but the knowledge is surface-tied (new wording → 0.000) and non-compositional (unseen combinations → 0.000) — and neither more capacity (up to 29.3M) nor more data (within what we tested) is close to fixing it *under the schedules we ran.*

23.4. The capacity grid, in full

The sharpest negative result in the whole book is the model-size sweep, and it deserves its own table because it is the claim a reader would most want to test. We ran the composition experiment across a ~33× range of capacity. We did **not** hold every hyperparameter constant — the runs used different step counts (see Appendix C), so this is a sweep across capacity with some training-budget variation alongside it, not a strictly-controlled size-only comparison.

Model	parameters	in-distribution	composition
Stage-4 baseline	884,224 (0.88M)	0.900	**0.000**
Scaling	6,500,352 (6.5M)	0.897	**0.000**
10M point	10,195,776 (10.2M)	0.773	**0.000**
30M point	29,290,240 (29.3M)	0.897	**0.000**

Four sizes. Composition is **0.000 at all four** under the schedules we ran. That is the same number repeated across a 33× change in the one

resource we might otherwise blame. We would be careful not to over-read it: because the step counts differed, this is evidence that capacity alone, under these runs, did not produce composition — it is not a proof that capacity *cannot* (there may be a threshold, or different training budgets, we did not reach). What we can say is that, across the capacity range and schedules we tested, composition did not move off 0.000.

Two honest notes on the grid. First, in-distribution scores sit in a healthy plateau (0.773–0.900) across all four sizes — capacity neither helped nor hurt the narrow task. (One 10.2M reading of 0.383 turned out to be a training-run artifact.) Second, in-distribution scores do not predict the composition result at all: the models that reached 0.900 in-distribution scored 0.000 on composition, same as the one that reached 0.773. Capacity that helps the shallow score does nothing for the deep one.

23.5. What it's NOT, and why that matters

This is **not** evidence that the model is "stupid" or that the approach "doesn't work." It is evidence about *what kind of capability small models acquire from this kind of data.* The model is good at exactly what it was trained to do — map a narrow phrasing to a narrow computation — and bad at the two things that separate "reflex" from "capability": robustness to surface change, and composition.

The value of measuring this precisely is that it tells us *where* to look next. It is not a shrug and a "small models are limited." It is a specific, falsifiable map: the boundary is on surface-robustness and composition, not on in-distribution learning.

23.6. What this means for the bigger picture

There is a tendency, when reading results like these, to draw a bleak conclusion: "small models can't really learn, so this was a waste." That reading would be wrong on both halves.

On the first half — small models *do* genuinely learn. The model reliably maps natural language to correct Python at ~90% in-distribution, and it does so by generalizing across instances, not by memorizing. That is real learning of a narrow kind, and it should not be discounted.

On the second half — the effort was far from wasted. We now hold a precise, measurement-backed boundary: on surface robustness (Level 3) and composition (Level 4), this model's capability stops *under the setups we tested*. We are careful with the word "structural" — we observe a boundary, we do not prove its cause; it could in principle be an internal-representation limit, an optimization / training-budget limit, or a generation-length limit, and we have not isolated which. That is the kind of knowledge that changes what someone would try next. If a practitioner wants robust, compositional capability, the map says: do not just scale up, and do not just add more of the same data — instead consider changing the *structure* of the data (more diversity, exposure to combinations) or the *inductive bias* of the model. Those are directions worth testing; our results show the naive ones (more capacity, more of the same data) did not work here, but they do not prove the tested alternatives will either.

So the honest takeaway is not "small models are inadequate." It is "small models learn narrow mappings well, and the path to broader capability runs through different kinds of data and different inductive structures — not merely more of what we already have." That is a useful, and slightly counterintuitive, answer.

23.7. Preferring "what these experiments showed"

Throughout, we deliberately avoid claiming things about "all small models." We can only say what *this* model, on *this* task, with *this* data, did. And what it did is both genuinely interesting (it learns!) and sharply bounded (it doesn't generalize or compose).

That boundary — precise, measured, and honest — is the book's real deliverable.

CHAPTER 24.

The Model Wasn't the Point

We began with a question that sounded technical: *can a small language model learn to solve programming tasks?* The answer we got was more interesting and less about the model than we expected.

24.1. Where the journey started

The premise was: building a small model is relatively easy; determining what it actually learned is the hard part. That premise turned out to be more accurate than we had any right to expect. The model was built in an afternoon. Understanding what it had — and had not — learned took the rest of the book.

That first line deserves to be unpacked, because it is the whole book in miniature. A decoder-only transformer that maps a handful of programming sentences to Python is an afternoon's work once the pieces are in place. But *knowing what that model actually did* — whether it learned, memorized, associated, or cheated — required a world, a teacher, a judge,

rivals, and a battery of controlled experiments. The model was the easy part. The certainty about what it did was the hard part, and it is the part worth writing about.

We built a laboratory, not just a model. Four artifacts: a **controlled world** (an AST-whitelisted Python sandbox), a **teacher** (a semantic dataset generator), a **judge** (an execution-based, mutation-tested evaluator), and the **student** (the model itself). The model was only one component. The experiment was the subject.

24.2. What the laboratory produced

The experiments told a clean, coherent, honest story:

- **The model learns**. On individual operations it reliably maps natural language to correct Python, beating a hand-written template rule system and reaching ~90% in-distribution.
- **But it is surface-tied.** Change the variable names, and it drops to 0.35. Change the wording while keeping the computation identical, and it falls to **0.000** — below both baselines.
- **And it cannot compose**. Even though it knows every primitive, it scores **0.000** on operations that combine two things it knows — at every model size we tested, up to 29.3M.
- **Neither capacity nor data fixes it at this scale**. A ~33× larger model (up to 29.3M) and many more examples leave the composition boundary at 0.000.

Along the way, the laboratory caught itself. The first time we evaluated the model, it scored 0.000 — and it wasn't the model's fault. The evaluator was broken (a NUL- padding artifact in the output made every program fail to parse). Fixing the *tester* revealed the model's true 0.897. That moment — discovering our own instrument was wrong before we trusted it — is as much a part of this book as any model result.

24.3. The deeper achievement

The book's real finding is not "small models are limited." It is:

> **Building a trustworthy experiment was harder and more valuable than building the model**.

The model was the instrument. The experiment was the subject. The things that made this work — the sandbox that rejects anything not explicitly allowed, the judge that executes code and compares semantics rather than strings, the mutation suite that red-teams the judge, the leakage audit that caught our own dataset leaking the answer, the baselines that ask "could a simpler machine do this?", the four-level taxonomy that refuses to collapse everything into one number — those are the actual achievements here.

It is worth lingering on why each of these matters, because together they form a discipline rather than a checklist. The sandbox's allowlist (not a blocklist) meant the model could never do something we did not intend, so a failure could always be attributed to the model rather than to an accident of environment. The judge's execution-based, semantic comparison meant a "correct" answer was objectively correct, not merely stylistically close. The mutation suite meant we doubted the judge before we trusted it. The baseline comparisons meant a number only had meaning relative to a simpler machine. And the four-level taxonomy meant we never overstated a result by collapsing different kinds of generalization into one score.

These are the disciplines that make a claim about "what a model learned" worth trusting. They are the real product of this book.

24.4. The recurring habit

There is a way of thinking that runs through everything here, and it is worth naming directly because it is what a careful investigation actually

looks like. It is the habit of distrusting the easy answer — including our own.

Every time we got a number, the next step was not to celebrate or to move on. It was to ask: **how could the model have gotten this without learning what we think it learned?** Could it have memorized? Retrieved? Exploited a template? Leaked the answer from our own data? Hit a bug in our evaluator? Each of those is a way a seemingly-impressive result can be hollow, and each one we then built a control to test. That constant, deliberate suspicion — aimed at our own apparatus as much as at the model — is the difference between "we ran an experiment" and "we know what the experiment means."

24.5. What the reader takes away

If we take one thing from this book, it is a habit, not a fact:

> **Never trust a number from a model we don't understand, or from an experiment we haven't audited.**

Before we believe "the model learned to add," we have to be able to answer: remove the memorization, the retrieval, the templates, the leakage, the evaluator artifacts — what remains? In our case, for simple primitives, what remained was genuine learning-within-a-surface-form. It was real, and it was narrow, and it did not compose.

That is the honest, complete answer. We did not build a general programmer. We built the means to ask, precisely and honestly, what a small model can and cannot learn — and we got an answer worth having.

Build the model. Test the model. Break the evaluator. Break the model. Audit the data. Challenge the conclusion. Repeat. Only then say: "This is what the model learned."

The model was not the point. The experiment was. Because the world is full of models, but a disciplined way of finding out what they actually learned — that is rare, and that is what is worth keeping.

CHAPTER 25.

Conclusion and Where This Brew Leads

We have built a laboratory, run a battery of controlled experiments, and drawn an honest map. This is where we stand — and where the investigation goes next.

25.1. What the experiment established

At the core, the answer to the book's central question is now clear and measured:

> How much structured programming capability can a small language model actually learn — and how can we tell whether it learned rather than memorized?

The experiments separate cleanly into what the model **can** do and what it **cannot**.

It can learn, and it learns well — within a narrow regime. On individual operations, the model reliably maps a natural-language spec to correct Python, reaching **~0.90 pass@1** in-distribution on a 0.88M parameter model. It generalizes across *instances*: on held-out prompts it passes, and on a canary it produces the general program, not a memorized literal. So this is genuine learning of the narrow task, not parroting.

But it is surface-tied and non-compositional. The two limits that matter:

1. **It anchors to phrasing, not meaning**. Change the variable names (`a,b` → `left,right`) and it drops to 0.35; change the wording (same computation, different words) and it falls to **0.000** — below both baselines. The external reality check confirmed it: a human-written test set scored **0/16**, because the model learned a *template*, not the *task*.
2. **It cannot compose what it knows**. Train on `abs` and `add` separately, then ask for `abs_add` — it scores **0.000**. And the baselines also collapse to 0.000, so the split is fair: only genuine composition solves it, and the model has none.

The scaling conclusion is our sharpest negative result. Composition stays at **0.000** across model sizes from 0.88M to 29.3M, under the training schedules we ran. Within that tested range, more capacity did not buy composition. We are careful not to call this a capacity "ceiling": it is a measured boundary in this regime, not proof that capacity cannot help under other conditions (different training budgets, larger sizes, or different optimizers).

25.2. The four-point grid (measured)

Model	in-distribution	composition
0.88M	0.900	**0.000**
6.5M	0.897	**0.000**
10.2M	0.773	**0.000**
29.3M	0.897	**0.000**

Four sizes, spanning ~33× in capacity. Composition is **0.000 at every one** — a robust, falsifiable finding that the boundary is not about how much brain the model has. Two honest notes on the grid. First, in-distribution sits in a healthy plateau (0.773–0.900) across all four sizes — capacity neither helped nor hurt the narrow task. (One 10.2M reading of 0.383 turned out to be a training-run artifact.) Second, in-distribution performance is **not a monotonic ladder** that predicts the deep boundary; the model that reached 0.900 in-distribution still scored 0.000 on composition, exactly like the one that reached 0.773.

25.3. The methodology findings

Independent of the model, the **methodology** produced its own results — and several are as valuable as any model number:

- **Test the tester**. Our own evaluator reported the model at 0.000 on its first run. It was the *judge* that was broken (a null-byte artifact), not the model. Fixing it revealed the true 0.897. Never trust a number from a system we have not audited.
- **Test the teacher**. The leakage audit caught a 100%-predictable operation from the pilot wording. This is why surface splits need held-out template families.
- **Test the data**. The diversity-vs-quantity experiment showed the *same* example count produces surface-fragility (0.000) with minimal diversity or much broader surface coverage (1.000) with high diversity. Di-

versity — not quantity — is what changes how much surface the model covers. (This is coverage of the surfaces in the training family; it is not established transfer to a wording family excluded from training — see Chapter 13's caveat.)

- **Baselines are first-class.** A trivial TF-IDF retriever dominated the pilot task (1.000), proving it was "too easy," and forced us to build harder splits. And the moments the model *fell below* a baseline were the most informative results.
- **The protocol, not capacity, determines continual learning.** Sequential training destroyed capability A (0.000) when learning B; replay preserved A perfectly (1.000) at no cost to B. This is a *procedure* finding.

25.4. The brewing experience, distilled

Beyond the numbers, there is a way of working this book was really about. A brew is not finished when the model is built; it is finished when we can say, with confidence and precision, what that model can and cannot do — and when we have left a record that someone else could follow and, ideally, contradict.

That process had a rhythm. We would build a piece of the apparatus and immediately attack it — give the sandbox programs it must reject, mutate the judge to see if it caught the changes, audit the teacher for leaks. We would run an experiment, get a number, and then spend as much effort asking *how the model might have cheated* as we had spent getting the number. And more than once, the interesting discovery was not in the model's behavior but in our own tools: the judge that mis-scored a working model, the data generator that leaked the operation, the pilot task that a lookup table could solve perfectly.

This is the honest shape of empirical work. It is seldom a straight line from hypothesis to clean result. It is a loop of building, testing the builder, being surprised, and revising. The surprise is not a detour from

the method; it is the method. And a book about science should show that loop, not hide it.

25.5. Where this brew leads

The honest map points at specific next experiments, not vague "more data / bigger model":

1. **Does *structural* diversity of combinations unlock composition?** We tested two things and both failed: (a) feeding a few composites during training (held-out still 0.000), and (b) diversifying the *surface* of the composite exposure (many phrasings × id-sets; held-out still 0.000). So the remaining open question is diversity of the *inner structure* — exposing the model to many *different* primitive pairings, not just different phrasings of the same few — to see whether that teaches a general compose-operator. That is the one lever we haven't isolated.
2. **Does the 10.2M in-distribution plateau replicate across configs?** The clean retrain measured 0.773 (a stable plateau, not the earlier 0.383 artifact). A proper training-duration and config sweep at 10M would confirm the plateau is robust rather than a single-seed coincidence.
3. **Does continuing to scale past 30M shift the boundary?** The grid stops where the brew lab's CPU becomes impractical. A larger-scale run (or a GPU) would probe whether the composition wall eventually moves.
4. **Does diversity fix composition?** Diversity fixed the *surface* problem (0.000 → 1.000). Whether the *same* lever helps composition is the natural next test of the "structural data" hypothesis.
5. **Human-style training**. The 0/16 human test set shows the model fails on naturally-phrased prompts. Training on diverse, human-style prompts — the lever that already fixed surface robustness — is the obvious path to real usability.

25.6. The honest ending

The model was the instrument. The experiment was the subject. We did not build a general programmer, and we will not pretend otherwise. What we *did* build is the means to ask, precisely and honestly, what a small model can and cannot learn — and we got an answer worth having:

> A small model learns narrow tasks well, but its knowledge is surface-tied and non-compositional, and within the tested range neither more capacity nor more data turns it into robust, compositional capability.

And the brakes we learned — test the tester, test the teacher, compare against baselines, audit shortcuts, preserve failures — are the durable part. That is the method worth keeping, and it's the reason the book's real subject was never the model. The brew was never only about the model; it was about learning to ask a question carefully enough that the answer could be trusted.

25.7. Where this sits: related work

None of the ideas here is new in isolation, and we want to say so plainly rather than imply novelty. This book's contribution is not inventing a technique; it is assembling a small, fully-controlled laboratory and being unusually honest about what the results do and do not establish. The underlying disciplines have long histories, and a reader who wants to go deeper should look there:

- **Compositional generalization**. The question of whether a model can combine known pieces in novel ways is a longstanding benchmark problem (e.g. the SCAN and COGS datasets). Our finding — that a tiny model does not compose under our tests — is a narrow case of a much-studied puzzle, studied far more thoroughly in that literature.

- **Execution-based code evaluation**. Running generated code against hidden tests is the standard for code-generation evaluation (the pass@k / HumanEval tradition). We borrowed that idea and applied it at a miniature scale, and our mutation-control work echoes the broader practice of testing the evaluator itself.
- **Shortcut learning**. "The model found an easier signal than the one we intended" is a well-documented phenomenon; our shortcut audit is a small, disciplined instance.
- **Catastrophic forgetting and rehearsal**. The effect we measured in Chapter 20 is a classic finding; the observation that keeping old data in the mix helps is also long-established.
- **Mutation testing**. Originating in software engineering, it is the source of our judge red-team.

We cite these traditions to place the book in context, not to claim we advanced them. If a reader wants the canonical treatments, the benchmarks and papers named above are the natural next step.

25.8. Reading further

- For compositional generalization benchmarks: SCAN, COGS (among others).
- For execution-based code evaluation: the pass@k / HumanEval literature.
- For shortcut learning and spurious correlation: the shortcut-learning literature.
- For continual learning and rehearsal: the catastrophic-forgetting literature.
- For mutation testing of test suites: the software-testing literature.

We did not benchmark against these in this book (the task and scale are far smaller), so we make no claim of competing with them — only of building a small, well-controlled instance and reporting it honestly.

25.9. Three short exercises

If you want to check that you have the book's discipline, try these. Answers are in the back of this chapter.

1. **Coverage or generalization?** You train a model on three wording families and evaluate it on a fourth that it has never seen. It scores 0.900. Is that a coverage result or a generalization result? What if instead you evaluated it on a *rephrasing within one of the three families it saw* — is *that* same interpretation?

2. **Distinguish two programs.** For the two functions `abs(a + b)` and `abs(a) + abs(b)`, find one input where they give the same answer and one where they differ. Why does a single passing test not distinguish them, and what kind of test does?

3. **What does a zero establish?** A model scores 0.000 on a held-out composition split. Write down what this does and does *not* establish — name at least two other causes of a zero that the result does not rule out, and one thing you could run to distinguish them.

(Answers: 1. The first is generalization to an excluded family; the second is coverage of a surface within the training family — only the first speaks to a never-seen family, and the book keeps them apart. 2. For `a = 2, b = 3` both give `5`; for `a = -3, b = 2` they give `1` and `5`. One test can be satisfied by both programs, so a distinguishing test must use a signed input where the two differ. 3. The zero shows that, under this decoding/evaluation procedure, no program passed every hidden test. It does not identify the cause (instruction interpretation, syntax generation, length handling, coverage, optimization, or a particular held-out operation), is not proof of a general capacity limit, and is not proof that the model "cannot compose" — a per-stage failure report or an all-composites control would narrow which cause holds.)

CHAPTER A.

The Generator, Every Piece

Chapter 5 describes the teacher conceptually. This appendix shows the actual source, so we can see — and reproduce — exactly how the data was made. All of this lives in `src/data_generator/` and `src/program_generator/`, `src/nl_renderer/`, `src/test_generator/` in the companion repository.

A note on what follows: this is a **condensed, annotated excerpt** of the real generator, not a complete runnable file. Where a section repeats in the same style for every operation, we mark it with an ellipsis (`...`). A few helpers that would be defined elsewhere in the source — the `OPERATIONS` table, `_rand`, and `_expected` — are used here as-is; the point is to show the *shape* of each part and how the semantic record flows through it, not to provide every line that would appear in a standalone module. The full, runnable source is in the repository.

A.1. The one line that ties it together

The whole generator is this single function. Given an operation and a seed, it builds the semantic intent and then generates everything else from it:

```
[] def make_task(operation, task_id, seed, id_set="letters", split="train", family=None): rng = random.Random(seed) sem = make_semantics(operation, rng, id_set) # the ground truth ref = generate_reference(sem, rng) # the correct Python prompt, fam_ name = render_nl(sem, family=family, rng=rng) # the English tests = generate_hidden_tests(sem, rng=rng) # the hidden tests return { "task_id": task_id, "semantics": sem, "reference_code": ref, "nl": {"prompt": prompt, "template_family": fam_name}, "hidden_tests": tests, "metadata": {"seed": seed, "identifier_set": id_set, "para- phrase_family": fam_name}, "spec_version": "python_capability_ vo_1", "dataset_version": "vo_1", "split": split, }
```

That's the whole architecture in ten lines. Everything below is one of the four building blocks it calls.

A.2. The semantics

```
[] IDENTIFIER_SETS = { "letters": ["a", "b", "c", "x", "y"], "words": ["left", "right", "first", "second", "value", "numbers", "items"], }

def make_semantics(operation, rng, id_set="letters"): n = OPER- ATIONS[operation] # arity names = list(IDENTIFIER_SETS[id_ set][:n]) # pick names from the set arg_type = ("list" if operation in ("list_sum", "list_len", "first_element") else "int") # Every oper- ation here returns an int scalar. Lists appear only as INPUTS # (a list of ints); list_sum/list_len/first_element all return an int, not # the list itself. So output_type is always "int"; only input_types vary. out- put_type = "int" args = [{"name": names[i], "type": arg_type, "role": "operand"} for i in range(n)] return {"operation": operation, "args": args, "input_types": [arg_type] * n, "output_type": output_type}
```

A.3. The reference program

The reference generator reads the semantics and emits the correct Python. A few cases:

```
[] def generate_reference(sem, rng=None):  op = sem["operation"];
names = [a["name"] for a in sem["args"]] if op == "add": return
(f"def solve({', '.join(names)}):\n" f" return {names[0]} + {names[1]}")
if op == "list_sum": return (f"def solve({names[0]}):\n total = 0\n"
f" for x in {names[0]}:\n total = total + x\n" f" return total") # ... and
so on for every operation, in the same style.
```

A.4. The natural-language renderer

The renderer phrases the same semantics several ways. Each **template family** is a function:

```
[] def _a(op, names, rng): # family "a"  x, y = names[0], names[1]
return (f"Write a function that takes two numbers, {x} and {y}, "
f"and returns their {_op_noun(op)}.")

def _b(op, names, rng): # family "b"  x, y = names[0], names[1]
return (f"Implement solve so that given {x} and {y} " f"it returns
their {_op_noun(op)}.")

def _c(op, names, rng): # family "c"  x, y = names return (f"Define
a function solve({x}, {y}) " f"that returns {x} {_op_sym(op)} {y}.")

# Each operation has its own set of families; _op_noun/_op_sym map
an operation # to the right word ("sum", "product", ...) and symbol
("+", "*", ...), so the # same families work for every operation, not just
addition. FAMILIES = {"add": [_a, _b, _c], "mul": [_a, _b, _c], ...}
```

The semantics is fixed; only the wording changes. That is the entire point.

A.5. The hidden tests

Tests are generated from the semantics, with deliberate edge-case coverage. The expected value is computed by the operation itself — not copied from the reference:

```
[] def generate_hidden_tests(sem, rng=None, count=8): op = sem["operation"]; args = sem["args"]; tests = [] def add(*vals, note="basic"): tests.append({"args": list(vals), "expected": _ expected(op, vals), "note": note}) add(_rand(rng), _rand(rng), note="basic") # ordinary add(0, _rand(rng), note="zero") # zero add(-_rand(rng), -_rand(rng), note="negative")# negative add(_ rand(rng, 900, 1000), _rand(rng), note="large") # large # … edge cases for the specific operation return tests[:count]
```

A.6. The leakage audit

Finally, the generator is checked. The audit trains a trivial classifier to see whether the wording leaks the operation:

```
[] vec = TfidfVectorizer() X = vec.fit_transform([t["nl"]["prompt"] for t in tasks]) clf = LogisticRegression(max_iter=1000) scores = cross_val_score(clf, X, y, cv=5) print(f"classifier predicts operation: {scores.mean():.3f}") # high = strong overlap; see note below
```

When we ran this on the first pilot renderer, it printed **1.000** — the wording perfectly predicted the operation. It is important to read this correctly: much of that is **task signal**, **not leakage**. The prompt must name the operation ("sum", "absolute value"), so a classifier can trivially recover it from the wording; that is the task being solvable as stated, not a defect. The audit's real purpose is to catch the *opposite* problem — where some *other* accidental cue (a duplicated example, a leaked hidden-test value, an identifier that reveals the answer) is what drives the score. So we treat a high score as a flag to inspect *which* words drive it, not as proof of leakage by itself.

A.7. How to run it

[] python scripts/generate_data.py --n-per-op 200 *# writes pilot_train/val/test + manifest* python scripts/audit.py *# runs the leakage audit*

That's the teacher. It's not magic — it's a controlled generator, audited for fairness, that makes the entire experimental program possible.

CHAPTER B.

The Failure Ledger

The design requires that we do not sanitize the journey into a success story. This is the honest record of everything that went wrong, was surprising, or was a failed attempt — because each of these taught us something, and several are as valuable as any success.

Each entry: what happened, what we learned, and what we did about it.

B.1. The evaluator reported the model at 0.000 — but the model was fine

What happened. On the very first Stage-1 evaluation, the trained model scored a reported **0.000** (syntax failure on every problem). This looked like a total failure of the approach.

What we found. The model was actually producing programs that were correct up to the end of the program — but our generation output ap-

pended null-byte padding, and Python's parser *correctly* rejects null bytes (they are not valid source). The fault was in our **extraction contract**: we handed the parser padded output without stripping the padding. So the **judge** was scoring what were, after truncation, valid programs as syntax errors.

What we learned. The most important lesson of the book, demonstrated on ourselves: *never trust a number from a system we have not audited.* Fixing the judge revealed the model's true 0.897. If we'd taken 0.000 at face value, we'd have wrongly concluded the approach failed.

Action. Fixed `generate_code` to truncate at the first null byte; reran; got 0.897.

B.2. The pilot wording was so predictive of the answer we had to look again

What happened. The leakage audit trained a trivial TF-IDF classifier to predict the operation from the natural-language prompt. It succeeded **100%** of the time.

What we learned. At first we recorded this as "the wording leaks the answer," but on reflection most of it is **task signal, not leakage**: the request is supposed to name the operation ("sum", "absolute value"), so a classifier recovering it from the wording is the request doing its job. What it does tell us is that a model "succeeding" might just be exploiting that phrasing rather than learning the computation — which is *why* the surface-generalization tests matter. We therefore treat the 100% score as a flag to inspect *which* words drive it (operation words vs. an accidental cue), not as proof of contamination.

Action. Recorded as finding-001 (with the interpretation corrected above); noted the pilot is fine for Stage 1 but must add wording-family holdouts before any surface test.

B.3. A bigger model did NOT fix composition

What happened. We scaled the model across a ~33× range (0.88M → 29.3M) expecting the composition boundary to move. It did not: composition stayed at **0.000** at all four sizes (0.88M, 6.5M, 10.2M, 29.3M).

What we learned. At ~1–30M parameters, under the training schedules we ran, compositional generalization stayed at 0.000 and was **not moved by scaling alone**. We are careful not to call this a capacity "ceiling" — it is a measured boundary in this regime, not proof that capacity cannot help under other conditions. It is a genuine negative result that redirects the investigation.

Action. Recorded in `docs/experiments/scaling_composition.md`.

B.4. Sequential training caused catastrophic forgetting (A → 0.000)

What happened. Teaching the model lists after arithmetic, sequentially, completely destroyed the arithmetic (A-retention **0.000**).

What we learned. Catastrophic forgetting is real and severe here — but it is a **procedure** problem, not a fate. Replay (interleaving A and B) retained A perfectly (1.000) with zero cost to learning B.

Action. Recorded; the chapter states this clearly — the protocol, not capacity, determines the outcome.

B.5. The pilot task was "too easy" (retrieval solved it)

What happened. The retrieval baseline scored **1.000** on the pilot test set, and the model scored 0.897. So a non-learning lookup did better than the model.

What we learned. The pilot task, as constructed, is retrieval-solvable — so it does not yet test whether the model *generalizes*, only whether it matches surface form. This is exactly the "task too easy" outcome the plan anticipated.

Action. We did not hide it; we used it to justify harder splits (surface, then composition), which produced the real findings above.

B.6. The human-written test set: 0/16

What happened. The trained model solved **0/16** naturally-phrased human prompts — even simple "add two numbers" phrased as a person would.

What we learned. This is the external, real-world corroboration of the surface finding. The model learned a *template*, not the *task*. The moment a real user phrases a request differently, it fails.

Action. Recorded as an external sanity check; it sharpens the book's honest conclusion.

B.7. The free-model proofread returned degraded output

What happened. We dispatched the 85K-char draft to two free models for a proofread. One leaked its chain-of-thought and broke into noise; the other timed out. No usable independent review was collected.

What we learned. Free endpoints are flaky for long documents, and we should not fabricate fixes from unusable output. We were honest about this rather than pretend a review happened.

Action. Noted honestly; the book's line-edit stands on its own merit, not a fabricated review.

B.8. Transient SIGILL crashes during training

What happened. Long CPU training runs occasionally crashed with signal 4 (SIGILL) on the AVX-only Ivy Bridge CPU — intermittently, one run succeeds and the next dies.

What we learned. Hardware quirks are real and part of the brew. The apparent fix was `TORCH_DISABLE_AVX2=1` plus a retry loop (and `OMP_NUM_THREADS=1` for larger models). We report this as a *workaround we used*, not as a confirmed root-cause repair: the crashes were intermittent, so a successful retry after setting the variable does not by itself prove the variable is what fixed it, and we did not independently verify that the installed torch build actually reads this environment variable. All we can claim is that wrapping runs in a retry loop made them succeed reliably.

Action. Wrapped all long runs in a retry loop; documented the workaround.

B.9. Parameter-math and naming mistakes (caught by testing)

What happened. Early writing had a parameter-count mis-derivation (didn't sum to the true 884K/884,224) and the human-test evaluator initially referenced a non-existent key.

What we learned. Always verify arithmetic against the real build, and let the "test the tester" discipline catch even our own book's numbers.

Action. Recomputed from the actual build; fixed the evaluator.

B.10. A diagnostic schema we propose but did not measure

Many of the failures above are reported as a single "0.000" pass@1 number. That is honest but blunt: it does not tell a reader *why* a run failed. A candidate program can fail at several distinct points — it never parses, it parses but the judge rejects it, or it passes the judge but is the wrong kind of program.

For each of the four judge stages we would like to record not just pass/fail but **failure-stage counts**: how many candidates fell over at the parse of the reference, at the numerical comparison, at the comparison of the mutation count, or at the pass/fail of every hidden test. That lets a reader see whether a 0.000 is a *syntax collapse* (the model emits garbage) or a *semantic near-miss* (the model emits a coherent program that is just wrong).

> We did **not** collect these per-stage counts for the experiments in this book — the evaluator recorded only the final verdict, not the intermediate stage at which each candidate stopped. Adding stage-level accounting is straightforward (the four judge stages already exist as functions, so we would count candidates at each one) and we leave the data and the figure as proposed work, not as a measured result.

The point of this ledger is the opposite of marketing. Every entry above, in its own way, made the final understanding sharper — and several of them — the broken evaluator, the wording leak, the task-too-easy result, the 0/16 human test — are among the most valuable findings in the book precisely because they forced us to be honest.

CHAPTER C.

The Experiment Registry

Every result in this book is traceable to a run, a configuration, a tokenizer, and a checkpoint. This registry is the single place to look up each one: what was tested, how it was run, where the numbers live, and what is *not* tested. Where a cell could not be measured we say "not tested" rather than "0" — an unmeasured quantity is not a zero.

C.1. How to read a run

Each entry names an **experiment** (what question it asks), a **run ID** (a stable label that appears in figures and tables), the **architecture** (`n_embd` / `n_layer` / `n_head`, all on a `block_size` of 128 and a `vocab_size` of 291), the **training steps**, the **seed**, and the **artifact** (manifest + checkpoint) that holds the record. The pass@1 figures are generated by the execution-based judge over a fixed number of held-out prompts; those counts are given where recorded.

C.2. The operation dictionary

Every operation has exactly one definition across renderer, oracle, reference, and tests. This table is the ground truth. All operations take either `int` inputs or a **list of ints**, and every one returns an `int` scalar. There are no list-returning operations in the task family.

ID	arity	input	reference implementation	phrase
`add`	2	int, int	`a + b`	sum of two numbers
`sub`	2	int, int	`a - b`	difference of two numbers
`mul`	2	int, int	`a * b`	product of two numbers
`max_of`	2	int, int	`max(a,b)`	maximum of two numbers
`min_of`	2	int, int	`min(a,b)`	minimum of two numbers
`abs_of`	1	int	`abs(a)`	absolute value
`list_sum`	1	list[int]	`sum(xs)`	sum of a list
`list_len`	1	list[int]	`len(xs)`	length of a list
`first_element`	1	list[int]	`xs[0]`	first element
`cond_abs_max`	2	int, int	`max(abs(a),abs(b))`	abs of the larger magnitude

ID	arity	input	reference implementation	phrase
`abs_add`	2	int, int	`abs(a) + abs(b)`	sum of two absolute values
`sub_abs`	2	int, int	`abs(a - b)`	absolute difference
`sum_of_abs`	1	list[int]	`sum(abs(x) for x in xs)`	sum of absolute values of a list
`max_add`	2	int, int	`max(a+b, a-b)`	larger of sum or difference
`double_first`	1	list[int]	`2 * xs[0]`	twice the first element
`len_minus_one`	1	list[int]	`len(xs) - 1`	one less than the length

Note on `abs_add` versus `abs(a+b)`: `abs_add` computes `abs(a) + abs(b)` — the sum of two absolute values — *not* the absolute value of the sum, `abs(a + b)`. These are different functions (for `a=-3, b=2` they give `5` and `1`). The book refers to `abs_add` as the sum of absolute values throughout. `sum_of_abs` is the list analogue: it takes a **list** and sums the absolute values of its elements, so it must not be confused with `abs_add` (which takes two ints).

C.3. Registry

Experiment	Run ID	Arch (emb/L/H)	Steps	Seed	Model params	Artifacts (manifest / checkpoint)	Key result
Stage 1 pilot	`stage1_seed0`	128/4/4	1500	0	884,224	stage1.json / stage1.pt	in-dist 0.897; loss 5.84→0.019
Stage 3 surface	`stage3_surface`	128/4/4	1500	0	884,224	stage3_surface.json / stage3_surface.pt	in-dist 0.900; new-IDs 0.350; held-out wording 0.000
Stage 4 composition baseline	`stage4_pilot`	128/4/4	1500	0	884,224	(uses pilot ckpt) / stage1.pt	in-dist 0.900; composition 0.000
Stage 4 — 6.5M	`stage4_6p5m`	256/8/8	1000	0	6,500,352	stage4_composition_6p5m.json / stage4_6p5m.pt	in-dist 0.897; composition 0.000
Stage 4 — 10.2M	`stage4_10m`	288/10/8	800	0	10,195,776	stage4_composition_10m.json / stage4_10m.pt	in-dist 0.773; composition 0.000
Stage 4 — 29.3M	`stage4_29m`	448/12/8	800	0	29,290,240	stage4_composition.json / stage4_composition.pt	in-dist 0.897; composition 0.000
Data scaling	`data_scaling`	128/4/4	800	0	884,224	data_scaling.json / —	in-dist 0.100→0.641 (100→3000)

Experiment	Run ID	Arch (emb/L/H)	Steps	Seed	Model params	Artifacts (manifest / checkpoint)	Key result
Quantity vs diversity	`diversity`	128/4/4	900	0	884,224	diversity.json / —	minimal 0.500/0.000; high 1.000/1.000
Memorization	`memorization`	128/4/4	1500	0	884,224	memorization.json / —	59/60 held-out; canary → general program
Continual learning	`continual`	128/4/4	700+700		884,224	continual.json / —	sequential A 0.000; joint A 1.000, B 0.889
Composition exposure	`composition_exposure_pilot`	128/4/4	1500	0	884,224	composition_exposure_pilot.json / stage4_exposure.pt	exposed 0.667; held-out 0.000; in-dist 1.000
Composition diversity	`composition_diversity_pilot`	128/4/4	1500	0	884,224	composition_diversity_pilot.json / stage4_compdiv_{minimal,high}_diversity_pilot.pt	exposed 0.550-0.592; held-out 0.000

C.4. The scaling run: an honesty note on "same protocol"

The model-size sweep is often summarized as "same training protocol, only size changed." That is **not strictly true**. The 6.5M run trained for **1000** steps while the 10.2M and 29.3M runs trained for **800** steps. The larger models also have a larger context budget (larger `block_size`-scaled em-

beddings) and a larger tokenizer-independent parameterization. We did **not** hold every training hyperparameter constant across the size grid. So the size comparison is *not* a single-variable, size-only experiment — training-budget and width/layer counts vary alongside size. We state the caveat here rather than imply a strictly-controlled sweep: what we can claim is that *composition stayed at 0.000 at every size under the schedules we ran and evaluated*. We cannot claim a clean size-isolate from the in-distribution differences (0.897 / 0.773 / 0.897), which may reflect the differing training budgets as much as the differing sizes.

C.5. The composition baseline: where the 0.88M number lives

The book's "Stage-4 baseline" (0.88M, in-dist 0.900, composition 0.000) comes from evaluating the **pilot checkpoint** (`stage1.pt`) on the composition split — not from `stage4_composition.json`. That manifest, despite being named "stage4_composition", is actually the **29.3M** run (n_embd=448, n_layer=12). So the small-model and large-model composition numbers come from different artifacts, and the registry above separates them explicitly. The 0.88M composition result should be read as the pilot model evaluated on unseen composite operations; the large-model runs are separate training runs.

C.6. Denominator and uniqueness notes

- **pass@1 denominators.** The pilot used 300 held-out test prompts (1400 train / 300 val / 300 test). Stage 4 used 300 held-out primitives for in-distribution and 240 composite tasks (6 ops × 40) for composition; the judge then scores each. The data-scaling rows use their own held-out split per size. Exact successes/totals are recorded in each manifest.
- **Unique model-visible prompts ≠ task IDs.** In the teacher, a task-ID seed changes the *hidden-test inputs*, so a new task ID can change the test values while the prompt and reference program stay identical. A

reported "held-out" prompt therefore is only meaningful if we have checked the *prompt string* (the model input) is new, not merely the task ID. We flag prompts-as-model-inputs explicitly where it matters; see the deduplication audit note below.

- **Tokenizer**. Tokenizers are trained on the combined train prompts + reference programs to a vocabulary of 291. The pilot tokenizer (`pilot_tok`) and the composition tokenizer (`comp_tok`) differ by a couple of tokens because the composition tokenizer was trained over composite reference programs too. A model must be evaluated with the tokenizer it was trained on; mixing tokenizers changes the embedding alignment. The registry entry's manifest records which tokenizer was used.
- **Training loop**. Fixed step count (no early stopping from validation loss); the reported loss is the final *training* loss, not a held-out measure.

C.7. What is NOT tested (explicitly)

- **Multiple seeds per size**. Each size was trained once, with one seed. We cannot attach error bars to the in-distribution spread (0.897 / 0.773 / 0.897) in Chapter

  12. Treat it as approximate, not an exact plateau.

- **Genuinely-held-out wording families for the *diversity* run**. The ch13 high-diversity model trained on *all* families × both id-sets, so its 1.000 surface score is coverage, not generalization to a never-seen family. The ch15 surface model, by contrast, trained on *all-but-the-last-family* per operation and was tested on that held-out last family — so its 0.000 is a genuine held-out-family result. These two are different situations and the book keeps them separate.
- **Conditionals, indexing, loops as isolated operations**. `list_sum`'s reference uses a loop and `first_element` uses indexing, but the *task family* does not test general loop or indexing competence as stan-

dalone operations; those appear only inside specific composites.

- **Composition beyond ~30M** (and the true "structural vs. above-our-ceiling" question) is not tested — the CPU lab becomes impractical beyond that.
- **A true replay-after-A-first** continual-learning design (vs the joint-training control we ran) is not tested; see the Chapter 20 caveat.
- **Mutation-testing acceptance** and a *surviving* mutant report showing the judge caught an error — a gap the book should close (see Chapter 17) before claiming the judge's reliability is demonstrated.

C.8. Reproducibility & the companion repository

Each run script lives in `scripts/` (e.g. `run_composition.py`, `run_diversity.py`, `run_composition_exposure.py`, `run_composition_diversity.py`, `run_surface.py`, `run_data_scaling.py`, `run_continual.py`, plus `train.py` and `eval_ckpt.py`). The recommended release tag is `v20261001`. Here is a concrete quick start:

1. **Clone the companion repository** (see the front matter for the URL) and check out the `v20261001` tag.
2. **Create an environment**. The runs were done on Python 3.11.16 with PyTorch 2.14.0+cpu (CPU build) and no GPU. A `requirements.txt` at the repo root pins the versions.
3. **Set the environment variable** `TORCH_DISABLE_AVX2=1` on this CPU lab (to avoid a SIGILL on the Ivy Bridge AVX-only silicon).
4. **Run a pilot**: `python scripts/train.py --steps 1500 --n-embd 128 --n-layer 4 --n-head 4 --bs 16 --seed 0` then `python scripts/eval_ckpt.py`. This trains and evaluates the 0.88M pilot and writes its manifest.
5. **Expected artifacts**: the run's manifest (e.g. `stage1.json`) and checkpoint (`stage1.pt`) under `artifacts/manifests/` and

`artifacts/checkpoints/`.

Runtime. Roughly 5-10 minutes per 1500-step pilot training on a single-core CPU; larger models (6.5M-29.3M) take substantially longer (minutes to low tens of minutes each). These are hardware-specific and will differ on your machine; the [MEASURED] numbers in the book are the ones from *our* runs.

Verified commands. The commands above and the evaluator used to produce the pass@1 results are the ones that ran for this book; the scripts are deterministic given the same seed and configuration (the run scripts fix the seed and hyperparameters in the registry).

A production note on the free-endpoint side. We do not rely on any external service for the results in this book; everything was trained and evaluated locally on the CPU lab described above, so the numbers do not depend on a hosted endpoint staying up or behaving.

CHAPTER D.

Glossary

Short definitions of the terms used in this book. Where a term has a standard meaning that we follow, we say so; where we use it in a deliberately narrow way, we flag it.

- **AST (Abstract Syntax Tree).** A tree representation of a program's structure, stripped of formatting. The sandbox builds an AST of the generated program and checks that it uses only allowed constructs.
- **BPE (Byte-Pair Encoding).** A tokenization method that repeatedly merges the most common symbol pairs into a single token. It turns text into a fixed-size vocabulary of subword pieces.
- **Baseline / rival.** A deliberately simple reference system (here, a keyword template-matcher and a TF-IDF retriever) used to interpret the model's score. A baseline tells us whether the task was already easy for a simpler machine.
- **Canary (held-out probe).** Here, a never-seen prompt used to check whether the model reproduces a stored value or produces a fresh gen-

eral program. We use the term in the held-out-probe sense, not the classic memorization-canary sense.

- **Checkpoint**. A saved copy of the model's weights at a point in training.
- **Composite operation**. An operation that is a combination of two primitives the model already knows (e.g. `abs_add = abs(a) + abs(b)`).
- **Continual learning**. Learning a new capability after an old one, and whether the old one is retained.
- **Distribution shift**. When the test inputs come from a different distribution than the training inputs. Surface change (new wording) is a mild case.
- **Greedy decoding**. Choosing, at each step, the single most likely next token, with no sampling. The book reports greedy pass@1.
- **Hidden tests**. Input/expected-output pairs used to grade a generated program. In this book they are evaluator-only artifacts: they never enter training.
- **In-distribution**. Tasks drawn from the same distribution as training (here, held out in *instances* but the same operation / surface family).
- **Judge**. The execution-based evaluator: it runs the generated program against hidden tests and compares outputs (with a semantic comparator and mutation control).
- **Mutation testing**. Generating deliberately-broken programs and checking that the hidden-test suite catches them, to measure how discriminating the tests are.
- **pass@1**. The fraction of evaluation prompts for which the model's (greedily decoded) program passes all hidden tests.
- **Primitive**. A single, non-composite operation (e.g. `add`, `abs_of`,

`list_sum`).

- **Sandbox / Gate A**. The software layer that runs generated programs and rejects any that use a construct outside an allowlist (e.g. `import`, `eval`), before they run.
- **Semantic comparator**. The rule that decides whether an output equals an expected value. Beyond Python's `==`, it adds float tolerance and strict bool/int handling.
- **Surface / surface form**. The exact wording and identifier names of a request, as opposed to its underlying meaning.
- **Train/val/test split**. The division of a dataset into the examples used to train, to monitor training, and to evaluate (held out). In this book the splits are semantically disjoint by operation or wording family.
- **Validator / Gate B**. The software layer that sits in front of the model and declines any incoming request that is not in the designed scope, before the model sees it.